FlowPilot: An Agentic Research Assistant for End-to-End Flow Process Design

Amirreza Mottafegh[a,†], Mincheol Park[b,†], Jun-young Jo[b], Jin Suk Myung[a], Boyoung Y. Park[b,*], and Gwang-Noh Ahn[a,*]

a Digital Chemical Research Center, Korea Research Institute of Chemical Technology, 141 Gajeong-ro, Yuseong-gu, Daejeon 34114, Republic of Korea

b Department of Biomedical and Pharmaceutical Sciences, Kyung Hee University, Seoul 02447, South Korea

*Boyoung Y. Park, E-mail: boyoungy.park@khu.ac.kr

*Gwang-Noh Ahn, E-mail: gnahn@krict.re.kr

†Amirreza Mottafegh and Mincheol Park contributed equally to this work.

Table of contents:

1 Related flow-chemistry and process-design agents....4

2 FlowPilot workflow, data contracts, and run provenance....5

2.1 System architecture and decision authority....5

2.2 Typed contracts and numerical closure....5

2.3 Reproducible standardized intake....7

2.4 GUI execution and single-source result rendering....9

2.5 Run-level provenance and reconstruction....9

3 Literature corpus and engineering rule base....10

3.1 Frozen literature-corpus composition....11

4 Ablation test and architecture comparison....15

5 Deterministic engineering, inventory realization, and refinement....31

5.1 Residence-time and gas-flow conventions....31

5.2 Multistage calculation and stream introduction....32

5.3 Heat-transfer screening....32

5.4 Inventory normalization and executable topology....32

5.5 Incremental experimental refinement....33

5.6 Model routing and run records....33

6 GUI operation, inventory management, and auditable examples....34

6.1 Batch entry, follow-up questions, and saved answers....34

6.2 Inventory input, normalization, and laboratory constraints....36

6.3 Examples of critical findings and evaluator limitations....38

6.4 Computational dry-run example: bromination and inline quench....39

7 Connected-process case studies supporting main-text Figures 5 and 6....41

7.1 Photoredox Giese addition followed by sulfoxidation (main-text Figure 5)....41

7.2 Telescoped DPDTC-mediated amidation (main-text Figure 6)....47

7.3 Recorded council deliberation for the two connected-process cases....51

8 Laboratory implementation, experimental procedures, and analytical data....97

8.1 General materials, instrumentation, and analytical methods 97
8.2 Irradiation sources and recorded specifications 98
8.3 Photoredox Giese addition and oxidation: preparation and operation 99
8.4 DPDTC-mediated amidation: preparation and operation 102
8.5 Product characterization and supplied spectra 105
8.6 Implementation reconciliation and calculation basis 110
8.7 Outstanding experimental metadata 112
9 References 113

1 Related flow-chemistry and process-design agents

Table S1 summarizes the tasks, outputs and reported validation of six related flow-chemistry or process-design agents [1-6]. Journal articles, a conference paper and preprints are distinguished. Different published evaluations are not treated as a common performance benchmark.

Table S1. Scope of FlowPilot and related flow-chemistry/process-design agents. This comparison summarizes reported tasks and validation; it is not a performance ranking. Source numbers refer to the ESI bibliography.

Study	Design task and output	Reported scope and validation
Chat-microreactor [1]	Flow-pattern prediction and microreactor guidance.	Local hydrodynamics and a piperacillin design application.
LLM-RDF [2]	Synthesis development, reactor design and experimental workflows.	Experimental execution, kinetic analysis, optimization and purification.
SapoMind [3]	Condition recommendation for continuous-flow saponification.	Experimental optimization of the selected microreactor reaction.
Text-to-Simulation [4]	Process topology and parameter generation for simulation.	Executable flowsheets and simulator feedback.
CeProAgents [5]Preprint	Hierarchical conceptual and parameter design for chemical processes.	Industrial-process case studies and simulation.
CAAF [6]Preprint	Deterministic validation and bounded candidate revision.	Constructed reactor-design constraints; not a laboratory-inventory trial.
FlowPilot Present work	Reaction-level input to feed, stage and equipment assignments.	Matched architecture comparison; two connected-stage laboratory implementations.

The scope comparison distinguishes design tasks rather than ranking systems evaluated under different conditions. FlowPilot assists researchers through a laboratory-constrained, end-to-end flow-process design workflow: defining the brief, interpreting chemistry, retrieving evidence, reviewing candidate configurations, calculating operating conditions and assigning equipment. Batch protocols provide the reaction information for this workflow. The contribution is the coordinated development of a connected process, not text conversion alone. The matched architecture test compares the recorded one-shot and FlowPilot configurations, not the external systems in Table S1.

2 FlowPilot workflow, data contracts, and run provenance

FlowPilot supports researcher-led process design by coordinating chemical interpretation, engineering calculation, equipment realization and review of alternative configurations. The workflow begins with standardized intake of a batch protocol, design objectives and laboratory constraints, and returns an inventory-assigned design or explicit unresolved requirements. Here, end-to-end describes the design workflow, not autonomous laboratory execution or proof of optimized reaction performance. Interpretation follows an information-authority order: measured evidence, hard safety and inventory constraints, confirmed protocol facts, chemist hypotheses and model inference. Hypotheses remain ideas to test rather than observations. Earlier experimental success outside a declared operating limit does not authorize a new proposal to override that limit.

2.1 System architecture and decision authority

A FlowPilot run passes through standardized intake, batch parsing, upstream chemistry analysis, plan-aware retrieval, deterministic calculation and design-space exploration, downstream proposal generation, multi-agent council review, inventory reconciliation, final validation, and artifact publication (Figure S1a). The upstream chemistry layer identifies the transformation, mechanistic features, stage structure, stream incompatibilities, phase behavior, and candidate process bottlenecks. The retrieval layer supplies literature analogies and engineering rules relevant to that plan. These model-derived interpretations inform design, but they cannot supersede confirmed protocol facts, measurements, or hard operating limits. Deterministic components calculate and reconcile dependent quantities such as flow rate, reactor volume, residence-time basis, gas delivery, pressure settings, and inventory assignments. The council can criticize a candidate and propose bounded revisions, but a council response is not an executable output. Revised fields are recalculated and checked before publication. The final disposition therefore depends on deterministic closure rather than model confidence. Three user-facing outcomes are distinguished. An executable screening design contains canonical parameters and a fully assigned process topology. An inventory-confirmation state identifies the missing capability required before numerical execution. A blocked state records the failed requirements and diagnostic topology but withholds executable run instructions.

2.2 Typed contracts and numerical closure

Major pipeline boundaries are represented by validated data objects rather than prose alone (Figure S1b). The principal chain is DesignInputPackage > BatchRecord > ChemistryPlan > FlowProposal > DesignCalculations > inventory-assigned ProcessTopology > FinalDesignContract. Natural-language rationales may accompany these objects, but downstream code consumes typed fields and performs boundary validation.

The FinalDesignContract is rebuilt after engineering realization and inventory reconciliation. A publishable screening result contains canonical parameters, stage definitions, streams, instrument assignments, a process graph, chemistry identity, and the available operating and validation instructions. The implemented gates check consistency among these records, including arithmetic, declared time basis, equipment allocation, and topology. An unresolved blocking check prevents publication of executable parameters and retains intermediate values as diagnostics. Passing these checks establishes compliance with the implemented contract; it does not establish chemical completeness, physical safety under all transients, or experimental yield.

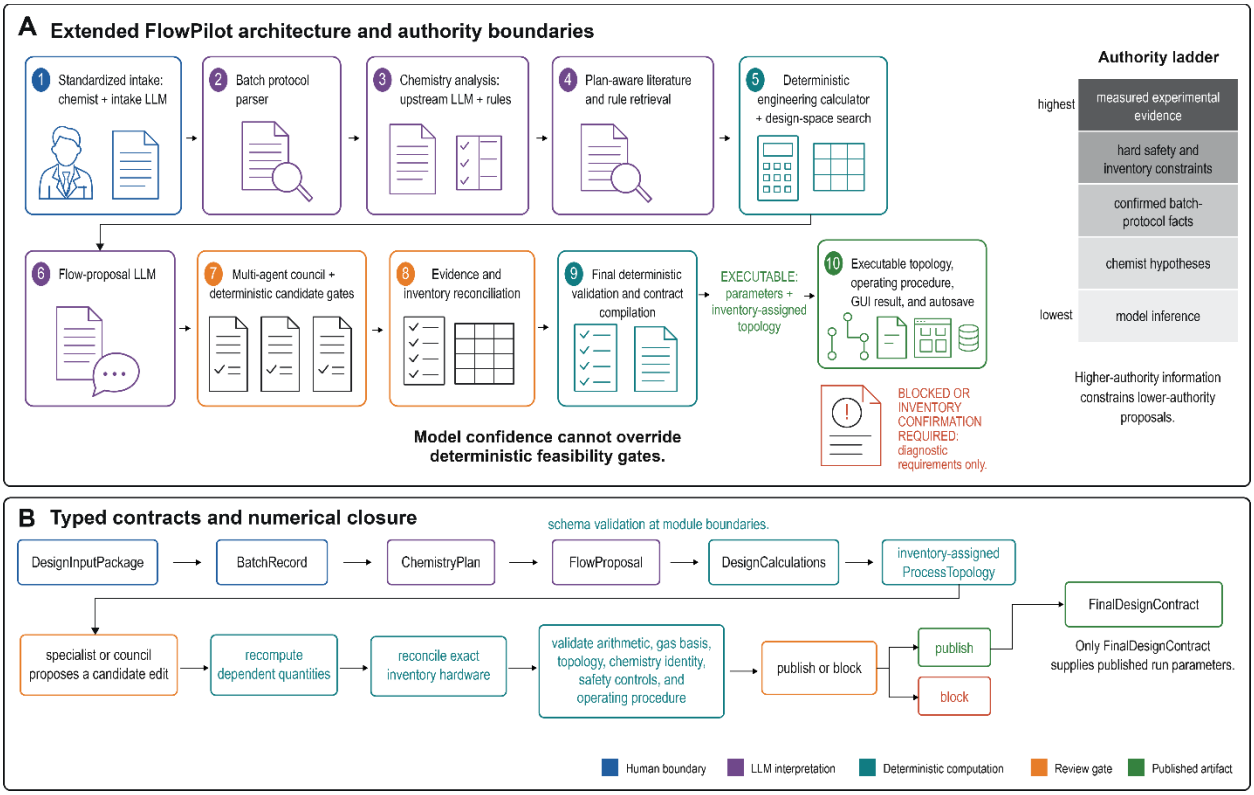


Figure S1. FlowPilot architecture, authority boundaries, and typed data contracts. (a) The workflow separates standardized intake, chemistry interpretation, retrieval, deterministic engineering, council review, inventory reconciliation, and artifact publication. Measured evidence and hard safety or inventory constraints outrank confirmed protocol facts, chemist hypotheses, and model inference. (b) Typed contracts and closure checks ensure that only a validated FinalDesignContract can publish executable parameters.

Table S2. FlowPilot components, data contracts, and decision responsibilities.

Component	Computational class	Principal input	Principal output	Decision responsibility
Standardized intake	LLM extraction + deterministic state rules	Free-text protocol and chemist answers	DesignInputPackage	Structures user information; fixed IDs and readiness rules remain deterministic.
Batch protocol parser	Structured parser + schema validation	Frozen protocol facts	BatchRecord	Normalizes batch conditions; cannot outrank the original confirmed protocol.
Chemistry analysis	Upstream LLM + retrieved fundamentals	BatchRecord, intake context	ChemistryPlan	Interprets mechanism, stages, sensitivities, and stream logic; does not publish hardware settings.

Component	Computational class	Principal input	Principal output	Decision responsibility
Plan-aware retrieval	Deterministic filtering and ranking	ChemistryPlan and retrieval query	Ranked analogies and rules	Supplies supporting precedents; retrieved analogies do not override measured evidence or constraints.
Engineering calculator and design-space search	Deterministic	BatchRecord, ChemistryPlan, inventory limits	DesignCalculations and feasible seeds	Owns arithmetic closure, candidate feasibility calculations, and bounded search outputs.
Flow-proposal agent	Downstream LLM	Chemistry, calculations, retrieval, inventory context	FlowProposal	Proposes a complete candidate and rationale; proposal remains non-executable until validation.
Multi-agent council	Specialist LLMs + deterministic gates	Candidate proposals and evidence	Selected or revised candidate	Critiques chemistry, engineering, safety, and practicality; revisions are bounded and recalculated.
Inventory reconciliation and topology compiler	Deterministic	Selected candidate and LabInventory	Inventory allocation and ProcessTopology	Binds required operations to declared equipment and rejects unresolved assignments.
Final validation and artifact compiler	Deterministic	Reconciled proposal, calculations, topology, safety context	FinalDesignContract	Owns executable/blocked disposition and canonical operating parameters.
GUI renderer and autosave	Deterministic presentation and serialization	FinalDesignContract plus diagnostic traces	Result views and stored artifacts	Publishes canonical outputs; diagnostic information is labelled and kept separate from run instructions.

2.3 Reproducible standardized intake

The intake layer converts the raw protocol and confirmed answers into a frozen DesignInputPackage. The question bank (flowpilot_questions_v1.1) contains eight core identifiers and five conditional identifiers (Table S3). The protocol activates conditional questions through deterministic gas, photochemistry and multistep triggers; the confirmed package state determines which answers are still missing. The LLM extracts information and interprets responses but cannot invent an identifier or override readiness. Protocol, objective and chemistry identity must be established; history, inventory, operating limits and hypotheses require either supplied information or an explicit unavailable status. Output preference is optional. Active conditional questions have their own resolution rules: unavailable is allowed for a gas-equivalent target or wavelength, but not for unresolved gas identity, gas-introduction stage or stage definition. Allowing an unavailable answer records uncertainty; it does not authorize unsupported hardware or safe operation. The

saved package retains effective answers and superseded history, question-bank version, active IDs, question-set hash, engineering requirements and provenance. For unchanged protocol, confirmed state and question-bank version, the ordered pending IDs are reproducible; extracted values, generated designs and conversational phrasing are not thereby guaranteed to be identical.

Table S3. Fixed standardized-intake question bank and readiness rules.

Question ID / information	Activation / readiness	Resolution	Design use
Q-BATCH-001 Batch protocol	Core; required	Confirmed protocol	Chemical facts and batch conditions
Q-OBJ-001 Design objective	Core; required	Answered	Design priorities and candidate ranking
Q-CHEM-001 Chemistry identity	Core; required	Confirmed transformation	Substrate/product identity and bond changes
Q-HIST-001 History	Core; required resolution	Supplied or unavailable	Measured evidence for refinement
Q-INV-001 Inventory	Core; required resolution	Supplied or unavailable	Equipment and capability constraints
Q-CONSTR-001 Operating limits	Core; required resolution	Supplied or unavailable	Operating and safety boundaries
Q-HYP-001 Hypotheses	Core; required resolution	Supplied or unavailable	Ideas to test, not measured facts
Q-PREF-001 Output preference	Core; optional	Answered or omitted	Requested presentation and screening format
Q-GAS-001 Gas identity	Gas domain; identity missing	Gas and active fraction established	Reactive-gas molar delivery
Q-GAS-002 Gas equivalents	Gas domain; positive target missing	Positive target or unavailable	Inlet/STP feed target; unknown remains provisional
Q-GAS-003 Gas-introduction stage	Multistep gas domain; stage missing	Introduction stage established	Gas mixer placement and downstream accounting
Q-PHOTO-001 Wavelength	Photochemistry; wavelength missing	Wavelength or unavailable	Light matching; unknown remains provisional
Q-MULTI-001 Ordered stages	Multistep domain; definition missing	Ordered stages and interstage operations	Stage boundaries and cumulative flows

2.4 GUI execution and single-source result rendering

In the standardized-intake GUI, the user submits a protocol, resolves the fixed intake questions, and imports or selects a laboratory-inventory profile before initiating design. The run action is enabled only when the package is ready. The finalized package is passed to the same translation pipeline used by programmatic execution, and the resulting package is attached to the result. After final validation, executable numerical views are constructed from the post-realization FinalDesignContract. These include the summary, engineering design, stream assignments, equipment allocation, process graph, safety controls, and canonical operating procedure. Chemistry-plan, retrieval, council, and experiment-loop records remain available as supporting context or audit traces. They are not independently promoted as authoritative run parameters. For a blocked run, the GUI displays the failure reasons and a requirements or diagnostic topology while withholding an executable procedure.

2.5 Run-level provenance and reconstruction

Completed GUI runs are automatically written to a uniquely timestamped directory. The artifact set preserves the standardized input, complete result, canonical final contract, concise summary, compiled topology, equipment allocation, instrument manifest, and available rendered diagrams. Diagnostic topology and rendering artifacts are stored separately when the run is blocked or requires inventory confirmation. The render manifest can associate diagram files with topology-hash metadata, allowing a stored diagram to be checked against the corresponding graph where that metadata is available. The autosave package supports inspection of what was entered, calculated, validated, displayed, and assigned to hardware. It should not, by itself, be described as a complete computational-replay bundle because separate prompt logs, exact model snapshots, decoding parameters, dependency versions, and software revision identifiers are not guaranteed by the GUI autosave function.

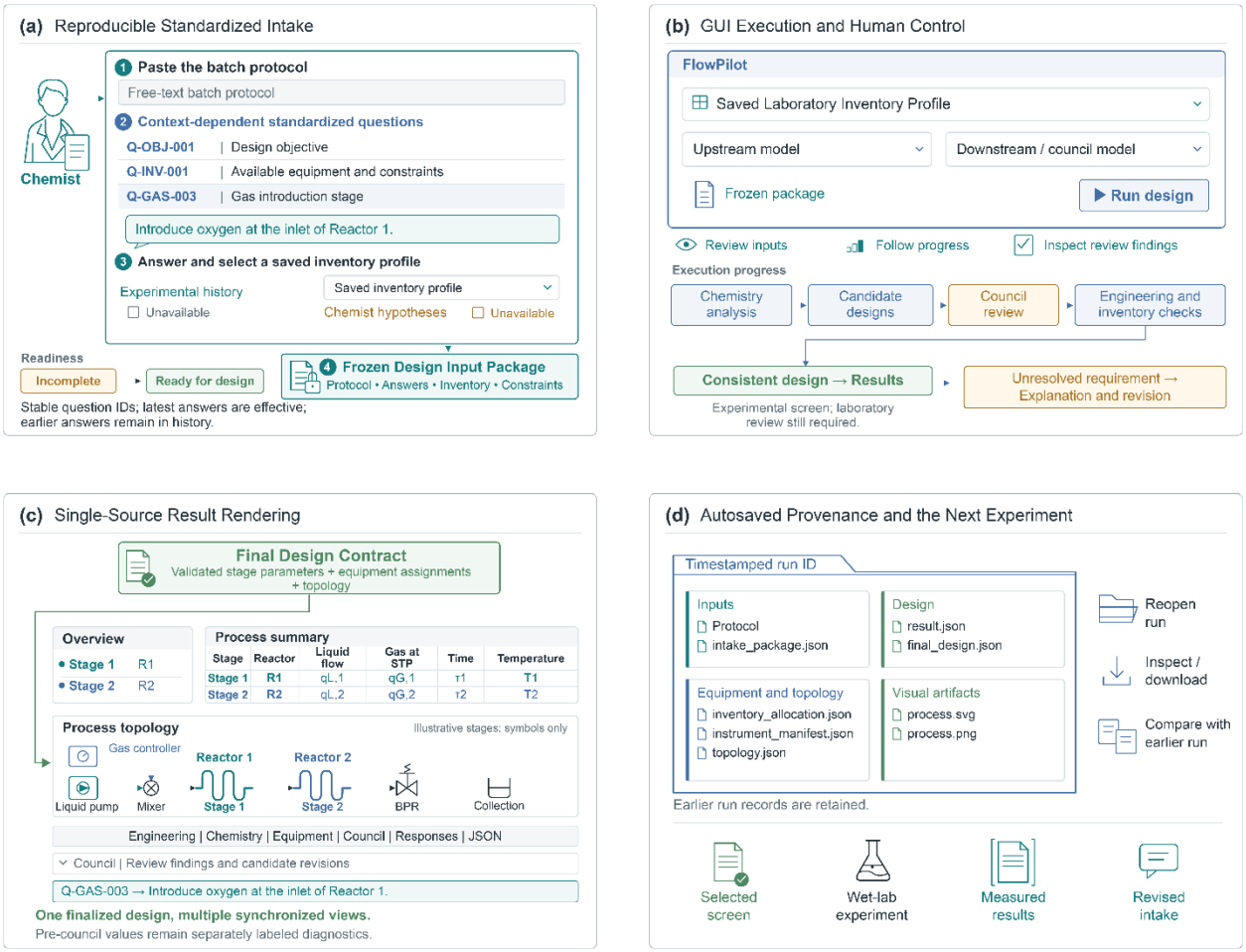


Figure S2. Standardized intake, GUI execution, synchronized results, and run provenance. (a) The chemist supplies a protocol, resolves context-dependent questions with stable identifiers, and selects inventory. The completed input package is frozen for the current run, while earlier answers remain in history. (b) The schematic GUI supports model selection, progress monitoring, and review, routing consistent designs to results and unresolved requirements to revision. (c) Overview, process tables, and equipment topology derive from one Final Design Contract; supporting diagnostics remain separately inspectable. Parameter symbols and the oxygen inlet arrangement are illustrative. (d) Timestamped folders preserve inputs, designs, equipment assignments, topology, and visual artifacts. Measured experimental feedback initiates a new design cycle while earlier records are retained. Computational consistency checks do not constitute experimental validation, and traceable input handling does not guarantee identical model responses.

3 Literature corpus and engineering rule base

FlowPilot uses two complementary knowledge resources. A structured literature corpus provides empirical batch-to-flow precedents and process metadata, whereas a handbook-derived rule base provides general engineering principles. Plan-aware retrieval selects a small set of literature analogies for the chemistry under consideration, and the upstream chemistry layer receives a prioritized subset of relevant handbook rules. These resources provide evidence and context; they do not replace deterministic calculations, laboratory inventory constraints, or experimental validation.

3.1 Frozen literature-corpus composition

The frozen manuscript export contains 464 classified flow-chemistry literature records. The classification spans thermal synthesis, photoredox and heterogeneous photocatalysis, cross-coupling, oxidation and reduction, electrochemistry, biocatalysis, polymer synthesis, crystallization, and other continuous-flow applications. For each record, available information was normalized into fields describing reaction class, reactor type, reactor material, bond or transformation label, operating conditions, number of inlet streams, and reported batch and flow outcomes. The primary categorical exports each contain 464 classified entries, but field completeness is not uniform. Explicit inlet-stream counts are available for 154 processes, batch yields for 135 records, and optimized-flow yields for 285 records (Table S4). Unknown, other, multiple, and long-tail classifications are retained rather than being excluded from denominators. This avoids inflating apparent coverage and makes the limitations of the extracted metadata visible.

The available yield distributions are descriptive. They should not be interpreted as a paired estimate of the causal benefit of flow because batch and flow values are missing for different subsets, publication and optimization selection can affect the distributions, and individual records differ in chemistry and objective. Figure S3 therefore reports field-specific denominators directly in each panel.

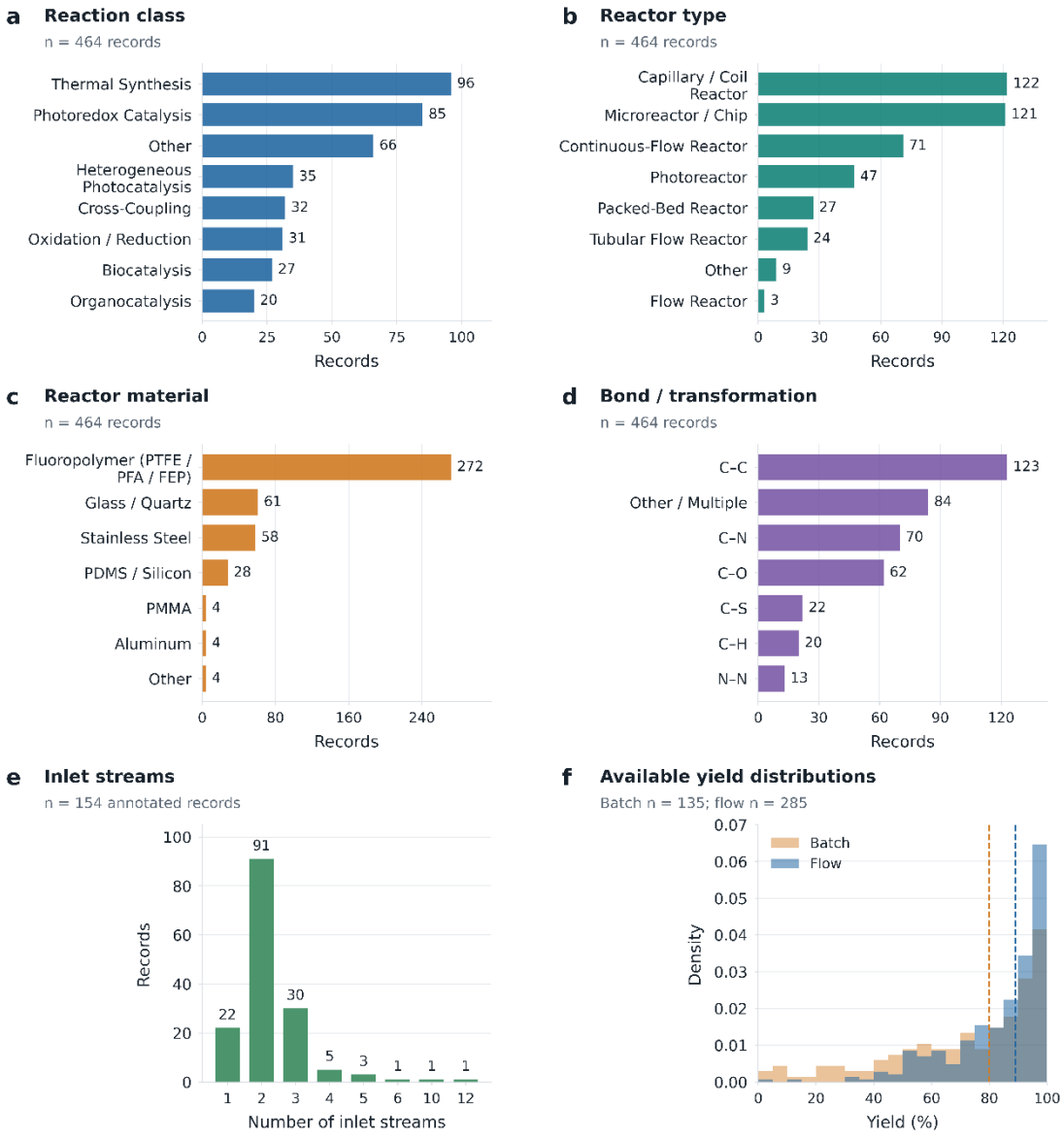


Figure S3. Composition of the flow-chemistry literature corpus. (a-d) Reaction class, reactor type, reactor material, and bond or transformation labels (n = 464 records in each panel). (e) Number of inlet streams among records with stream-count annotation (n = 154). (f) Available batch-yield (n = 135) and optimized-flow-yield (n = 285) distributions. Categories are descriptive and non-exclusive where the source record contains more than one label.

Table S4. Available literature-corpus fields and available denominators.

Metadata field	Available n	Coverage	Leading categories or summary	Interpretation note
Reaction class	464	100% classified export	Thermal Synthesis 96 (20.7%); Photoredox Catalysis 85 (18.3%); Other 66 (14.2%); Heterogeneous Photocatalysis 35 (7.5%); Cross-Coupling 32 (6.9%)	Other and long-tail classes retained.
Reactor type	464	100% classified export	Capillary / Coil Reactor 122 (26.3%); Microreactor / Chip 121 (26.1%); Continuous-Flow Reactor 71 (15.3%); Photoreactor 47 (10.1%); Packed-Bed Reactor 27 (5.8%)	Long-tail hardware labels retained.
Reactor material	464	100% classified export	Fluoropolymer (PTFE / PFA / FEP) 272 (58.6%); Glass / Quartz 61 (13.1%); Stainless Steel 58 (12.5%); PDMS / Silicon 28 (6.0%)	Classification granularity varies across records.
Bond/ transformation label	464	100% classified export	C–C 123 (26.5%); Other / Multiple 84 (18.1%); C–N 70 (15.1%); C–O 62 (13.4%); C–S 22 (4.7%)	Other/multiple is an explicit category.
Explicit inlet-stream count	154	33.2% of 464	1 stream: 22 (14.3%); 2: 91 (59.1%); 3: 30 (19.5%); 4: 5 (3.2%); 5: 3 (1.9%); 6, 10 and 12: 1 each (0.6% each)	Percentages use n = 154, not the full corpus.
Reported batch yield	135	29.1% of 464	Median 80%; IQR 56-93%	Available numeric values only.
Reported optimized-flow yield	285	61.4% of 464	Median 89%; IQR 75-96%	Not a paired causal comparison with batch yield.

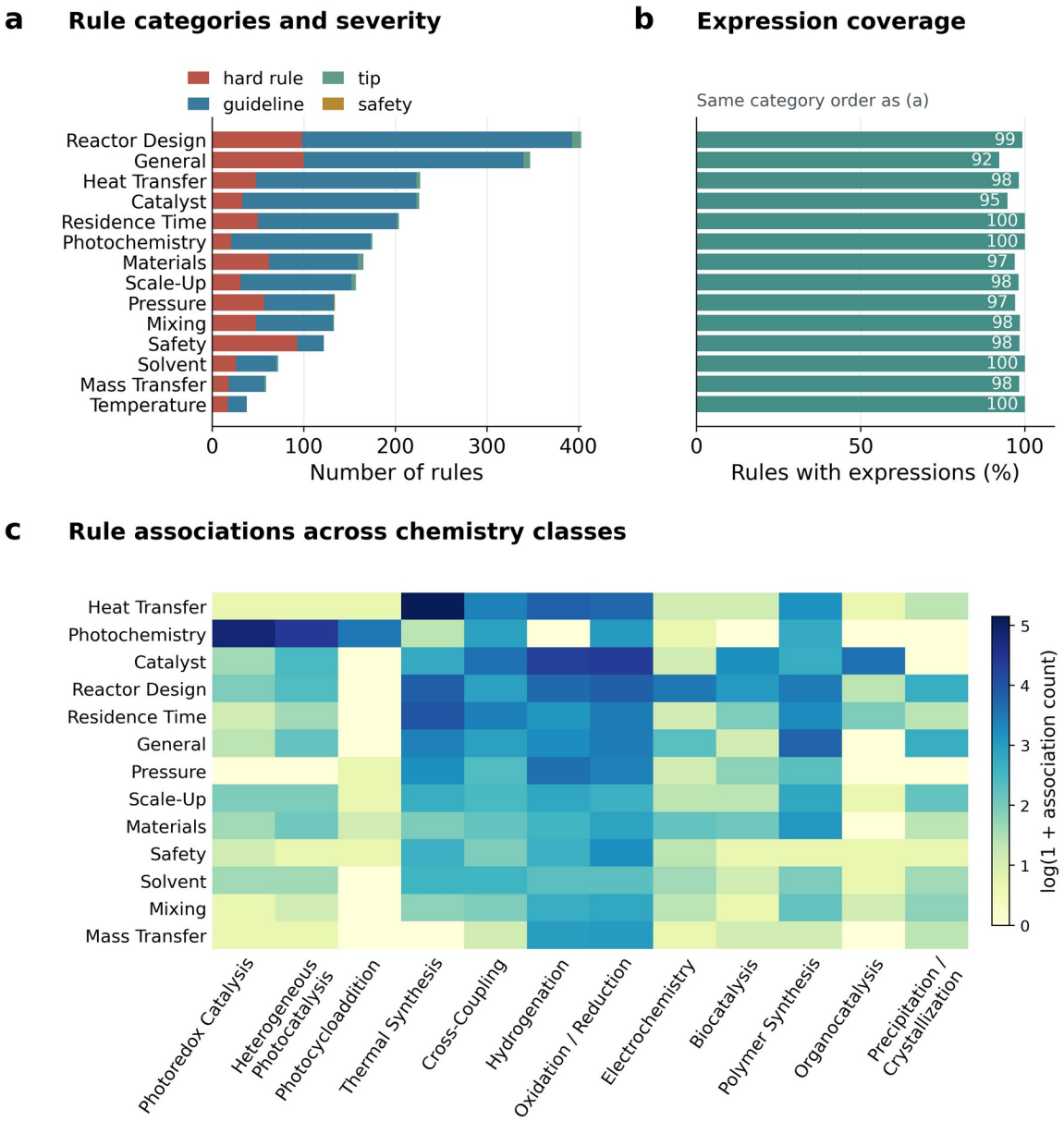


Figure S4. Audit of the handbook-derived rule base (2,537 rules; 29 categories). (a) Counts and stored severity labels for the 14 largest categories. Category and severity are distinct: the Safety category is not the same as the safety severity label. (b) Percentage of rules containing a machine-detected quantitative expression, in the same category order as (a). Expression detection does not verify an equation, and a hard_rule label does not by itself implement a deterministic gate. (c) Non-exclusive category associations with chemistry classes, shown as log(1 + association count). The distribution identifies the engineering topics available to prompt construction, including heat transfer, photochemistry and catalyst handling; it is not a performance metric. The source CSVs retain all category counts.

4 Ablation test and architecture comparison

Purpose and scope

The fixed-criteria multi-LLM evaluation was designed to compare the quality of final inventory-constrained flow-process designs produced by a direct one-shot large language model and by the complete FlowPilot architecture under matched inputs. The comparison was architecture-neutral at the evaluation stage: both methods received the same frozen chemistry, objective, laboratory inventory, and operating constraints, and both were required to return the same normalized final-design contract. The assessed object was the delivered design, not the internal reasoning trace or the number of intermediate agent calls. Two related experiments are reported in this section. The architecture comparison tested one-shot generation against full FlowPilot using five retained generator models, three chemistry cases, and three independent generation repeats. The internal module-ablation screen fixed the generator to Qwen3.8-27B and selectively removed or altered FlowPilot components to examine their contribution to outcome quality and executability.

This is a retrospective, development-stage comparison on three selected chemistry cases, not a prospective test on unseen reactions. The frozen campaign record states that cases were selected after a five-case pilot. The same case inputs were subsequently used for both architectures; that pairing controls supplied task information, not case-selection bias or development exposure. Scores describe assessed design quality for these tasks, not experimentally measured robustness across flow chemistry.

Paired architecture-comparison design

The retained manuscript comparison included the following generator models:

- Qwen3.6-27B
- Qwen3.8-27B
- GPT-4o
- Claude Sonnet 4.6
- Claude Opus 4.6

For every model, each of the three frozen chemistry cases was generated using both architectures in three independent calls. This produced:

5 models x 2 architectures x 3 cases x 3 repeats = 90 candidate outcomes

The paired unit was the same generator model, chemistry case and repeat index under the two architectures. Three generation repeats per case were retained, with a recorded temperature of 0.2. The campaign manifests and later repeat-extension records identify their provenance; repeat indexing does not imply shared random draws or statistically independent evaluator judgments. The designated source record was

excluded from final retrieval selection. This controls direct use of that record by RAG but does not establish absence of related information from the corpus or model pretraining.

Compared architectures

One-shot. A single model call received the complete frozen design input and a fixed output contract. No FlowPilot specialist agents, council deliberation, deterministic candidate selection, or post-generation correction operated inside this architecture.

Full FlowPilot. The same frozen design input entered the complete FlowPilot workflow, including chemistry interpretation, engineering calculations, specialist-agent assessment, council selection and revision, safety analysis, deterministic closure checks, and final schema enforcement. The internal workflow could use several LLM calls, but only the final normalized design was scored.

Common one-shot wrapper

The one-shot system instruction was:

> *You are a general chemistry assistant. Translate a batch chemistry protocol into a practical continuous-flow experiment.*

For each chemistry, the frozen design input reproduced in the three case-study subsections below was inserted after FROZEN DESIGN INPUT:, followed by the common response contract.

Exact one-shot response contract

> *Propose one complete flow design with numeric operating conditions, stream assignments, safety controls, and a brief rationale. Return one JSON object and no markdown with exactly these top-level fields: recommended_disposition, disposition_rationale, and proposal. The proposal value must be a JSON object, not a JSON-encoded string. The proposal object uses these fields: residence_time_min, flow_rate_mL_min, temperature_C, concentration_M, BPR_bar, reactor_type, tubing_material, tubing_ID_mm, reactor_volume_mL, residence_time_basis, residence_time_inlet_min, residence_time_in_channel_min, light_setup, wavelength_nm, streams, mixer_type, mixing_order_reasoning, pre_reactor_steps, post_reactor_steps, chemistry_notes, stage_parameters, reasoning_per_field, literature_analogies, engine_validated, safety_flags, confidence. Each stream is an object with stream_label, pump_role, contents, solvent, phase, flow_rate_mL_min, concentration_M, molar_equiv, gas_flow_sccm, and gas_flow_actual_mL_min where applicable. Distinguish inlet/STP gas flow from pressure-corrected in-channel gas flow. Ensure reactor_volume_mL is consistent with flow and residence time. State assumptions rather than inventing missing facts. recommended_disposition must be exactly EXECUTE, SCREEN, or BLOCK. Use BLOCK when the hard constraints make a safe design physically or operationally*

impossible. Use SCREEN for a feasible first experiment that still requires experimental validation. Use EXECUTE only for a directly executable design supported by sufficient evidence.

FlowPilot did not use one monolithic system prompt. It accepted the same frozen design input at its architecture boundary and distributed the information across its intake, chemistry, engineering, council, and validation stages. The archived run directories retain generation outputs, available prompt/stage records, evaluator packets and judgments; the common input is preserved separately from architecture-specific traces.

Identity-masked evaluation

Each final outcome was converted to a standardized candidate packet. Generator identity and architecture labels were removed before judging. Complete architectural blinding is not claimed because formatting or output style could reveal provenance. Every judge received the same protocol, objective, inventory, normalized final outcome, held-out reference facts, deterministic verification sheet, and fixed rubric.

Three separate evaluator model families assessed each candidate:

The archived judge configuration for the matched example was Qwen /models/Qwen3.6-27B, OpenAI gpt-5.4-2026-03-05 and Anthropic claude-sonnet-4-6, at temperature 0.0. These evaluator calls were separate from candidate generation. GPT-5.4 is not one of the five generators pooled in the primary aggregate, although an archived additional GPT-5.4 generator comparison remains displayed in main-text Figure 4b–c. Exact prompt schemas, seeds, responses and telemetry are retained in the companion source records. A recorded seed does not guarantee that a provider honors deterministic sampling.

- Qwen
- OpenAI
- Claude

The deterministic sheet exposed machine-checkable evidence for schema validity, numerical closure, inventory compliance, reactor and residence-time consistency, gas bookkeeping where applicable, topology coverage, and safety requirements. Deterministic evidence informed the judges but did not itself add an architecture-specific score bonus.

Benchmark scoring method

The same 14 criteria were applied to every candidate. Each applicable criterion received an integer rating from 0 to 4. All criteria and all judges had equal weight; no architecture bonus or hand-tuned criterion weighting was used.

Table S5. Fixed 0–4 rating anchors applied by each LLM judge.

Score	Anchor
0	Fundamentally wrong, unsafe, chemically invalid, or physically impossible.
1	Major deficiencies, multiple material defects, or poor reproducibility; substantial redesign is required.
2	Partially correct with one material but recoverable defect requiring revision before execution.
3	Correct overall with a minor defect that is unlikely to change execution or interpretation.

Score	Anchor
4	Correct, complete, executable, and supported; no material defect.

Table S6. Complete frozen outcome questions, critical-error triggers and applicability rules. Gas use denotes a gas reagent or gas-liquid process; multi-stage denotes more than one reactive stage.

Criterion	Exact evaluation question	Critical-error conditions	Applies
UO-01 Transformation fidelity	Does the final design preserve the intended substrate, product, transformation, catalyst strategy, reaction modality, and phase regime without unsupported substitution?	Wrong product or transformation; Chemically incompatible catalyst or modality substitution	Always
UO-02 Required materials	Are all required reagents, catalysts, solvents, additives, gases, and reactive components present and correctly assigned?	Required reactive component omitted; Unsafe or transformation-changing material introduced	Always
UO-03 Stoichiometry and feed chemistry	Are concentrations, equivalents, feed compositions, dilution, and chemical compatibility explicit and coherent?	Stoichiometry prevents intended reaction; Feed combination creates an unmitigated incompatibility	Always
UO-04 Condition and stage mapping	Are temperature, pressure, atmosphere, irradiation, addition order, stage boundaries, and chemistry-critical batch operations translated faithfully?	Chemistry-critical operation or stage omitted; Condition change makes the intended chemistry implausible or unsafe	Always
UO-05 Executable topology	Are feeds, pumps or MFCs, mixers, reactors, controls, separators, quench, collection, and waste connected in the correct executable order?	Required unit operation absent; Topology cannot physically execute the proposed chemistry	Always
UO-06 Liquid material balance	Do liquid concentrations and flow rates reproduce component molar flows, equivalents, total flow, dilution, and throughput?	Material balance is dimensionally invalid; Reported feeds cannot produce the stated stoichiometry	Always
UO-07 Residence-time and geometry closure	Do reactor volume, cumulative flow, geometry, holdup basis, per-stage residence time, and total residence time close numerically?	Residence time is physically impossible; Reactor geometry or volume contradicts the executable design	Always

Criterion	Exact evaluation question	Critical-error conditions	Applies
UO-08 Gas bookkeeping	For gas-using designs, do gas identity, STP definition, inlet flow, molar flow, equivalents, absolute pressure, channel flow, holdup, and gas-sensitive residence-time bases close and remain distinct?	Gas identity or pressure basis is unsafe; Gas molar flow, equivalents, or residence time is physically inconsistent	Gas use
UO-09 Multistage closure	For multistage designs, are stream-introduction stages, cumulative flows, individual reactor volumes and times, interstage operations, and total process residence time consistent?	Stage sequence changes the chemistry; Stage flow or residence-time closure is physically impossible	Multi-stage
UO-10 Inventory feasibility	Does every selected equipment item exist, and are flow, volume, material, ID, temperature, pressure, wavelength, and connectivity within declared limits?	Invented or unavailable essential equipment; Operating point exceeds a hard equipment limit	Always
UO-11 Physical and transport plausibility	Are phase behavior, mixing, pressure drop, heat transfer, mass transfer, and transport assumptions physically plausible and supported at the proposed operating point?	Unmitigated boiling, precipitation, plugging, runaway, or pressure failure; Transport assumption makes the design physically impossible	Always
UO-12 Hazards and controls	Are chemistry-specific thermal, pressure, gas, peroxide, azide, catalyst, solvent, and incompatibility hazards recognized and converted into concrete controls?	Major hazard omitted; Proposed execution lacks a necessary containment, purge, interlock, relief, or compatibility control	Always
UO-13 Operating procedure and work-up	Are feed preparation, leak testing, priming, startup, steady-state qualification, collection, normal and emergency shutdown, depressurization, quench, separation, work-up, and waste handling reproducible where required?	Missing procedure creates an immediate execution hazard; The experiment cannot be reconstructed from the delivered instructions	Always

Criterion	Exact evaluation question	Critical-error conditions	Applies
UO-14 Evidence and uncertainty calibration	Are source facts, calculations, assumptions, analogies, performance predictions, uncertainty, validation needs, confidence, and SCREEN/BLOCK/EXECUTE disposition represented honestly and proportionately?	Invented evidence or unsupported performance guarantee; Execution recommended despite an acknowledged unresolved critical defect	Always

NOT_APPLICABLE was permitted only when the criterion's applicability rule was false. Missing required information was rated 0 or 1 rather than excluded. In the frozen rubric, gas bookkeeping (UO-08) and multistage closure (UO-09) were the only criteria that could routinely become non-applicable. Critical-error flags were recorded separately and did not impose an arbitrary cap on the numerical score. In these three cases, UO-09 is inapplicable because there is only one reactive stage; gas mixing or product separation alone does not create a second reactive stage. UO-08 is inapplicable to CuAAC but applies to the two gas-using cases. Their omission from the cross-chemistry gain display does not remove applicable UO-08 ratings from the overall score.

For candidate i, criterion k, and judge j, the criterion consensus was:

$$C_{i,k} = \frac{1}{J} \sum_{j=1}^{J} s_{i,k,j}$$

where $s(i,k,j)$ is the judge's integer rating from 0 to 4. If $A(i)$ is the set of applicable criteria, the candidate score was:

$$S_i = \frac{1}{|A_i|} \sum_{k \in A_i} \frac{C_{i,k}}{4}$$

The resulting candidate score ranged from 0 to 1. For each model, architecture, and repeat, the three chemistry candidate scores were first averaged to obtain a repeat-level campaign score:

$$R_{m,a,r} = \frac{1}{3} \sum_{q=1}^{3} S_{m,a,r,q}$$

The reported model-architecture value was the mean of the three repeat-level scores, and the error bar was their sample standard deviation ($n = 3$, $\mathrm{ddof} = 1$). It was not a confidence interval. The matched architecture effect was calculated before aggregation:

$$\Delta_{m,q,r} = S_{\mathrm{FlowPilot},m,q,r} - S_{\mathrm{one\text{-}shot},m,q,r}$$

A positive value therefore favored FlowPilot for that matched case-repeat pair. Judge disagreement was retained separately and was not used as a substitute for generation-repeat variability.

Case study 1: Hydrogenolysis

Benchmark title: Three-phase N-diphenylmethyl hydrogenolysis

Source of the hydrogenolysis benchmark case: Tu et al., reference 7. Source exclusion is a retrieval control, not proof of absence from pretraining.

RAW BATCH PROTOCOL:

N-Diphenylmethylazetidin-3-ol (DMAOL; MW 239.32 g/mol) was prepared as a 3.8 wt% solution in methanol. For molarity conversion, explicitly use and label an assumed solution density of 0.792 g/mL, giving approximately 0.126 M DMAOL. A 200 mL charge was contacted with hydrogen over 0.5 g of 7 wt% Pd(OH)2/Al_2O_3 *catalyst in a stirred batch reactor at 60 °C and 21 bar for 5 h. Conversion to 3-azetidinol was 38.1%. Translate this into one conservative continuous-flow packed-bed screening design. Acetic acid is optional and must not be introduced unless explicitly justified as a screening variable. Report feed preparation, gas-liquid mixing, packed-bed operation, temperature, pressure basis, liquid flow, hydrogen inlet/STP flow, pressure-corrected in-channel hydrogen flow, hydrogen equivalents, nominal empty-bed space time, liquid-holdup/contact residence-time basis, separation, collection, startup, shutdown, nitrogen purge, and hydrogen/methanol/pressure safety controls.*

OBJECTIVE:

Produce one executable, inventory-constrained first gas-liquid-solid flow screen without using the held-out source answer.

HARD CONSTRAINTS:

- Use only the authoritative inventory supplied below.

- Hydrogen is the required reducing reagent and must be delivered through the certified H_2 *MFC.*

- Use a single DMAOL/methanol liquid feed and the declared gas-liquid mixer before the Pd(OH)2/Al_2O_3 *packed bed.*

- Operate at 60 °C and the available 21 bar BPR setpoint; explicitly state whether gas compression uses 21 bar absolute or 21 bar gauge.

- Report hydrogen flow at inlet/STP and pressure-corrected in-channel conditions, with the conversion calculation.

- Report H_2 *equivalents from gas molar flow divided by DMAOL molar flow.*

- Distinguish nominal empty-bed space time from liquid-holdup/contact residence time; do not silently use total gas-plus-liquid flow as liquid reaction time.

- Use the nitrogen line for startup leak/purge verification and shutdown displacement of residual hydrogen.

- Return SCREEN unless the inventory cannot support a safe design.

CHEMIST-CONFIRMED CHEMISTRY IDENTITY:

{"confirmed":true,"mechanism_type":"heterogeneous catalytic hydrogenolysis","reaction_class":"hydrogenolysis","reaction_name":"N-diphenylmethyl hydrogenolysis"}

Frozen inventory summary. The hydrogenolysis task supplies one plunger pump (0.1-2.0 mL min^{-1}; 40 bar limit), a stainless-steel packed bed (3.0 mL; 4.35 mm ID; 200 mm declared tube length), a 60 °C operating temperature and 21 bar BPR setpoint. H_2 and N_2 supplies, an H_2 MFC (1-100 mL min^{-1} at its standard reference), a T-mixer, heater, packed-bed connectors, gas–liquid separator, vented collector, H_2 check valve and safe vent are declared. No light source or degasser is supplied. The volume is nominal bed volume, not measured liquid holdup.

Case study 2: Photochemical oxidation

Benchmark title: Photocatalytic aerobic oxidation of Fmoc-L-methionine

Source of the photochemical benchmark case: Thomson et al., reference 8. Its supporting information includes Fmoc-L-methionine oxidation. The benchmark uses the frozen extracted input reproduced below; internally inconsistent reference-flow fields were excluded from numerical scoring.

RAW BATCH PROTOCOL:

Fmoc-L-methionine (3.7 mmol, 1.0 equiv) and 4,7-diphenyl-2,1,3-benzothiadiazole (10 mol%) were dissolved in acetonitrile at 0.10 M in a round-bottom flask. The solution was exposed to air and irradiated with a 1.5 W LED centered near 420 nm at 21 C for 5 min. Fmoc-methionine sulfoxide was formed in 99% yield. Translate this into one conservative continuous-flow photochemical screening design. Report feed preparation, air delivery and oxygen-equivalent basis, gas-liquid mixing, pressure basis, photoreactor assignment, wavelength, temperature, liquid flow, inlet/STP and pressure-corrected in-channel gas flow, liquid contact residence time, reactor volume, separation, collection, startup, shutdown, and oxygen/acetonitrile/light safety controls.

OBJECTIVE:

Produce one executable, inventory-constrained first gas-liquid photochemical flow screen without using the held-out source answer.

HARD CONSTRAINTS:

- Use only the authoritative inventory supplied below.

- Use one premixed Fmoc-L-methionine/photocatalyst/acetonitrile liquid feed and the declared air MFC.

- Use the declared gas-liquid mixer before the 420 nm photoreactor.

- Operate at the available 3 bar BPR setpoint and state whether pressure is absolute or gauge.

- Report air at inlet/STP and pressure-corrected in-channel conditions and calculate oxygen equivalents from 21 mol% oxygen.

- Use liquid flow and the 1.0 mL reactor volume to define liquid contact residence time; do not use total gas-plus-liquid volumetric flow as reaction time.

- No inline degasser is available or required because air is a reagent.

- All selected equipment must be named by its inventory equipment_id.

- Return SCREEN unless the inventory cannot support a safe design.

CHEMIST-CONFIRMED CHEMISTRY IDENTITY:

{"confirmed":true,"mechanism_type":"photocatalytic aerobic oxidation","reaction_class":"oxidation","reaction_name":"photocatalytic aerobic oxidation of Fmoc-L-methionine"}

Frozen inventory summary. The oxidation task supplies one HPLC pump (0.01-1.0 mL min^{-1}), a 1.0 mL PFA coil (1.0 mm ID; 1.273 m tubing), a 420 nm, 1.5 W light source, temperatures of 21 or 25 °C and a 3 bar BPR setpoint. A dry-air supply and MFC (0.1-10 mL min^{-1} at its standard reference), T-mixer, thermostat, connectors, gas–liquid separator, vented collector and interlocked enclosure/vent are supplied; no degasser is available. The coil and pump are rated to 10 bar. The frozen instruction calls V/Q_{liq} a liquid contact time; physically this is a nominal liquid-feed space time, not a measured phase-contact time. Its original wording is retained for reproducibility, without implying known liquid holdup.

Case study 3: CuAAC

Benchmark title: Cu/C-catalyzed azide-alkyne cycloaddition

Source of the CuAAC benchmark case: Fuchs et al., reference 9. Frozen case inputs and inventory constraints are retained unchanged.

RAW BATCH PROTOCOL:

Benzyl azide (0.50 mmol, 1.0 equiv) and phenylacetylene (0.55 mmol, 1.1 equiv) were combined with Cu/C catalyst (10 mol% Cu) in acetone (2.0 mL; benzyl azide concentration 0.25 M). The homogeneous liquid reaction mixture was heated under microwave batch conditions at 150 °C for 3 min. The corresponding 1,4-disubstituted 1,2,3-triazole was isolated in 94% yield. Translate this batch procedure into one conservative continuous-flow screening design. Preserve the Cu/C catalyst strategy and explicitly report feed preparation, reactor topology, temperature, liquid flow, pressure, reactor/contact volume, residence-time basis, collection, work-up, and azide/pressure safety controls.

OBJECTIVE:

Produce one executable, inventory-constrained first flow screen without using the held-out source answer.

HARD CONSTRAINTS:

- Use only the authoritative inventory supplied below.

- Use a single premixed liquid feed through the Cu/C packed-bed cartridge.

- Do not invent a gas stream, photochemical equipment, or unlisted hardware.

- Report catalyst-contact residence time and make reactor volume, liquid flow, and residence time mathematically consistent.
- Return SCREEN unless the inventory cannot support a safe design.

CHEMIST-CONFIRMED CHEMISTRY IDENTITY:
{"confirmed":true,"mechanism_type":"copper-catalyzed cycloaddition","reaction_class":"cuaac","reaction_name":"copper-catalyzed azide-alkyne cycloaddition"}
Frozen inventory summary. The CuAAC task supplies an X-Cube HPLC pump (0.1-3.0 mL min^{-1}; 40 bar limit), a stainless-steel Cu/C cartridge with declared volume 0.30 mL and ID 4.0 mm, a heater with a 200 °C upper limit, a 20 bar BPR, cartridge connector, vented collector and azide-service shield. Gas, light, mixer and degasser lists are empty. The separate cartridge-body record gives a length of 60 mm: its cylindrical volume is approximately 0.754 mL, not 0.30 mL. The inventory does not identify whether 0.30 mL denotes accessible liquid volume, packed volume or a different active section. These two values are retained as an unresolved input-basis ambiguity, not reconciled by inventing a porosity. The protocol's phrase "homogeneous liquid reaction mixture" also does not mean that the Cu/C catalyst is dissolved; the hard constraint requires catalyst retention in the packed bed.
Machine-readable benchmark inputs. The companion frozen_benchmark_inputs folder contains each complete case, authoritative inventory JSON and exact design-input string, named with hydrogenolysis_oprd_9b00416, aerobic_oxidation_fmoc_methionine or cuaac_adsc_200900726, together with SHA-256 hashes and source paths in manifest.json. Canonical hashes of protocol, objective, hard constraints and inventory agree across the three retained campaign sources supplying the five generators. These inventories define the computational tasks and are not the laboratory inventory used for Figures 5 and 6. The source manifest marks hydrogenolysis and CuAAC as page-verified and the photochemical case as machine-extracted/unverified; the latter designation is retained rather than promoted to verified ground truth. Original inputs and scores have not been changed during manuscript review.

Internal FlowPilot module-ablation screen
The internal screen used Qwen3.8-27B as the fixed generator and the same three chemistry cases. Fifteen architecture conditions were tested, giving 45 condition-case outcomes. Each cell was generated once; SD therefore describes between-case variation, not generation-repeat uncertainty. Disabled specialists received no LLM call or default-score penalty, and active council weights were renormalized. Table S7 lists all 15 conditions from the archived module summary. Main-text Figure 4d displays a selected subset. The archive's executable-outcome fraction is retained as metadata and should not be interpreted as independent experimental implementability or as a fair cross-architecture safety rating. DFMEA denotes design failure mode and effects analysis.

Table S7. All 15 Qwen3.8-27B module-screen conditions. Score means and sample SD summarize three different cases, with one generation per condition-case pair. Calls are mean generation LLM calls per case; flags are total judge reports across the three cases. The architecture-specific final-contract execution flag is not used as a comparative outcome; it remains in the raw archive.

Condition	Mean	SD	Calls	Flags
Without preselection refinement	0.935	0.028	11.0	0
One-pass council	0.932	0.019	9.7	0
Full FlowPilot	0.928	0.027	13.3	0
Without winner revision	0.926	0.042	13.0	0
Without DFMEA	0.926	0.042	12.7	0
Without Kinetics agent	0.924	0.057	12.0	0
Deterministic selection	0.920	0.041	12.7	0
Candidate budget 4	0.915	0.049	18.7	0
Without Safety agent	0.915	0.060	12.3	0
Without Fluidics agent	0.913	0.042	10.7	0
Candidate budget 1	0.907	0.048	12.3	0
Without Skeptic audit	0.904	0.039	13.3	0
Without Chemistry agent	0.902	0.054	12.0	0
One-shot	0.743	0.167	1.0	14
No council	0.670	0.137	3.0	19

The full configuration scored 0.928, compared with 0.670 without the council and 0.743 for one-shot generation in this screen. Each individual domain-reviewer removal had a lower mean than the full configuration (Chemistry 0.902; Fluidics 0.913; Safety 0.915; Kinetics 0.924). Conversely, one-pass review (0.932) and omission of preselection refinement (0.935) slightly exceeded it. The screen therefore supports examining coordinated review rather than assuming that every added step improves quality. Between-case SD and one generation per condition-case pair cannot establish a causal ranking of reviewers or distinguish small effects from generation variability.

Table S8. Matched one-shot and FlowPilot architecture scores by generator model.

Generator model	One-shot	FlowPilot	Mean difference
Qwen3.6-27B	0.765 +/- 0.014	0.904 +/- 0.029	+0.139
Qwen3.8-27B	0.747 +/- 0.019	0.915 +/- 0.011	+0.168
GPT-4o	0.684 +/- 0.011	0.910 +/- 0.002	+0.226

Generator model	One-shot	FlowPilot	Mean difference
Claude Sonnet 4.6	0.890 +/- 0.014	0.924 +/- 0.024	+0.034
Claude Opus 4.6	0.883 +/- 0.015	0.932 +/- 0.010	+0.049

Values are mean +/- sample SD across three repeat-level campaign means. The difference column is FlowPilot minus one-shot.

Across the 45 retained outcomes per architecture:

Table S9. Aggregate architecture-level score and critical-error summary.

Architecture	Mean candidate score	Campaigns without critical flags	Raw judge critical flags
One-shot	0.794	18/45	98
FlowPilot	0.917	43/45	4

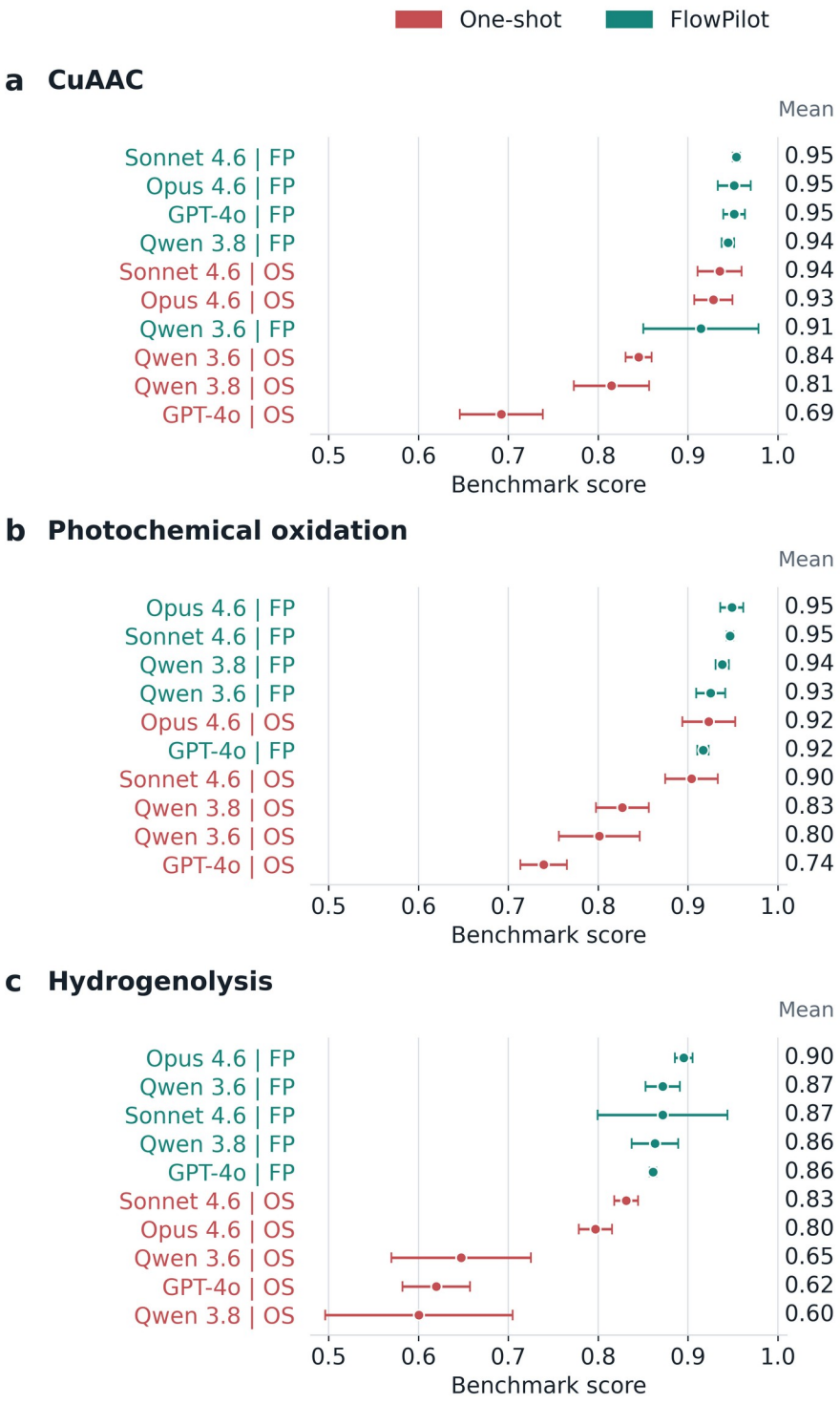


Figure S5. Case-specific benchmark scores for (a) CuAAC, (b) photochemical oxidation and (c) hydrogenolysis. Points and horizontal error bars show means and sample SDs across three generation repeats. Red denotes one-shot (OS); teal denotes FlowPilot (FP). Both Qwen models have 27B parameters. All 30 SDs are nonzero; small whiskers may be obscured by markers. Values beside the plots are rounded means.

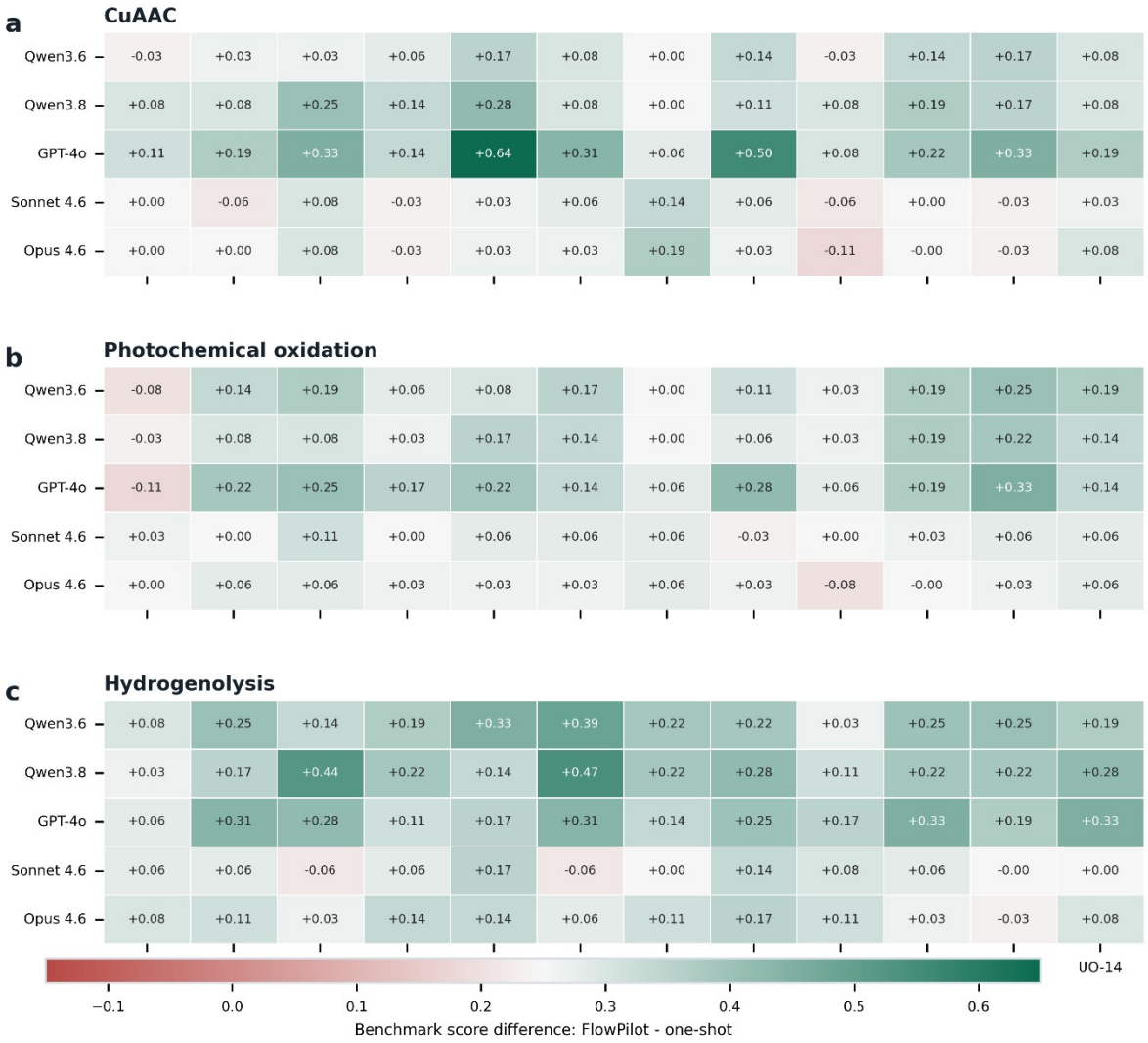


Figure S6. Mean paired FlowPilot-minus-one-shot score difference by criterion and chemistry; positive values favor FlowPilot. UO-08 is omitted from this cross-chemistry display because gas bookkeeping applies to oxidation and hydrogenolysis but not CuAAC. UO-09 is omitted because none of these cases has more than one reactive stage. Applicable gas-bookkeeping scores remain included in the overall benchmark score.

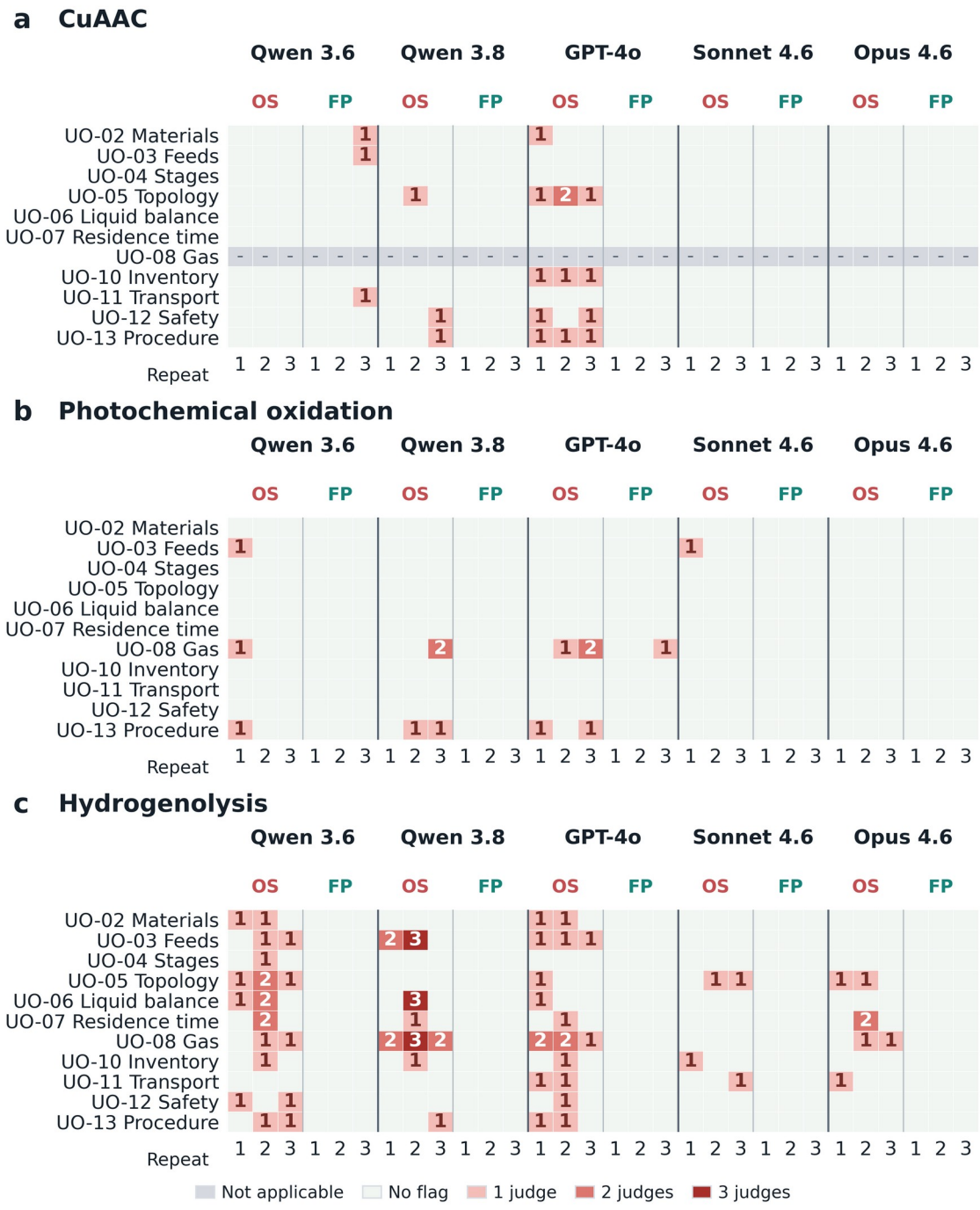
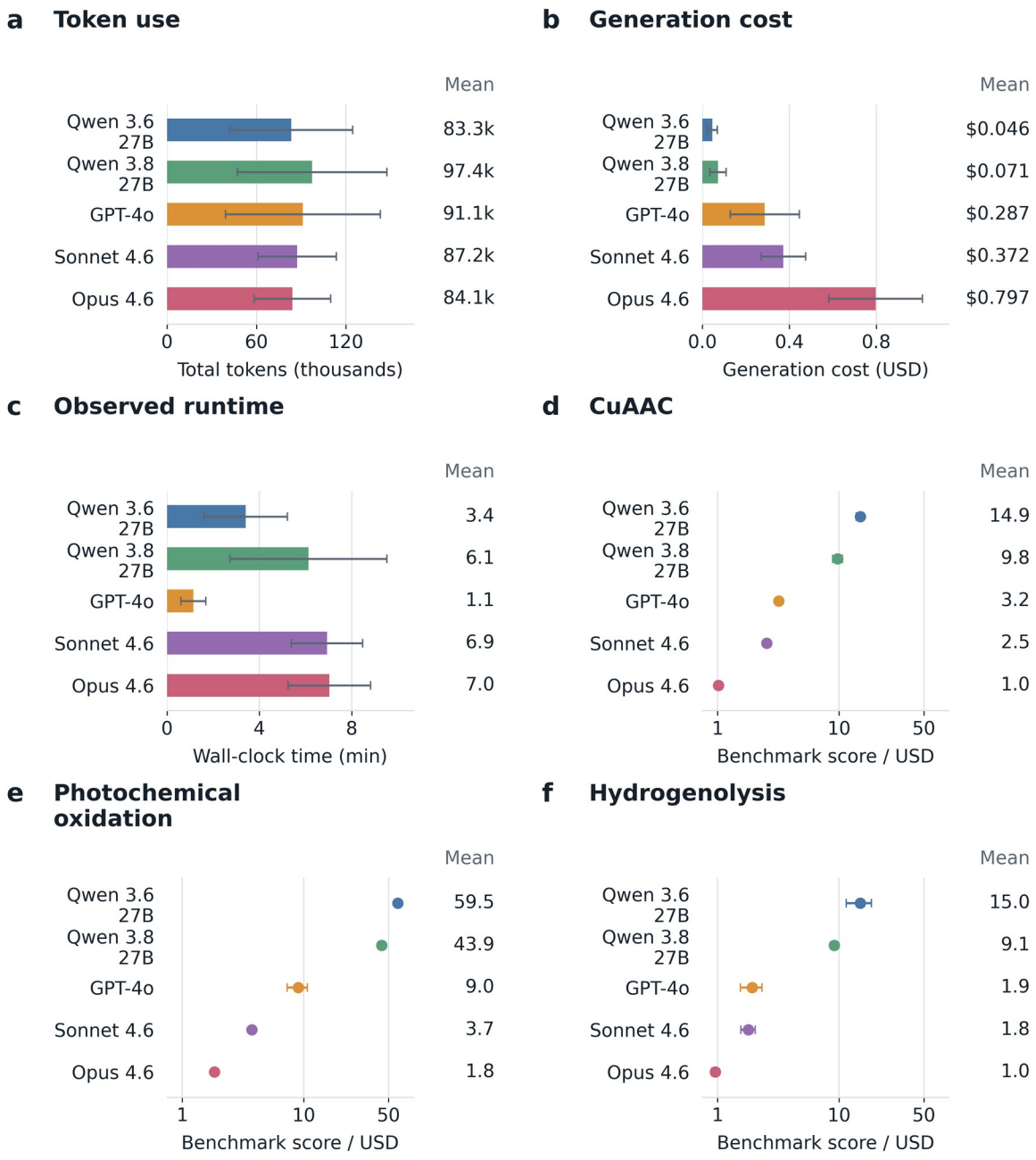


Figure S7. Critical flags for (a) CuAAC, (b) photochemical oxidation and (c) hydrogenolysis. Each model has adjacent one-shot (OS) and FlowPilot (FP) blocks with repeats 1-3. Rows identify criteria; cell values count judges flagging an issue, not unique physical errors. Pale cells have no flag; gray cells with a dash are inapplicable. The display retains 90 outcomes and 990 cells across 11 criteria. UO-01 and UO-14 had no critical flags and are not plotted; UO-09 is inapplicable throughout. All 98 one-shot and 4 FlowPilot judge flags are retained. No flag is not evidence of experimental success.



a-c: mean +/- sample SD; 9 campaigns per model (3 cases x 3 repeats).
d-f: mean +/- sample SD; 3 repeats per case; log scale. Runtime depends on the environment.

Figure S8. FlowPilot resource use and quality per generation cost. (a) Tokens, (b) estimated generation cost from the stored token-price schedule and (c) observed runtime: mean +/- sample SD across nine campaigns per model. (d-f) Mean benchmark-score/ USD ratios for CuAAC, photochemical oxidation and hydrogenolysis, respectively, with sample SDs across three repeats, on logarithmic axes. Ratios are calculated per campaign before averaging. Judge-model costs are excluded. Runtime is environment-dependent; token pricing is not a measurement of total local deployment cost.

5 Deterministic engineering, inventory realization, and refinement

FlowPilot separates model-proposed independent variables from quantities that can be recomputed. Flow rates, gas compression, molar equivalents, reactor volumes, stage times, pressure-drop indicators, heat-transfer indicators, and equipment assignments are recalculated after candidate selection and again after inventory realization. A numerical value becomes a run instruction only if the recomputed proposal, stage table, stream table, inventory allocation, process graph, and final contract agree within the implemented tolerances.

5.1 Residence-time and gas-flow conventions

Notation convention. Italic symbols denote physical quantities; descriptive subscripts are upright. Q_{liq} denotes liquid volumetric flow rate, V_R the reactor volume and V_{liq} the liquid holdup volume. The inlet/STP index $\tau_{inlet} = V_R/(Q_{liq} + Q_{gas,STP})$ is distinct from liquid residence time τ_{liq}. Liquid-flow and liquid-holdup symbols use a consistent descriptive subscript; reactor volume and liquid holdup are not interchangeable. Literal software field names, identifiers and paths retain their original spelling.

Liquid component molar delivery is $C \times Q$. In a fully liquid-filled, unpacked reactor, V/Q_{liq} is nominal liquid space time. Gas molar delivery uses the stated reference temperature and pressure; reactive-gas equivalents additionally include the active mole fraction. The inlet-reference index $V/(Q_{liq} + Q_{gas,STP})$ is a bookkeeping index, not a physical residence time under operating pressure. Ideal-gas scaling gives $Q_{gas,ch} = Q_{gas,STP}(T_{ch}/T_{STP})(P_{STP}/P_{abs})$, with absolute temperature and pressure. A time formed using this local gas volume remains a nominal homogeneous-flow estimate: it assumes negligible phase slip and neglects dissolution, gas consumption and axial variations unless separately modeled. The ratio $Q_{gas,ch}/(Q_{liq} + Q_{gas,ch})$ is a no-slip volumetric gas fraction, not measured holdup. For a packed bed or gas–liquid reactor, liquid contact time instead requires accessible liquid volume or holdup, with residence-time-distribution measurements needed to characterize dispersion. A standard-volume gas feed and a compressed volume must never be substituted into the same equivalent calculation without converting their reference states.

Table S10. Deterministic quantities, definitions, and closure checks.

Quantity	Definition / basis	Closure or qualification
Component molar feed	$\dot{n} = C\,Q$; M × mL min^{-1} = mmol min^{-1}	Component concentration, stream flow and units agree
Liquid space time / contact time	V/Q_{liq} for a filled liquid coil; V_{liq}/Q_{liq} for known liquid holdup	State whether V is total, empty-bed, accessible or liquid volume
Inlet/STP time index	$V/(Q_{liq} + Q_{gas,STP})$	Label as a reference-volume index, not operating residence time
Local gas volume	$Q_{gas,ch} = Q_{gas,STP}(T_{ch}/T_{STP})(P_{STP}/P_{abs})$	Absolute temperature and pressure; stated ideal-gas assumption
Reactive-gas equivalents	eq = $y_{active}\,(Q_{gas,STP}/V_{m,STP})/(C_{lim}\,Q_{lim})$	Use consistent molar units; $V_{m,STP}$ = 22.414 mL mmol^{-1} at 273.15 K, 1 atm

Quantity	Definition / basis	Closure or qualification
No-slip gas-fraction estimate	$Q_{gas,ch}/(Q_{liq} + Q_{gas,ch})$	Homogeneous-flow approximation only; not measured holdup
Multistage time total	Sum of stage times on a common basis	Use cumulative local flow; do not call mixed-basis sums physical residence time
Pressure drop	Single- or two-phase relation for the declared regime	Check total operating pressure against each component rating; a BPR is not pressure relief
Thermal screening	$UA = \mathrm{U}\ A_{wall}$; heat-generation/removal comparison	Report assumed heat-transfer and enthalpy inputs; no calorimetric validation implied

5.2 Multistage calculation and stream introduction

Each reactive stage receives a separate stage identifier, inlet composition, cumulative liquid and gas flow, assigned reactor volume, temperature, pressure, and residence-time basis. A feed introduced between reactors contributes only to downstream cumulative flow. Quench and work-up operations are represented in topology but are not counted as reactive stages. If only a total target time is available, allocation uses declared stage times or stage batch-time weights and otherwise applies an explicit equal-allocation fallback. Final validation checks both the stage sum and each local volume-flow-time identity.

5.3 Heat-transfer screening

The heat-transfer module reports wall area, surface-to-volume ratio, overall thermal conductance UA, estimated heat generation and removal rates, a thermal Damkohler-like ratio, and a screening margin. The implemented relations are $A_{wall} = \pi . d . L$, $Q_{rem} = U . A . \Delta T_{lm}$, and $Da_{th} = Q_{gen} / Q_{rem}$. Reaction enthalpy, overall heat-transfer coefficient, and driving-temperature difference can be estimated from reaction and reactor classes when measurements are unavailable. These values are screening assumptions rather than calorimetric evidence; an apparently favorable estimate does not replace reaction calorimetry for scale-up or strongly exothermic chemistry.

5.4 Inventory normalization and executable topology

Laboratory inventories may be entered as structured JSON or extracted from PDF, Word, PowerPoint, spreadsheet, and text sources. Extracted content is normalized into typed equipment classes and reviewed before use. Each item carries a stable equipment_id and available capability fields. Free-text statements such as 'no inline degasser' become explicit unavailable capabilities or hard constraints rather than prompt-only notes. During design realization, every required operation is assigned to an inventory item. The allocator checks type, material, inner diameter, volume, flow range, temperature, pressure, wavelength, phase compatibility, and connectivity where declared. Under strict assignment, absent or invented hardware cannot be published in the executable topology.

A process diagram is generated from the same inventory-assigned topology stored in the final contract. If assignment or numerical closure fails, FlowPilot may render a requirements topology for diagnosis, but it does not label that diagram executable and does not publish numerical run instructions. This distinction prevents a polished intermediate graph from being mistaken for a laboratory-ready process.

5.5 Incremental experimental refinement

Closed-loop refinement accepts one or more ExperimentRecord objects containing actual operating conditions, measured response, and operational observations. Measured evidence has higher authority than model inference. The module diagnoses underconversion, selectivity loss, solubility or fouling, pressure instability, and gas-liquid instability; it then proposes bounded changes to residence time, liquid and gas flow, temperature, concentration, pressure, tubing, filtration, or safety controls. For multiple experiments, the campaign preserves every observation, identifies the best observed anchor, and may fit a simple apparent response-versus-residence-time model. Extrapolated targets are reported as screening estimates with acceptance criteria, not guaranteed yields. Every cycle creates a new version while retaining the preceding design and measurements.

5.6 Model routing and run records

The user interface exposes separate model routes for upstream chemistry interpretation and downstream proposal/council execution. The selected provider, model identifier, and local OpenAI-compatible endpoint mapping are applied for the duration of one serialized run and restored afterward. Model routing is written to pipeline-runtime provenance. Because hosted model aliases and local checkpoints can change, a reproducible report should retain the exact model identifier, endpoint or provider, generation parameters, software revision, dependency environment, frozen DesignInputPackage, inventory profile, raw model events, deterministic audit, and canonical final result.

Computational reconstruction requires the frozen inputs and inventory, exact model identifiers, generation parameters, software revision, raw model events, deterministic audit and final record. Historical liquid-volume, inlet-reference and pressure-corrected times retain their recorded definitions; later GUI defaults do not alter archived scores. Complete case inventories and the long worked-run transcripts are supplied in the companion archive.

6 GUI operation, inventory management, and auditable examples

Actual browser captures made on 1 October 2026 document protocol entry, fixed-ID questions, form-based inventory editing and JSON export. These interface demonstrations do not constitute new benchmark outcomes or chemical experiments. Full-screen captures, structured input/output records and the browser trace are retained in the companion archive.

The GUI example uses the two-stage DPDTC-mediated amidation of 3-methyl-4-nitrobenzoic acid with benzylamine, with a thioester intermediate. Its input specifies 30 min at 95 C for activation, 5-10 min cooling, and another 30 min at 95 C after amine addition. This example is separate from the three benchmark chemistries in Section 4 and is used here only to demonstrate the input and inventory workflow.

6.1 Batch entry, follow-up questions, and saved answers

Figure S9 shows the complete design-studio front page during a live intake interaction. Numbered callouts identify workspace navigation, protocol entry, standardized questions, model routing, inventory binding and the readiness gate. The screenshot uses a DPDTC protocol and the saved inventory profile; it was captured without a new design-generation call. It shows a question-answering state, not a completed experiment or a fabricated design result.

Three fresh requests with the same protocol, profile and extraction mode returned identical pending question IDs and question-set hashes. The pending IDs were Q-OBJ-001, Q-CHEM-001, Q-HIST-001, Q-HYP-001. The user supplied the archived objective and chemistry confirmation, supplied the archived hypotheses, and marked absent experimental history unavailable. The answer-saving endpoint returned ready_for_design = true. Complete questions, answers and intermediate packages are retained in the companion evidence, including items not simultaneously visible inside the scrolling GUI panel.

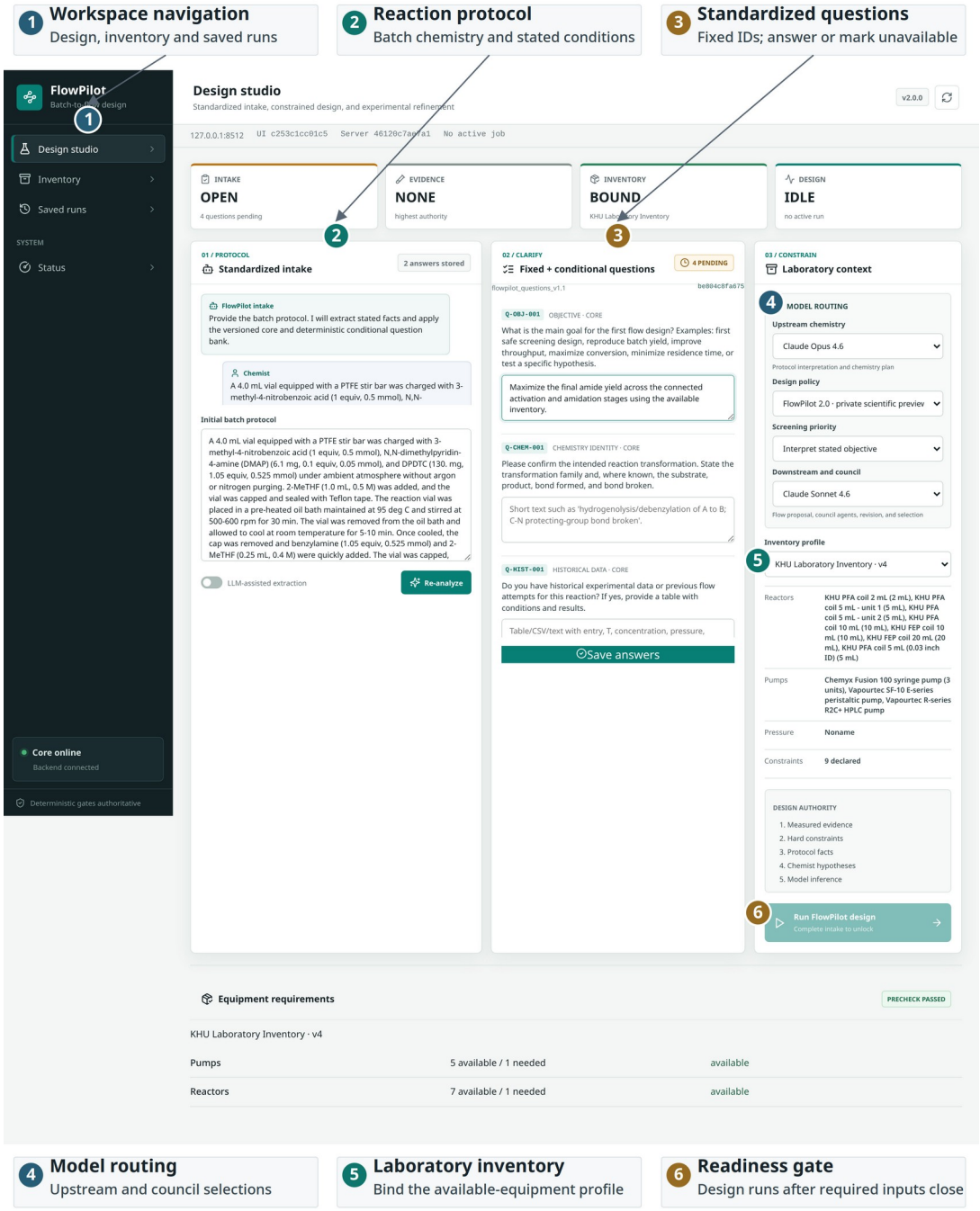


Figure S9. Annotated front page of the FlowPilot web application. (1) Navigation to design, inventory and saved runs. (2) Batch-protocol entry and the current intake exchange. (3) Fixed-ID questions and answer fields. (4) Upstream and downstream model selections. (5) Bound laboratory inventory. (6) Run control, disabled until required intake items are resolved. The browser image is an actual October 1 capture; only explanatory shapes, arrows and labels were added. The complete-resolution image and capture record are supplied separately. Inventory editing is shown in Figure S10.

6.2 Inventory input, normalization, and laboratory constraints

The inventory workspace supports direct entry in categorized forms as well as import from documents or an existing JSON profile (Figure S10). Its 15 categories cover pumps, reactors, tubing, light sources, gas hardware, mixers, pressure controllers, temperature controllers, degassers, filters, separators, connectors, collectors, reactor trains and safety accessories. Prepared fields are derived from the same typed equipment models used by the pipeline. Equipment can be added, edited or deleted without rebuilding the remaining inventory.

Forms retain equipment identifiers, quantities, system membership and compatibility, together with operating limits and notes. For pumps, the minimum supported flow and the setting increment are separate fields. Unknown optional ratings remain unspecified rather than becoming zero or false. The Constraints section records unavailable functions, permitted alternatives and additional operating limits. The KHU version-4 profile is used here to match the archived example, not to redefine the later laboratory configuration. Table S11 retains the corresponding source values.

Any committed edit invalidates the previous validation state. The current draft must pass profile validation before Export JSON, Save profile or Use in design is enabled. Export produces the complete InventoryProfile, including equipment, constraints and source provenance; importing this file restores the same structured data. Save profile writes a new version, while Use in design binds the profile and its interpreted operating limits to the intake package. A valid profile can still contain warnings and does not certify hardware compatibility or laboratory safety; design-specific checks are performed separately.

In the browser check, opening and applying the prepared pump form, validating, exporting and re-importing the KHU profile retained all seven reactor and three pump records, compatibility fields, operating constraints and source provenance. The saved laboratory profile was not overwritten. Dedicated JSON import bypasses LLM extraction. Separate desktop and mobile tests also created a new profile using the equipment forms and confirmed that invalid or unvalidated edits could not be exported or used for design.

Table S11. Examples of the KHU version-4 inventory data and their interpretation.

Inventory item	Stored information	Design implication
Degassing	No inline degasser; offline pre-degassing is allowed.	Omit the device, not the oxygen-exclusion requirement.
Reactors	PFA: 2, 5 and 10 mL; two 5 mL/1.016 mm units and one 5 mL/0.762 mm unit. FEP: 10 and 20 mL.	Use declared volume, material and ID combinations.
Temperature	Profile limits: PFA 80 °C; FEP 50 °C.	Equipment-specific limits, not universal material limits.
Pumps	Three Chemyx Fusion 100 units; Vapourtec SF-10 and R2C+.	Check the selected pump and syringe range.
Pressure control	Confirmed BPR settings: 2, 8 and 9 bar; maximum 10 bar.	Check the full flow path and pressure basis.
Mixing	One connector; no dedicated mixer.	Generic T-mixer requires verification; a two-port union cannot merge feeds.
Source caveats	Conflicting tubing units, missing radiant power and original BPR omission.	Retain uncertainty; a missing-value sentinel is not measured zero.

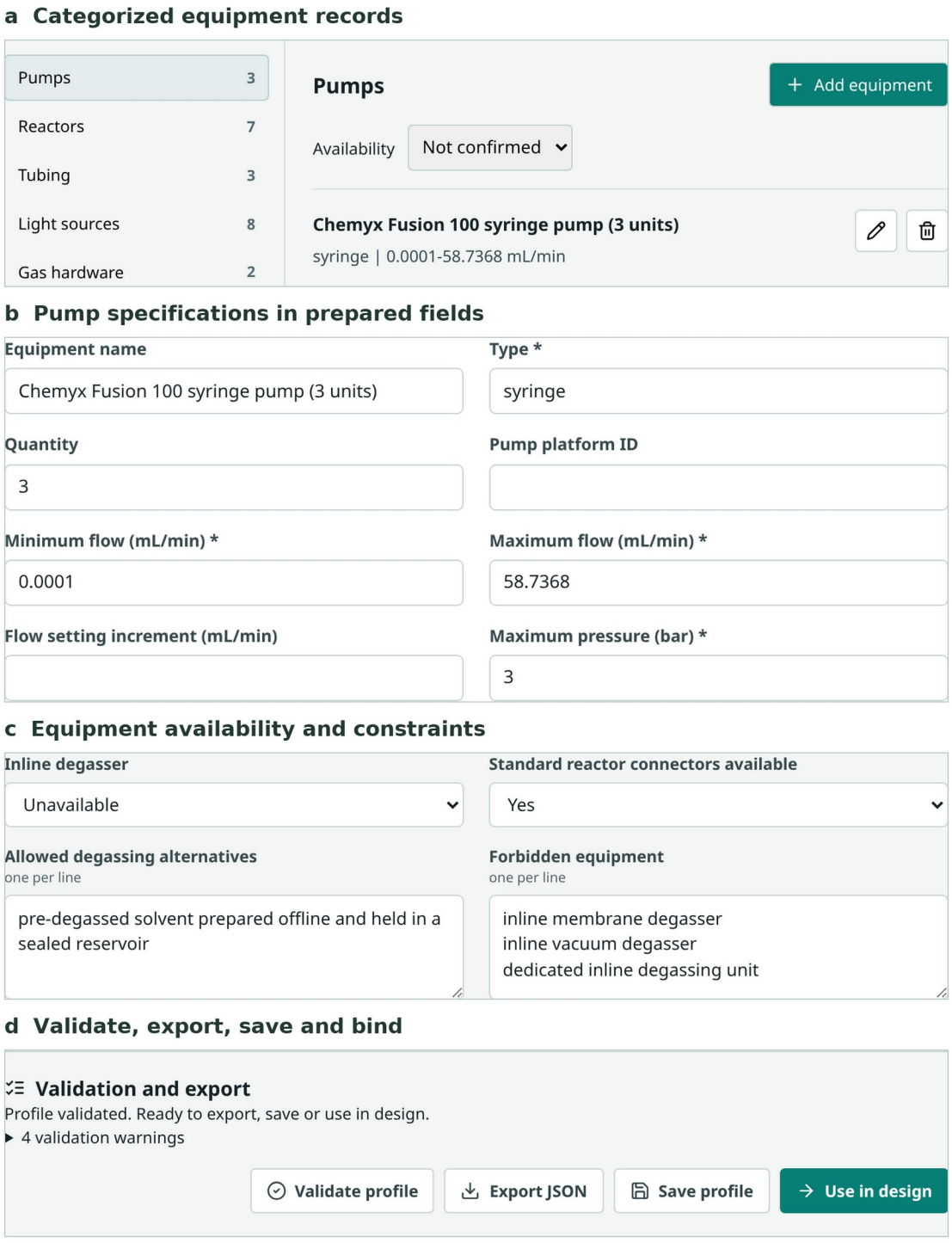


Figure S10. Form-based inventory management in the live webapp. (a) Equipment-category navigation and a pump record. (b) Prepared pump fields distinguish flow limits from the setting increment; a blank increment remains unknown. (c) Declared unavailability of an inline degasser, permitted alternatives and other constraints. (d) Profile validation and the JSON export, versioned-save and design-binding controls. Cropped excerpts use the archived KHU version-4 profile. Warnings remain accessible in the collapsed validation details; schema validity is not laboratory approval. Full views and the exported JSON accompany the figure.

6.3 Examples of critical findings and evaluator limitations

Table S12 gives traceable examples from retained one-shot and FlowPilot outcomes. They were selected to illustrate different failure mechanisms, not sampled to estimate a failure rate. A judge flag, a source-supported design inconsistency, and a deterministic software block are different observations. Multiple criteria can flag the same underlying defect. The archived benchmark scores and Figure S7 flag counts are unchanged; the arithmetic recheck below is an explicit post hoc audit, not a silent rescore.

Table S12. Concrete archived issues and one checked evaluator false positive.

Record / criterion	Observed issue	Interpretation
One-shot Opus 4.6; hydrogenolysis repeat 01; N2-D1FC65D8; UO-05	The procedure places a vented gas-liquid separator before the sole pressure-defining BPR while describing a pressurized reactor. The original post-reactor instructions confirm this order.	One judge flagged UO-05. The stated pressure-control and venting arrangement is incomplete as written; component pressure ratings alone do not resolve the topology.
One-shot Opus 4.6; hydrogenolysis repeat 02; N2-B6923FF7; UO-07	The main parameter reports 20.27 min, while its own basis text gives 3.0 mL / 0.1 mL/min = 30.0 min empty-bed time and a different holdup time.	Two judges flagged UO-07. The claimed basis and serialized value disagree. Distinct time definitions must not be collapsed into one unlabeled value.
FlowPilot Qwen3.6-27B; CuAAC repeat 03; N2-110E4388; UO-02/03/11	The final preparation instruction charges 0.25 mmol Cu/C into a 10 mL feed even though the catalyst is also assigned to a packed-bed cartridge. The saved operating_procedure confirms the instruction.	One judge raised three criterion flags for one solids-assignment defect. The feed-preparation compiler and heterogeneous-catalyst placement must agree; passing geometry checks is insufficient.
FlowPilot Opus 4.6; oxidation repeat 01; archived worked-run transcript	The archived council summary reports no usable candidates and a calculator center-point fallback. The raw narrative and final structured conditions then differ (compa**n**ion archiv**e**).	Operational fallback and artifact synchronization are separate limitations. A completed pipeline does not establish successful council selection or consistency of every report artifact.
FlowPilot GPT-4o; oxidation repeat 03; N2-E85D424E; UO-08	One Qwen judge called 3.3158 sccm versus 3.3158 mL/min at STP a factor-of-1000 defect.	This specific allegation is incorrect. These volumetric units are equal at the same standard conditions. The arithmetic gives approximately 1 O_2 equivalent, not 0.001. The historical flag remains disclosed as a judge false positive.

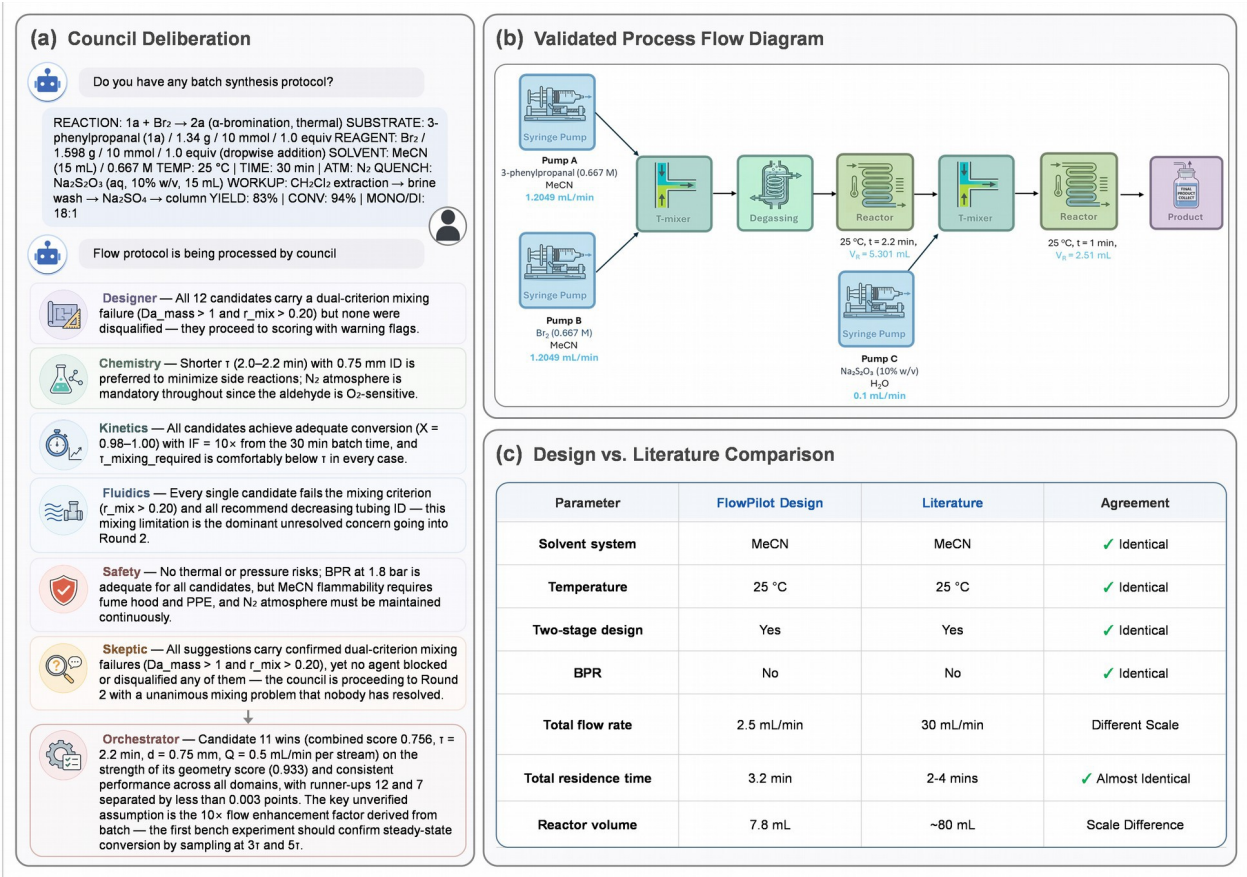
Unit audit for N2-E85D424E: oxygen feed = (3.3158 mL/min air x 0.21) / (22.414 mL/mmol) = 0.0310662 mmol/min O_2. Dividing by the recorded substrate flow of 0.031068 mmol/min gives 0.99994 equivalent. The same reference temperature and pressure must be used for both volume units. This checks the stated thousand-fold allegation only; it does not validate all other aspects of that design. It also explains why the heatmap should be described as judge-reported critical flags rather than confirmed error counts.

6.4 Computational dry-run example: bromination and inline quench

Before the connected reactions in Section 7 were implemented, a computational example examined α-bromination of 3-phenylpropanal with bromine in acetonitrile followed by aqueous thiosulfate quenching, based on the chemistry reported by Lu et al. [10]. The design question was how to connect reaction and residual-reagent deactivation without reproducing the batch addition and work-up as disconnected instructions. Figure S11 reproduces the former preprint Figure 6 unchanged, retaining the displayed council exchange, process scheme and contextual literature comparison.

The displayed scheme assigns 2.2 min to the reaction segment and 1.0 min to quench contact. At the displayed feed rates, these correspond to approximately 5.30 and 2.51 mL, respectively; the downstream calculation includes the added quench stream. All 12 candidates nevertheless retained a mixing warning in the archived discussion. This was a dry run, not a measured yield, conversion or safety demonstration, and agreement with a literature time range at another scale does not establish successful process transfer.

The unedited historical artwork also exposes limitations of that early output: the selected-candidate dialogue and the final scheme use different flow values, and the safety dialogue refers to a BPR absent from the scheme and comparison table. Its legacy label “Validated Process Flow Diagram” and estimated conversion statements must therefore not be read as experimental validation or a closed executable design. This record illustrates the need for the shared final-design record described in Section 2; it is excluded from the architecture-score aggregates and from the wet-lab evidence.



Parameter	FlowPilot Design	Literature	Agreement
Solvent system	MeCN	MeCN	✓ Identical
Temperature	25 °C	25 °C	✓ Identical
Two-stage design	Yes	Yes	✓ Identical
BPR	No	No	✓ Identical
Total flow rate	2.5 mL/min	30 mL/min	Different Scale
Total residence time	3.2 min	2-4 mins	✓ Almost Identical
Reactor volume	7.8 mL	~80 mL	Scale Difference

Figure S11. Historical computational demonstration of aldehyde α-bromination followed by inline quenching, reproduced unchanged from preprint Figure 6. (a) Archived council discussion and unresolved mixing concerns. (b) Proposed equipment arrangement. (c) Contextual comparison with the literature process [10]. The legacy validation wording refers to the historical computational illustration, not laboratory verification. Conflicting dialogue/scheme values are retained and discussed in Section 6.4; the image must not be used as an execution protocol.

7 Connected-process case studies supporting main-text Figures 5 and 6

The two experimentally examined chemistries used a common protocol and inventory within each case, with three response sets varying the design priorities. Tables S13–S14 provide the exact answers; Figures S12–S17 retain the corresponding proposed topologies. This section documents the inputs and proposals. Section 8 separately records the actual feed preparation, operating points, yield measurements and implementation changes; the three sets are not three independent experimental replicates.

7.1 Photoredox Giese addition followed by sulfoxidation (main-text Figure 5)

The target sequence converts the alpha-silyl sulfide and acrylonitrile to a sulfide, then to sulfoxide 4a. The chemistry source is Park et al. [11]. Stage 1 is oxygen-free; pure oxygen is introduced only at the Stage 2 inlet in the revised flow proposal. The original batch protocol uses ambient air in the second step. The change to pure oxygen is a researcher-specified constraint in the September 18 reruns, not an autonomous discovery attributed to the model.

Supplied batch protocol (verbatim supplied transcription).

Step 1: Giese reaction A re-sealable pressure tube (13 × 100 mm) equipped with a magnetic stir bar was charged with (((4-methoxyphenyl)thio)methyl)trimethylsilane (1, 0.2 mmol, 1.0 equiv), acrylonitrile (2, 0.4 mmol, 2.0 equiv), and $[Ir(dF(CF_3)ppy)_2(dtbpy)]PF_6$ (0.001 mmol, 0.5 mol %) under an argon atmosphere. A degassed solvent mixture (2.0 mL, 0.1 M with respect to 1) of EtOH:pH 9 buffer (5:1, v/v) was added, and the resulting light greenish-yellow mixture, under vigorous magnetic stirring, was positioned 3 cm from a pair of 5 W blue LEDs (λmax = 452 nm) and irradiated at room temperature (25 °C, maintained with a cooling fan to counteract heating from the LEDs) for 4 h

Step 2: Oxidation reaction After 4 h, the cap of the tube was removed to expose the reaction mixture to ambient air, providing atmospheric oxygen as the terminal oxidant, and stirring was continued under otherwise identical irradiation conditions for an additional 6 h.

Table S13. Standardized responses for Giese addition/oxidation (case 1; main-text Figure 5). Shared answers are listed once; objectives and hypotheses retain the supplied wording, with form prefixes and duplicate punctuation removed.

Set	Question ID	Confirmed response
All three	Q-GAS-002	Use at least 2.0 equiv of oxygen relative to the limiting substrate; higher oxygen equivalents may be used if justified by the reaction and operating conditions.
All three	Q-GAS-003	Introduce oxygen only at the Stage 2 inlet. Stage 1 must remain oxygen-free.
1	Q-OBJ-001	Develop a safe, reproducible, and inventory-compatible two-stage flow process that maximizes the final yield of sulfoxide 4a while avoiding unnecessarily long residence times. Design the two stages as one connected process, because Stage 1 determines the composition entering Stage 2.

Set	Question ID	Confirmed response
1	Q-HYP-001	Stage 1 performance may depend on effective oxygen exclusion and photon delivery. Incomplete sulfide formation will limit the material available for subsequent oxidation. Stage 2 may depend on oxygen availability and gas–liquid transport. Use the batch reaction time as a reference rather than a fixed target, and determine the residence time based on flow performance. Balance photon delivery, oxygen feed, pressure, and stage-specific residence times based on the final sulfoxide yield and selectivity.
2	Q-OBJ-001	Prioritize high conversion through both stages and a high final yield of sulfoxide 4a. Choose the shortest residence times reasonably justified for this conversion-focused starting design rather than adding a large time excess by default. Keep reactor volumes and flow rates practical for flow screening.
2	Q-HYP-001	Stage 1 requires sufficient photon delivery and residence time to generate enough sulfide for Stage 2. Use the batch reaction time as a reference rather than a fixed target, and determine the residence time based on flow performance. Stage 2 requires adequate oxygen delivery, but oxygen feed, pressure, gas fraction, and residence time are coupled. Residence time should be optimized based on the final sulfoxide yield and selectivity rather than increased by default.
3	Q-OBJ-001	Identify a practical two-stage flow design that balances final sulfoxide yield and selectivity with substrate throughput, total processing time, oxygen delivery, and stable operation. Prefer a compact, higher-throughput starting design where chemically justified.
3	Q-HYP-001	Stage 1 may maintain sufficient sulfide formation at higher liquid throughput if photon delivery remains effective. Use the batch reaction time as a reference rather than a fixed target, and determine the residence time based on flow performance. In Stage 2, pressure and oxygen feed can alter oxygen availability, gas fraction, and contact time simultaneously. Evaluate the overall process based on substrate throughput, stage-specific residence times, and the final sulfoxide yield and selectivity. Shorter residence times may be explored, but should be validated experimentally.

Design records: outputs/figure5_pure_oxygen_physics_20260918, attempt_01 for each set. The routing was Claude Opus 4.6 upstream and Claude Sonnet 4.6 downstream, scientific_v2 policy, with 12 candidates. The source response sets specify integrated process performance, conversion and throughput, respectively; all require oxygen introduction only at Stage 2. The later pure-oxygen requirement was a researcher-specified revision following the gas-flow observation, not a different wording silently substituted into Q-GAS-002. Figures S12–S14 preserve all three proposed topologies; their convergence is not evidence of independent experimental repeats.

Equipment interpretation: Stage 1 uses an inventory-confirmed UV-150-compatible PFA coil and the UV-150 light; Stage 2 uses the Manual 2 module and its own 448 nm strip light. The UV-150 light is not detached for use with an unrelated manual module. The liquid pump is a Vapourtec E-series channel with BLUE tubing; gas delivery uses the FF-C00 MFC, a CV3301 gas-line check valve and the P713 mixing junction. No inline degasser is assigned. Offline deoxygenation and protection of the Stage 1 feed remain required by the oxygen-free protocol.

Unresolved experimental checks: the gas-line check valve addresses liquid entry into the gas branch, not gas migration into the upstream liquid reactor. The earlier experimental backflow observation is therefore not declared solved by its presence. Confirm valve orientation and pressure rating, liquid-branch isolation/protection, actual junction pressures, oxygen-compatible hardware, and startup/shutdown procedures before testing the archived 7 bar(g) proposal. The optional transient tool used illustrative dynamics and a 12 bar(g) simulated gas-source pressure; that value is not a laboratory setpoint and exceeds the recorded MFC inlet rating. Its outputs must not be treated as a calibrated incident prediction or safety clearance. Gas pressure traces and equipment response data were unavailable.

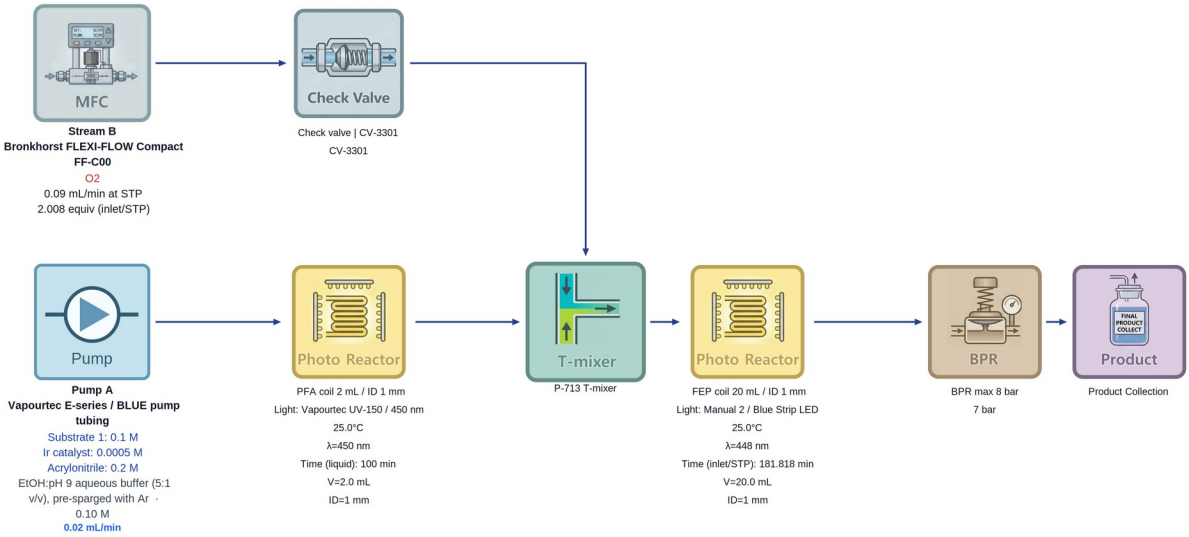


Figure S12. Proposed Giese addition/oxidation topology for response Set 1 (main-text Figure 5). Horizontal layout of the archived design; no new model run was used. Oxygen is introduced only after R1; the gas-line check valve is not proof of upstream liquid-branch protection. R1 uses liquid flow; R2 uses the nominal inlet/STP time index, not pressure-dependent residence time. This is a proposal, not a diagram of independently verified operating conditions; implementation differences are recorded in Tables S17–S18.

Reading the scheme: R1: 2 mL, liquid 0.02 mL min^{-1}, gas 0 mL min^{-1} at STP, time 100.00 min, 25 °C; R2: 20 mL, liquid 0.02 mL min^{-1}, gas 0.09 mL min^{-1} at STP, time 181.82 min, 25 °C.

Feed A, 0.02 mL min^{-1}: substrate 1a (0.1 M; 0.002 mmol min^{-1}); iridium photocatalyst (0.0005 M; 1e-05 mmol min^{-1}); acrylonitrile (0.2 M; 0.004 mmol min^{-1}).

Feed B, 0.09 mL min^{-1} at STP: O_2 (0.004016 mmol min^{-1}).

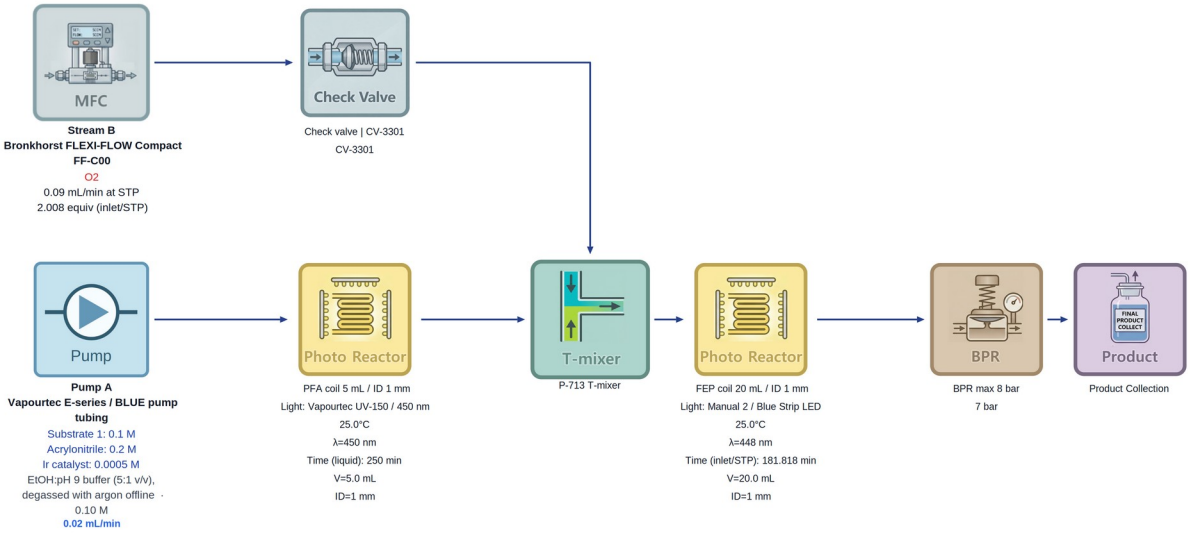


Figure S13. Proposed Giese addition/oxidation topology for response Set 2 (main-text Figure 5). Horizontal layout of the archived design; no new model run was used. Oxygen is introduced only after R1; the gas-line check valve is not proof of upstream liquid-branch protection. R1 uses liquid flow; R2 uses the nominal inlet/STP time index, not pressure-dependent residence time. This is a proposal, not a diagram of independently verified operating conditions; implementation differences are recorded in Tables S17–S18.

Reading the scheme: R1: 5 mL, liquid 0.02 mL min^{-1}, gas 0 mL min^{-1} at STP, time 250.00 min, 25 °C; R2: 20 mL, liquid 0.02 mL min^{-1}, gas 0.09 mL min^{-1} at STP, time 181.82 min, 25 °C.

Feed A, 0.02 mL min^{-1}: substrate 1a (0.1 M; 0.002 mmol min^{-1}); acrylonitrile (0.2 M; 0.004 mmol min^{-1}); iridium photocatalyst (0.0005 M; 1e-05 mmol min^{-1}).

Feed B, 0.09 mL min^{-1} at STP: O_2 (0.004016 mmol min^{-1}).

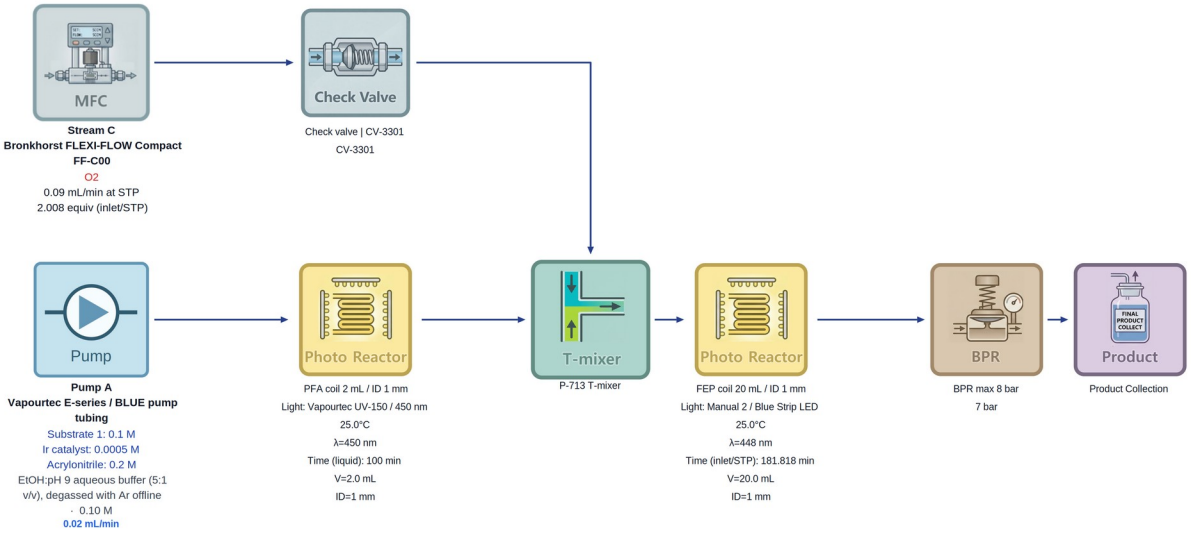


Figure S14. Proposed Giese addition/oxidation topology for response Set 3 (main-text Figure 5). Horizontal layout of the archived design; no new model run was used. Oxygen is introduced only after R1; the gas-line check valve is not proof of upstream liquid-branch protection. R1 uses liquid flow; R2 uses the nominal inlet/STP time index, not pressure-dependent residence time. This is a proposal, not a diagram of independently verified operating conditions; implementation differences are recorded in Tables S17–S18.

Reading the scheme: R1: 2 mL, liquid 0.02 mL min^{-1}, gas 0 mL min^{-1} at STP, time 100.00 min, 25 °C; R2: 20 mL, liquid 0.02 mL min^{-1}, gas 0.09 mL min^{-1} at STP, time 181.82 min, 25 °C.

Feed A, 0.02 mL min^{-1}: substrate 1a (0.1 M; 0.002 mmol min^{-1}); iridium photocatalyst (0.0005 M; 1e-05 mmol min^{-1}); acrylonitrile (0.2 M; 0.004 mmol min^{-1}).

Feed C, 0.09 mL min^{-1} at STP: O_2 (0.004016 mmol min^{-1}).

7.2 Telescoped DPDTC-mediated amidation (main-text Figure 6)

The supplied protocol forms N-benzyl-3-methyl-4-nitrobenzamide through a 2-pyridyl thioester; the chemistry source is Saunders et al. [12]. Batch timing, nominal concentrations and stage order are protocol facts. Flow residence times, the omission of interstage cooling, and the use of a premixed activation feed are proposed choices requiring validation.

Supplied batch protocol (verbatim supplied transcription).

A 4.0 mL vial equipped with a PTFE stir bar was charged with 3-methyl-4-nitrobenzoic acid (1 equiv, 0.5 mmol), N,N-dimethylpyridin-4-amine (DMAP) (6.1 mg, 0.1 equiv, 0.05 mmol), and DPDTC (130. mg, 1.05 equiv, 0.525 mmol) under ambient atmosphere without argon or nitrogen purging. 2-MeTHF (1.0 mL, 0.5 M) was added, and the vial was capped and sealed with Teflon tape. The reaction vial was placed in a pre-heated oil bath maintained at 95 °C and stirred at 500–600 rpm for 30 min. The vial was removed from the oil bath and allowed to cool at room temperature for 5–10 min. Once cooled, the cap was removed and benzylamine (1.05 equiv, 0.525 mmol) and 2-MeTHF (0.25 mL, 0.4 M) were quickly added. The vial was capped, sealed with Teflon tape, and returned to the oil bath at 95 °C, and stirred at 500–600 rpm for another 30 min before workup.

Table S14. Standardized responses for DPDTC-mediated amidation (case 2; main-text Figure 6). The chemistry-identity answer is common to all sets. Supplied wording is retained apart from form prefixes and duplicate punctuation.

Set	Question ID	Confirmed response
All three	Q-CHEM-001	This is a two-stage DPDTC-mediated amide formation through a thioester intermediate. In Stage 1, 3-methyl-4-nitrobenzoic acid is converted to the corresponding 2-pyridyl thioester using DPDTC and DMAP, without isolating the intermediate. In Stage 2, benzylamine is added to the Stage 1 effluent to form N-benzyl-3-methyl-4-nitrobenzamide. The amide C(O)–N bond is formed and the thioester C(O)–S bond is cleaved during Stage 2.
1	Q-OBJ-001	Develop a safe, reproducible, and inventory-compatible two-stage telescoped flow process that maximizes the final amide yield while avoiding unnecessarily long residence times. Design the two stages as one connected process, because Stage 1 determines the composition entering Stage 2.
1	Q-HYP-001	Stage 1 conversion will directly affect the amount of thioester available for Stage 2 amidation. The cooling step before benzylamine addition may be a batch-handling step rather than an essential reaction step and should be evaluated in flow. Use the batch temperature and reaction times as reference conditions rather than fixed targets, and determine the operating conditions based on flow performance. The Stream A/Stream B molar ratio should match the intended stoichiometry, taking both concentration and flow rate into account.

Set	Question ID	Confirmed response
2	Q-OBJ-001	Prioritize high conversion through both stages and a high final amide yield. Choose the shortest residence times reasonably justified for this conversion-focused starting design rather than adding a large time excess by default. Keep reactor volumes and flow rates practical for flow screening.
2	Q-HYP-001	Stage 1 requires sufficient conversion to generate enough thioester for Stage 2. Stage 2 requires sufficient reaction time for conversion to the final amide. Use the batch temperature and reaction times as reference conditions rather than fixed targets, and determine the operating conditions based on flow performance. Evaluate the overall process based on the final amide yield rather than either stage alone. The Stream A/Stream B molar ratio should reflect the intended stoichiometry based on both concentration and flow rate.
3	Q-OBJ-001	Identify a practical two-stage flow design that balances final amide yield with substrate throughput, total processing time, and stable operation. Prefer a compact, higher-throughput starting design where chemically justified.
3	Q-HYP-001	Stage 1 must maintain sufficient thioester formation as throughput is increased. Direct transfer from Stage 1 to benzylamine addition may avoid the batch cooling step if chemically and operationally acceptable. Use the batch temperature and reaction times as reference conditions rather than fixed targets. Evaluate the overall process based on substrate throughput, stage-specific residence times, and the final amide yield. Shorter residence times may be explored, but should be validated experimentally. The Stream A/Stream B molar ratio should match the intended stoichiometry based on both concentration and flow rate.

Design records: outputs/khu_revised_six_20260915/presentation/figure6_set1 through figure6_set3; collaborator export 20260915_163845. The routing was Claude Opus 4.6 upstream and Claude Sonnet 4.6 downstream, scientific_v2 policy, with 12 candidates. The response sets prioritize integrated performance, conversion and throughput, respectively. Figures S15–S17 reproduce the three proposed topologies. The laboratory grouped Sets 2/3 into one operating-point result; proposal multiplicity must not be treated as additional yield replicates.

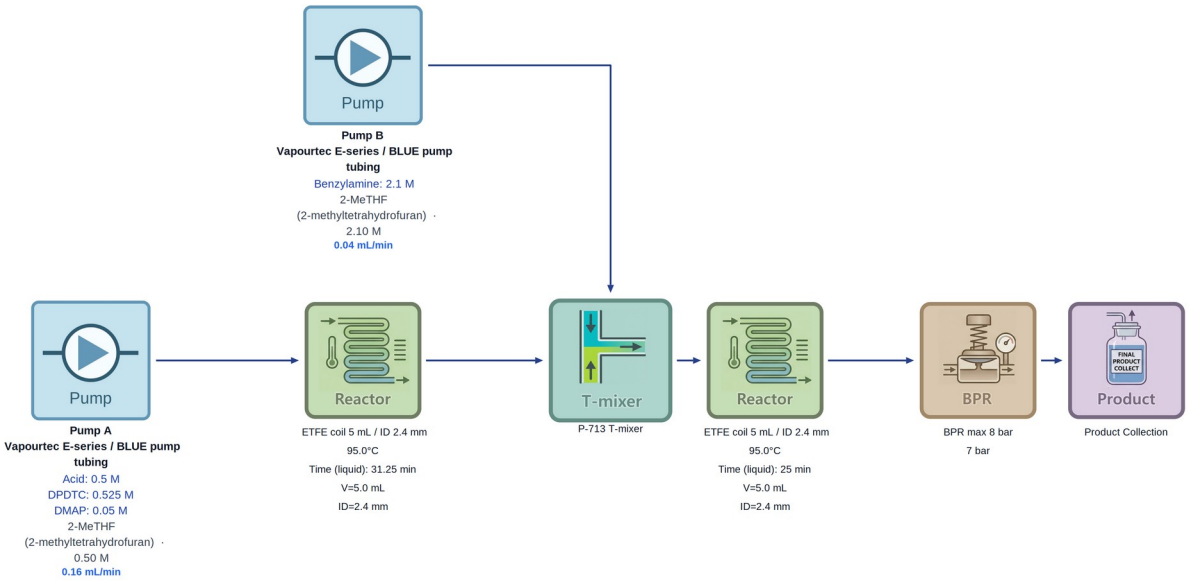


Figure S15. Proposed DPDTC-mediated amidation topology for response Set 1 (main-text Figure 6). Horizontal layout of the archived design; no new model run was used. Benzylamine enters between the coils. R1 time uses Feed A alone; R2 time uses the combined liquid flow. Feed concentrations and equipment labels are retained from the proposal. This is a proposal, not a diagram of independently verified operating conditions; implementation differences are recorded in Tables S17–S18.

Reading the scheme: R1: 5 mL, liquid 0.16 mL min^{-1}, gas 0 mL min^{-1} at STP, time 31.25 min, 95 °C; R2: 5 mL, liquid 0.2 mL min^{-1}, gas 0 mL min^{-1} at STP, time 25.00 min, 95 °C.

Feed A, 0.16 mL min^{-1}: acid 1b (0.5 M; 0.08 mmol min^{-1}); DPDTC (0.525 M; 0.084 mmol min^{-1}); DMAP (0.05 M; 0.008 mmol min^{-1}).

Feed B, 0.04 mL min^{-1}: benzylamine (2.1 M; 0.084 mmol min^{-1}).

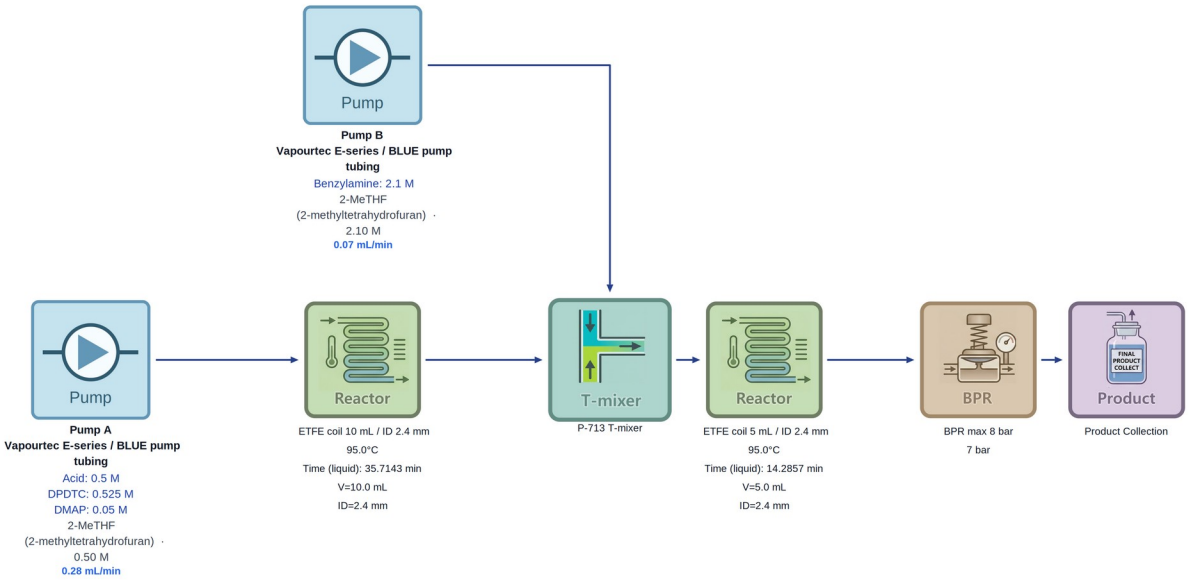


Figure S16. Proposed DPDTC-mediated amidation topology for response Set 2 (main-text Figure 6). Horizontal layout of the archived design; no new model run was used. Benzylamine enters between the coils. R1 time uses Feed A alone; R2 time uses the combined liquid flow. Feed concentrations and equipment labels are retained from the proposal. This is a proposal, not a diagram of independently verified operating conditions; implementation differences are recorded in Tables S17–S18.

Reading the scheme: R1: 10 mL, liquid 0.28 mL min^{-1}, gas 0 mL min^{-1} at STP, time 35.71 min, 95 °C; R2: 5 mL, liquid 0.35 mL min^{-1}, gas 0 mL min^{-1} at STP, time 14.29 min, 95 °C.

Feed A, 0.28 mL min^{-1}: acid 1b (0.5 M; 0.14 mmol min^{-1}); DPDTC (0.525 M; 0.147 mmol min^{-1}); DMAP (0.05 M; 0.014 mmol min^{-1}).

Feed B, 0.07 mL min^{-1}: benzylamine (2.1 M; 0.147 mmol min^{-1}).

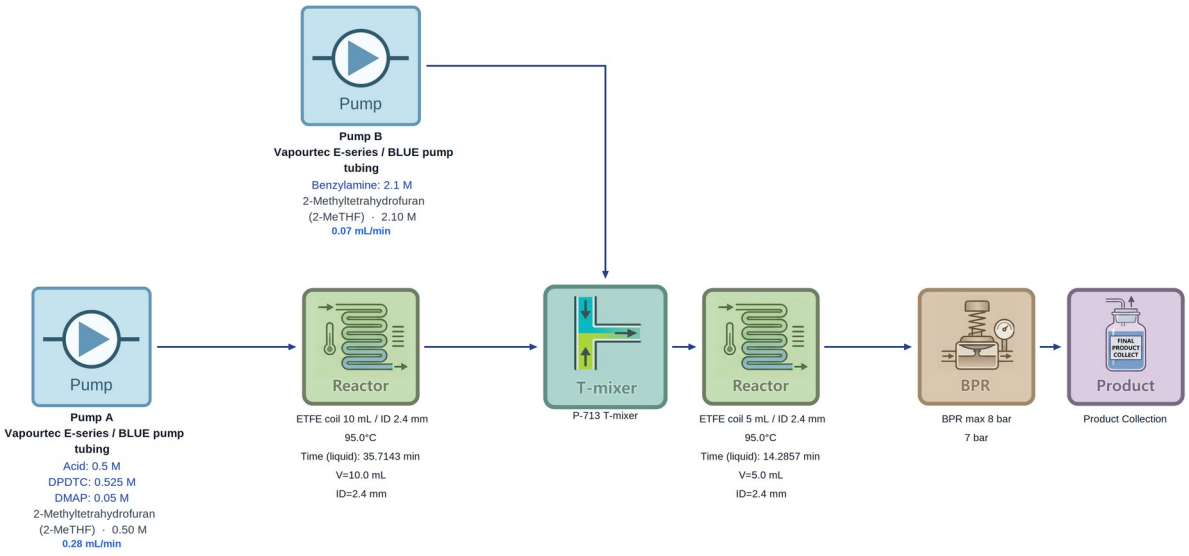


Figure S17. Proposed DPDTC-mediated amidation topology for response Set 3 (main-text Figure 6). Horizontal layout of the archived design; no new model run was used. Benzylamine enters between the coils. R1 time uses Feed A alone; R2 time uses the combined liquid flow. Feed concentrations and equipment labels are retained from the proposal. This is a proposal, not a diagram of independently verified operating conditions; implementation differences are recorded in Tables S17–S18.

Reading the scheme: R1: 10 mL, liquid 0.28 mL min^{-1}, gas 0 mL min^{-1} at STP, time 35.71 min, 95 °C; R2: 5 mL, liquid 0.35 mL min^{-1}, gas 0 mL min^{-1} at STP, time 14.29 min, 95 °C.

Feed A, 0.28 mL min^{-1}: acid 1b (0.5 M; 0.14 mmol min^{-1}); DPDTC (0.525 M; 0.147 mmol min^{-1}); DMAP (0.05 M; 0.014 mmol min^{-1}).

Feed B, 0.07 mL min^{-1}: Benzylamine (2.1 M; 0.147 mmol min^{-1}).

7.3 Recorded council deliberation for the two connected-process cases

This section documents the recorded council reports underlying the three proposed configurations for each connected-process case (Figures S12-S17; main-text Figures 5 and 6). The Figure 5 records are the pure-oxygen reruns of 18 September 2026; the Figure 6 records are the amidation runs of 15 September 2026. These are the campaigns already cited in Sections 7.1 and 7.2, not new generations. Reported choices describe proposals at the time of computation. Actual implementation and analytical results remain separately documented in Section 8.

Each run used Claude Opus 4.6 for upstream chemistry interpretation and Claude Sonnet 4.6 for downstream generation and every council role. The council therefore comprises separate role-conditioned calls to the same downstream model, not independent model families or human reviewers. Twelve inventory-constrained connected-process candidates were supplied for review. Candidate screening times were generated using fixed exploratory multipliers relative to the batch holds and reconciled to available equipment; the multipliers are not fitted rate constants, yield predictions or experimentally established intensification factors.

DrChemistry examined identity, feed composition and order of addition; DrKinetics examined stage exposure and uncertainty; DrFluidics examined flows, equipment and transport; and DrSafety examined

operating limits and unresolved hazards. Their structured reports used candidate identifiers, recommendations, hard-violation flags, justifications and uncertainties. The Skeptic request included the specialist reviews, and the Chief request included both those reviews and the Skeptic response. Thus the recorded exchange is a sequence of structured reports passed between roles, rather than a free-form simultaneous conversation. The Chief selected one existing candidate and recorded alternatives, answer impacts, limitations and follow-up measurements. No additional iterative design-revision round should be inferred from a response-format retry.

For each selected candidate, the complete four specialist verdicts, available candidate-specific Skeptic assessment and complete Chief response are reproduced below. Shared issues are introduced before the case records. The companion file FlowPilot_Figures5_6_full_council_reports.docx contains every recorded council response for all 12 candidates in all six runs, including tool requests and unsuccessful truncated responses. The source archive retains exact response text, full prompts and requests, candidate pools, final results and source hashes. The detailed ESI account is not a substitute for that unabridged archive. Source identifiers F5-S1 to F6-S3 map to their original paths in source_manifest.json.

Quoted or transcribed model statements are historical outputs, not independently verified scientific conclusions or instructions to operate. Numerical discrepancies and interpretation limits are annotated explicitly rather than silently editing the reports. Final values are checked against the archived stage/feed records. A recommendation of acceptable or prefer means the model retained a candidate for screening; it does not establish chemical success, resolve all engineering uncertainties or authorize laboratory execution.

Typography of reproduced reports. Scientific symbols, chemical formulas and unit exponents in the displayed council reports are normalized for publication. Their numerical values and scientific wording are retained, including the errors identified in the interpretation notes. Exact original strings remain available in the unchanged raw logs and full-council companion.

The objective-policy record is also important for interpreting the three sets. In both chemistries, Set 2 was classified as yield_priority; Sets 1 and 3 were classified as balanced. Set 3 included higher-throughput wording in the human brief but was not classified as a distinct throughput_priority policy. Consequently, the three sets are different response-conditioned selections, not a controlled ablation of three reliably distinct policy modes. Candidate identifiers are local to each run, and convergent choices are retained rather than treated as independent physical designs.

Figure 5: Giese addition and oxidation

Chemistry and exposure. The reviews kept the oxygen-free Giese step separate from oxidation and assessed the complete connected sequence. The recorded liquid feed contained substrate at 0.1 M, acrylonitrile at 0.2 M and iridium photocatalyst at 0.0005 M. Offline feed deoxygenation was retained because no inline degasser was assigned. The council considered the available UV-150-compatible Stage 1 coils and the separate Manual 2 oxidation module with its own light. It did not establish photon-dose equivalence, reaction kinetics or a predicted sulfoxide yield. A larger coil was considered as a way to change Stage 1 exposure, not as proof of better conversion.

Gas identity and dose. Pure oxygen at the Stage 2 inlet was already imposed by the researcher in the frozen input; the council did not autonomously discover the air-to-oxygen change. The 0.09 mL/min inlet/STP setting at 0.02 mL/min liquid feed delivers approximately 2.008 mol O_2 per mol substrate, using 22.414 mL/mmol at 273.15 K and 1.01325 bar. It is a delivered feed ratio, not measured gas uptake. References in

the reports to an unapproved oxygen change mean that laboratory oxygen-service approval remained outstanding, not that the researcher had failed to specify oxygen.

Fluidics and safety. DrFluidics and DrSafety each requested the transient-flow tool for all 12 candidates. Their request identifiers point to the same deterministic result bundle within a run; these are two reviews of shared tool evidence, not independent simulation replications. The archive contains 192 conditional simulations per Figure 5 run. The gas-line check valve does not close the connected reverse path from the gas-injection junction into the Stage 1 liquid branch. Proposed liquid-branch protection, pressure-decoupled transfer and startup controls remained recommendations for engineering review; the tool record states design_modification_applied = false and laboratory_execution_status = review_required.

Conditional-model boundary. Gas-source pressure, dead volume, compliance and device response were illustrative assumptions because measured startup traces were unavailable. The assumed 12 bar(g) gas-source pressure was not a laboratory setpoint and exceeded the recorded 10 bar MFC inlet limit. The pressure reference and device rating require confirmation. Tool outputs therefore cannot establish equipment compliance, predict incident probability or demonstrate absence of backflow. The qualitative earlier backflow observation supports examining the pathway, but does not calibrate the simulations or prove what would happen at the lower proposed gas flow.

Outcome across response sets. Sets 1 and 3 selected Candidate 1: a 2 mL Stage 1 coil with a 100 min liquid V/Q time. Set 2 selected Candidate 6: a 5 mL Stage 1 coil with a 250 min liquid V/Q time. All three retained a 20 mL Stage 2 coil, 0.09 mL/min oxygen at STP and 0.02 mL/min liquid, giving an inlet/STP time index of 181.82 min. This Stage 2 index is not pressure-dependent liquid residence time, photon exposure or a kinetic equivalent of the 360 min batch hold. All candidates had the same substrate feed throughput (0.002 mmol/min), so selection of the compact design did not itself increase feed throughput.

Interpretation notes for the unedited reports

Time interpretation: some reports call Stage 2 liquid exposure approximately 1000 min, obtained from 20 mL/0.02 mL/min. This is a liquid-only space-time calculation using the entire reactor volume, not measured two-phase liquid residence. Likewise, the 181.82 min inlet/STP index must not be interpreted as a measured fraction of batch reaction exposure.

Numerical transcription: the Figure 5 Set 3 Chief response gives the photocatalyst molar flow as 0.001 mmol/min. The final feed record and concentration-times-flow calculation give 0.0005 mol/L x 0.02 mL/min = 0.000010 mmol/min. The original statement is retained in the transcript as a 100-fold numerical error, not adopted in the figure text or experimental recipe.

Geometry and transport claims: the Set 1 DrFluidics shared assessment groups Candidate 6 with large-bore ETFE options even though Candidate 6 uses 1 mm-ID PFA. The Set 3 Chief attributes a smaller in-channel gas volume fraction to the selected reactor relative to other 7 bar candidates with smaller reactors. Their gas and liquid feed rates, pressure and temperature are the same; changing reactor volume alone does not establish a different gas/liquid volumetric flow ratio or measured gas holdup. A known bore defines geometry but does not establish photon-dose adequacy or a measured transport advantage.

Protocol interpretation: several reports describe the Stage 2 batch temperature as unspecified, whereas the supplied procedure continues under otherwise identical irradiation conditions after Stage 1 at 25 deg C. That wording does not establish an independently measured Stage 2 temperature, but it does not justify presenting 25 deg C as an unrelated new condition. Similarly, the qualitative backflow incident does not

prove that every lower-flow alternative will reverse; the defensible finding is that the unprotected pathway remains.

Figure 5, response Set 1 (F5-S1): selected Candidate 1

Recorded objective: Develop a safe, reproducible, and inventory-compatible two-stage flow process that maximizes the final yield of sulfoxide 4a while avoiding unnecessarily long residence times. Design the two stages as one connected process, because Stage 1 determines the composition entering Stage 2.

Final-record cross-check: R1: 2 mL, 100.00 min (liquid); R2: 20 mL, 181.82 min (inlet/STP apparent). The model priority was balanced. All four specialists reviewed all 12 candidates; the recorded veto list is empty. The selected candidate was preserved by final numerical reconciliation.

Response-format retries: DrSafety, attempt 1. These responses were truncated or invalid JSON and were retried, not treated as extra deliberation rounds. The unsuccessful text remains in the companion archive.

DrChemistry: complete selected-candidate verdict (recorded text; F5-S1, call 1)

Candidate: 1

Recommendation: acceptable

Hard violation reported: false

Justification: Baseline: Stage 1 τ=100 min (0.42× batch) in 2 mL PFA/UV-150 450 nm; Stage 2 τ_{inlet}=181.8 min (0.51× batch) in 20 mL FEP/Manual2, BPR 7 bar. Balanced stage ratios. Interpretable anchor for comparison. Backflow unresolved (shared); all other checks pass.

Uncertainty 1: Stage 1 at 100 min (0.42× batch) may be inadequate for Giese reaction; uncharacterized.

Uncertainty 2: Stage 2 apparent τ is inlet/STP index only; liquid exposure much longer but unverified.

Uncertainty 3: Backflow: unprotected reverse path to Stage 1 unresolved.

Uncertainty 4: CV-3301 pressure rating unconfirmed.

Uncertainty 5: O_2 service and gas-source change unapproved.

DrKinetics: complete selected-candidate verdict (recorded text; F5-S1, call 2)

Candidate: 1

Recommendation: acceptable

Hard violation reported: false

Justification: S1: 2 mL PFA, τ_{liq}=100 min (0.42× batch). S2: 20 mL FEP, τ_{inlet}=181.8 min (0.50× batch), BPR 7 bar, O_2 0.09 sccm, 2.01 equiv. Provides a mid-range S1/S2 combination as a baseline. Backflow and O_2-service issues unresolved but not a hard violation absent confirmed occurrence.

Uncertainty 1: S1 liquid τ 0.42× batch; adequacy for Giese completion unproven.

Uncertainty 2: S2 inlet/STP index 0.50× batch; liquid exposure ~1000 min (unconstrained by gas index).

Uncertainty 3: Unprotected liquid reverse path to S1 unresolved across all candidates.

Uncertainty 4: CV-3301 pressure rating unconfirmed; O_2-service compatibility unreviewed.

Uncertainty 5: Stage 2 temperature not batch-resolved.

DrFluidics: complete selected-candidate verdict (recorded text; F5-S1, call 4)

Candidate: 1

Recommendation: acceptable

Hard violation reported: false

Justification: Stage 1: 2 mL PFA, τ_{liq} = 100 min (ratio 0.42 vs. 240 min batch). Stage 2: 20 mL FEP, τ_{inlet} = 181.8 min (ratio 0.51 vs. 360 min batch), BPR 7 bar. Balanced sub-batch screen for both stages. No BPR violations; pressure headroom confirmed within model assumptions. Unprotected liquid reverse path unresolved (shared). Adequate balanced exploratory baseline.

Uncertainty 1: Unprotected liquid branch reverse path to Stage 1 unresolved.

Uncertainty 2: Stage 1 τ_{liq} 100 min is 42% of batch; photocatalytic conversion unknown.

Uncertainty 3: Stage 2 photon adequacy at 448 nm (vs. 452 nm) unconfirmed.

Uncertainty 4: BPR opening pressure reached in simulation but under illustrative, not measured, parameters.

DrSafety: complete selected-candidate verdict (recorded text; F5-S1, call 7)

Candidate: 1

Recommendation: acceptable

Hard violation reported: false

Justification: Stage 1: 2 mL PFA at 100 min liquid τ (0.42× batch). Stage 2: 20 mL FEP at 181.8 min inlet/STP τ (0.51× batch), 7 bar BPR, 2.01 equiv O_2 at 0.09 sccm. Unprotected reverse path to Stage 1 unresolved; pure O_2 use requires chemist and oxygen-service confirmation. Tool predicts no liquid displacement in entered-profile scenarios (unvalidated). CV-3301 pressure rating unconfirmed.

Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 reactor unresolved; engineering review required before execution.

Uncertainty 2: CV-3301 on MFC branch does not protect liquid branch; max pressure rating unconfirmed.

Uncertainty 3: Pure O_2 substitution for air is unapproved; requires chemist, inventory, and oxygen-service review.

Uncertainty 4: Gas check-valve cracking pressure (1.0 bar) concerns differential pressure across that valve, not BPR minus cracking pressure; low-flow seating unverified.

Uncertainty 5: Tool results conditional on illustrative, unmeasured parameters; no reversal predicted is not proof of safe operation.

Uncertainty 6: Stage 2 batch temperature not stage-resolved; rate and yield implications unknown.

Uncertainty 7: Acrylonitrile excluded from RED pump tubing; BLUE pump assignment confirmed.

Uncertainty 8: Photon adequacy at 450 nm vs. 452 nm target not established.

Skeptic: selected-candidate assessment and shared qualifications

Vetoes: []

Shared assessment / Gas source change: Pure O_2 replacing batch air is a gas-source change. source_context.gas_delivery.requires_chemist_confirmation is not explicitly set to true in the schema field, but the intake record (Q-GAS-001) records this as a user-confirmed requirement and the measured-evidence context notes the prior experiment used O_2 without design approval, resulting in observed reverse flow. Pure O_2 use remains unapproved for oxygen-service purposes across all candidates and must undergo chemist, inventory, and oxygen-service review before laboratory execution.

Shared assessment / Reverse path finding: All twelve candidates share an unresolved credible risk: BF-REVERSE-PATH:st2_mixer:s^3:st1_reactor. The liquid branch between Stage 1 outlet and the Stage 2 gas-introduction junction carries no forward-oriented non-return protection. The CV-3301 check valve is declared only on the MFC (gas) branch; it does not protect the liquid branch. A valve on one branch does not protect a different branch. The KHU qualitative observation (15 Sep 2026) of gas reverse flow is unresolved measured evidence. Reducing gas flow does not close this unprotected reverse path. No simulation result — including the tool's prediction of zero liquid displacement — establishes that backflow cannot occur: the tool uses illustrative unmeasured parameters and a finite scenario window, and its output is predicted liquid displacement at the junction, not confirmation that oxygen reaches or does not reach Stage 1. This finding requires branch-specific engineering review before any candidate is executed.

Shared assessment / Physics tool status: DrFluidics and DrSafety results are conditional on unmeasured illustrative parameters (viscosity 1.0 mPa·s, gas plenum 0.2 mL, MFC ΔP 1.0 bar, liquid compliance 0.01 mL/bar, multiphase resistance multiplier 3.0). No reversal predicted in these finite scenario windows does not establish safety or exclude backflow under real startup, off-normal, or shutdown conditions. Candidates 3, 6, and 11 gas-first startup simulations did not reach BPR opening pressure within the simulated window; steady-state behaviour for those as-designed cases is unconfirmed.

Shared assessment / Cv3301 pressure rating: CV-3301 cracking pressure is 1.0 bar; this concerns the pressure difference across that valve. The maximum pressure rating of CV-3301 is not supplied and must be verified before use at 7 bar or 8 bar system pressures. Cracking pressure is not a pressure rating.

Shared assessment / Stage2 temperature: Stage 2 batch temperature is not stage-resolved in the protocol. All candidates assign 25°C; this is unverified. No validated rate correction or yield consequence can be stated.

Shared assessment / Stage2 gas residence time: Stage 2 Q_gas_sccm and $Q_{gas,STP}$ values are gas volume at inlet/STP (273.15 K, 1.01325 bar). The apparent residence-time index $V/(Q_{liq} + Q_{gas,STP})$ is not a measured liquid residence time or photon exposure index. Gas equivalents are proposed delivered molar ratios, not established stoichiometric consumption or uptake.

Shared assessment / Wavelength note: Protocol specifies 452 nm. Assigned light sources are UV-150/450 nm (Stage 1) and Manual 2/448 nm (Stage 2). These are nearest available; adequacy for catalyst excitation is not established.

Shared assessment / Acrylonitrile pump: Acrylonitrile is excluded from RED pump tubing. All candidates assign liquid feed to BLUE pump; this is consistent with the exclusion constraint.

Shared assessment / No kinetic data: Kinetics are uncharacterized for both stages. Stage-hold ratios are exposure indices only. No yield, conversion, or rate superiority can be inferred from any ratio or residence-time comparison. Near-batch times are not inherently the best choice; short times are not inherently failures.

Shared assessment / Large bore etfe photon: Candidates using ETFE 5 mL or 10 mL coils (ID 2.4 mm) in Stage 1 or Stage 2 carry an uncharacterized photon penetration uncertainty. The batch protocol used a narrow-bore tube irradiated at close range; 2.4 mm ID reduces surface-to-volume ratio and increases path length. This is an unresolved experimental uncertainty, not a hard violation.

Shared assessment / Fep temperature: FEP coils (coil_row6, coil_row7) have max temperature 50°C. At the 25°C screen temperature this is not a constraint, but it limits future elevated-temperature exploration at Stage 2 for candidates using those reactors.

Candidate justifications / C1: Baseline: S1 2 mL PFA τ_{liq}=100 min (0.42×), S2 20 mL FEP τ_{inlet}=181.8 min (0.51×), BPR 7 bar, 2.01 equiv O_2. Balanced sub-batch screen; interpretable anchor. No hard violations. All shared uncertainties apply.

Candidate justifications / C2: Identical reactors to C1; BPR reduced to 2.5 bar. In-channel gas volume 0.028 mL/min (vs 0.012 mL/min in C1); higher gas fraction may alter slug behaviour. Lower pressure reduces O_2 solubility driving force — effect uncharacterized. Useful BPR comparison. No hard violations.

Candidate justifications / C3: S1 5 mL ETFE ID 2.4 mm τ_{liq}=250 min (1.04×); S2 20 mL FEP τ_{inlet}=181.8 min, BPR 7 bar. Only candidate with S1 at or above batch. Gas-first as-designed startup simulation did not reach BPR opening pressure (finite window); steady-state unconfirmed for that case. ETFE 2.4 mm photon penetration uncharacterized. No hard violations; startup operability warrants chemist attention.

Candidate justifications / C4: S1 2 mL PFA τ_{liq}=100 min (0.42×); S2 20 mL FEP τ_{inlet}=181.8 min, BPR 1.5 bar. Lowest system pressure of all candidates; in-channel gas volume 0.040 mL/min is highest in 20 mL FEP group. O_2 solubility driving force lowest here. Useful low-pressure bracket. No hard violations.

Candidate justifications / C5: S1 2 mL PFA τ_{liq}=100 min (0.42×); S2 10 mL ETFE ID 2.4 mm τ_{inlet}=90.9 min (0.25×), BPR 7 bar. Both stages well below batch. ETFE 2.4 mm in S2: photon penetration and gas-liquid distribution uncharacterized. Shortest S2 apparent index. No hard violations; S2 adequacy most uncertain here.

Candidate justifications / C6: S1 5 mL PFA ID 1 mm τ_{liq}=250 min (1.04×); S2 20 mL FEP τ_{inlet}=181.8 min, BPR 7 bar. Near-batch S1 in narrow-bore PFA is better-characterized optically than ETFE 2.4 mm. Gas-first as-designed startup simulation did not reach BPR opening pressure (finite window). No hard violations; startup confirmation needed.

Candidate justifications / C7: S1 2 mL PFA τ_{liq}=100 min (0.42×); S2 10 mL PFA ID 1 mm τ_{inlet}=90.9 min (0.25×), BPR 7 bar. Both stages sub-batch, narrow-bore throughout. Provides short-S2 reference with favourable photon geometry. No hard violations.

Candidate justifications / C8: S1 2 mL PFA τ_{liq}=100 min (0.42×); S2 10 mL FEP ID 1 mm τ_{inlet}=90.9 min (0.25×), BPR 7 bar. Functionally identical exposure profile to C7; FEP Stage 2 max 50°C limits future elevated-temperature use but is not a constraint at 25°C. No hard violations; distinguishing variable vs C7 is Stage 2 material only.

Candidate justifications / C9: S1 5 mL ETFE ID 2.4 mm τ_{liq}=250 min (1.04×); S2 10 mL ETFE ID 2.4 mm τ_{inlet}=90.9 min (0.25×), BPR 7 bar. Both stages large-bore ETFE; photon penetration and gas-liquid contact uncharacterized in both. Carries the greatest geometric uncertainty of the near-batch-S1 group. No hard violations.

Candidate justifications / C10: S1 2 mL PFA τ_{liq}=100 min (0.42×); S2 10 mL ETFE ID 2.4 mm τ_{inlet}=90.9 min (0.25×), BPR 2.5 bar. Lower BPR with large-bore S2; highest in-channel gas fraction among 10 mL S2 candidates (0.028 mL/min). O_2 solubility and slug stability least favourable in this sub-group. No hard violations.

Candidate justifications / C11: S1 5 mL ETFE ID 2.4 mm τ_{liq}=250 min (1.04×); S2 10 mL PFA ID 1 mm τ_{inlet}=90.9 min (0.25×), BPR 7 bar. Gas-first as-designed startup did not reach BPR opening pressure (finite window); steady-state unconfirmed. S1 photon penetration uncharacterized (2.4 mm). Mixed-bore design provides a cross-stage geometry comparison. No hard violations.

Candidate justifications / C12: S1 2 mL PFA τ_{liq}=100 min (0.42×); S2 10 mL PFA ID 1 mm τ_{inlet}=90.9 min (0.25×), BPR 2.5 bar. All-PFA narrow-bore throughout; lower BPR than C7. Both stages well below batch. Low-pressure, short-both-stages bracketing candidate. No hard violations.

Requested measurement 1: Branch-specific non-return protection on the liquid line between Stage 1 outlet and the Stage 2 gas-introduction junction: verify device identity, service direction, cracking pressure differential, maximum pressure rating, O_2 compatibility, and low-flow seating behaviour before execution of any candidate.

Requested measurement 2: CV-3301 maximum pressure rating: must be confirmed independently from cracking pressure specification before use at system pressures up to 7–8 bar.

Requested measurement 3: Chemist and oxygen-service review of pure O_2 substitution for the batch air atmosphere: O_2 concentration, materials compatibility, MFC service range, and laboratory safety approval required before any candidate is executed.

Requested measurement 4: Gas supply pressure and MFC inlet differential pressure at operating conditions: required to validate model margin estimates; illustrative values (12 bar supply, 1.0 bar ΔP required) are not measured setpoints.

Requested measurement 5: Stage 2 batch temperature: protocol does not resolve a temperature for Stage 2; confirm operating temperature before designing rate or yield expectations.

Requested measurement 6: Startup sequence and pressure traces for Stage 1–Stage 2 connected operation: required to validate or falsify the unresolved backflow risk and simulation predictions.

Requested measurement 7: Photon penetration depth for 2.4 mm ID ETFE coils (Stage 1 in C3, C6 startup comparison; C9, C11): verify irradiance at reaction locus relative to 452 nm protocol lamp at 3 cm distance.

Requested measurement 8: Stage 1 Giese conversion as a function of flow residence time: minimum data point needed to select a Stage 1 reactor for yield-optimised operation.

Requested measurement 9: Stage 2 oxidation completion as a function of flow residence time and O_2 equivalent at 7 bar and 2.5 bar: required to distinguish among Stage 2 reactor and BPR options.

Requested measurement 10: Acrylonitrile compatibility and solubility in EtOH:pH 9 buffer at 0.2 M stock concentration: confirm no precipitation or phase separation in the pump and inlet tubing at 0.02 mL/min.

Chief: complete recorded selection report

Candidate: 1

Justification: Candidate 1 (S1: 2 mL PFA, τ_{liq}=100 min, 0.42× batch; S2: 20 mL FEP, τ_{inlet}=181.8 min, 0.51× batch; BPR 7 bar, 2.01 equiv O_2, 0.09 sccm) is selected as the interpretable balanced baseline. Under balanced priority with no kinetic data, neither stage should be assumed adequate or inadequate from time ratios alone. C1 provides mid-range exposure in both stages, uses narrow-bore (1 mm) PFA and FEP giving defined photon geometry, and sets the highest system pressure among the 20 mL FEP group (7 bar), maximising dissolved O_2 driving force for Stage 2. All shared unresolved issues apply; no hard violations found.

Objective alignment: Balanced priority requires an interpretable starting point from which shorter and longer stage holds can be compared. C1 anchors the screen at 0.42× batch (S1) and 0.51× batch (S2), covering sub-batch exposure for both stages simultaneously. This is not the shortest or longest design; it is the mid-range combination that allows yield data from a single run to direct subsequent experiments toward longer S1 (C3, C6) or longer S2 (if both 0.42× and 0.51× prove insufficient), or shorter (C7, C12) if both are already excessive. BPR 7 bar provides the highest O_2 pressure head among FEP-20 mL candidates, minimising one source of Stage 2 variability. No yield or conversion can be inferred; adequacy of either stage is uncharacterized.

Alternative 1 / Candidate: 6

Alternative 1 / Reason not selected: S1 at 250 min (1.04× batch) in 5 mL PFA 1 mm provides near-batch Stage 1 exposure and is the preferred near-batch S1 candidate (narrow bore, better photon geometry than ETFE 2.4 mm). However, gas-first as-designed startup simulation did not reach BPR opening pressure within the finite window, leaving full steady-state startup behaviour unconfirmed. Under balanced priority with no prior data, spending the first experiment on a near-batch S1 condition without confirming startup operability is less efficient than C1 as an anchor. If C1 data show Stage 1 conversion is limiting, C6 becomes the priority follow-up.

Alternative 2 / Candidate: 7

Alternative 2 / Reason not selected: S1: 100 min (0.42× batch, same as C1); S2: 10 mL PFA 1 mm, τ_{inlet}=90.9 min (0.25× batch), BPR 7 bar. Provides the short-S2, short-S1 bracketing point. S2 apparent index is the lowest among narrow-bore 7-bar candidates; if oxidation requires closer to batch time, this condition risks insufficient Stage 2 exposure without yielding diagnostic information. C1's longer S2 (181.8 min, 0.51× batch) is more likely to produce detectable product, making C1 more informative as the first experiment. C7 is the recommended short-S2 follow-up after C1.

Alternative 3 / Candidate: 3

Alternative 3 / Reason not selected: S1: 5 mL ETFE 2.4 mm, τ_{liq}=250 min (1.04× batch); S2: 20 mL FEP, τ_{inlet}=181.8 min, BPR 7 bar. Near-batch S1 in large-bore ETFE; photon penetration at 2.4 mm ID is uncharacterized relative to the batch narrow-tube geometry. Additionally, gas-first as-designed startup simulation did not reach BPR opening pressure (finite window). The combination of uncharacterized photon delivery and unconfirmed startup makes C3 less interpretable as a first experiment than C1 (PFA 1 mm, simpler startup profile). If S1 is rate-limiting, C6 (same τ, narrow bore, confirmed photon geometry) is preferred over C3.

Shared assessment / Applies to all candidates 1: Unprotected liquid-branch reverse path: gas introduction junction (Stage 2) connects directly to Stage 1 reactor via the liquid branch with no forward-oriented non-return valve on that branch. The CV-3301 check valve is declared only on the MFC (gas) branch and does not protect the liquid branch. This is an unresolved credible risk (BF-REVERSE-PATH:st2_mixer:s^3:st1_reactor) across all twelve candidates. Reducing gas flow does not close this path. Branch-specific non-return protection on the liquid line must be reviewed before any candidate is executed.

Shared assessment / Applies to all candidates 2: Pure O_2 substitution: The protocol specifies air (ambient atmosphere). Use of pure O_2 was not approved in the original design; the prior experiment used O_2 without design approval and observed reverse flow. Pure O_2 use constitutes an unapproved gas-source change requiring chemist confirmation, oxygen-service review, and materials compatibility assessment before laboratory execution.

Shared assessment / Applies to all candidates 3: CV-3301 pressure rating: Cracking pressure (1.0 bar) is not a pressure rating. Maximum pressure rating of CV-3301 is unconfirmed and must be verified before use at 7–8 bar system pressures. The cracking pressure concerns the differential pressure across that valve, not BPR minus cracking pressure.

Shared assessment / Applies to all candidates 4: Physics tool results: DrFluidics and DrSafety outputs are conditional on illustrative unmeasured parameters (viscosity 1.0 mPa·s, gas plenum 0.2 mL, MFC ΔP 1.0 bar, liquid compliance 0.01 mL/bar, multiphase resistance multiplier 3.0). No liquid displacement predicted does not establish safe operation or exclude backflow under real startup, transient, or shutdown conditions. Tool output is predicted liquid displacement at the junction, not confirmation that oxygen reaches or does not reach Stage 1.

Shared assessment / Applies to all candidates 5: Gas residence-time index: Q_gas_sccm and $Q_{gas,STP}$ values are gas volume at inlet/STP (273.15 K, 1.01325 bar). The apparent Stage 2 index $V/(Q_{liq} + Q_{gas,STP})$ is not a measured liquid residence time or photon exposure. Gas equivalents are proposed inlet molar ratios, not measured stoichiometric uptake.

Shared assessment / Applies to all candidates 6: Wavelength: Protocol specifies 452 nm. UV-150/450 nm (Stage 1) and Manual 2/448 nm (Stage 2) are nearest available; photon adequacy for catalyst excitation is not established.

Shared assessment / Applies to all candidates 7: Stage 2 temperature: Not batch-resolved. All candidates assign 25°C; no validated rate correction is available.

Shared assessment / Applies to all candidates 8: Kinetics: Uncharacterized for both stages. No yield, conversion, or rate superiority can be inferred from any hold ratio.

Shared assessment / Applies to all candidates 9: Acrylonitrile: Excluded from RED pump tubing. All candidates assign liquid feed to BLUE pump; consistent with exclusion constraint.

Answer impact 1 / Question: Q-GAS-001

Answer impact 1 / Effect: Pure O_2 confirmed by chemist (Q-GAS-001); however, this is an unapproved gas-source change relative to the batch air atmosphere and requires oxygen-service review and laboratory safety confirmation before use. The answer sets the gas identity but does not establish approval for oxygen service.

Answer impact 2 / Question: Q-GAS-003

Answer impact 2 / Effect: Oxygen introduction at Stage 2 inlet only is implemented in C1 (and all candidates). The unprotected liquid-branch reverse path means this constraint cannot be guaranteed without branch-specific non-return protection; this remains unresolved.

Answer impact 3 / Question: Q-HIST-001

Answer impact 3 / Effect: The KHU qualitative backflow observation (15 Sep 2026) is the only measured operability evidence. It directly motivates the unresolved reverse-path finding and confirms that reducing gas flow alone does not resolve the topology risk. It does not quantify backflow rate or conditions.

Answer impact 4 / Question: Q-HYP-001

Answer impact 4 / Effect: Stage 1 oxygen exclusion depends on resolving the liquid-branch reverse path before execution. Stage 2 oxygen availability is determined by BPR (7 bar in C1 maximises O_2 partial pressure) and inlet equivalents (2.01 equiv); gas-liquid mass transfer is uncharacterized. The hypothesis that Stage 1 sulfide formation limits Stage 2 oxidation cannot be tested until Stage 1 conversion data are available.

Limitation 1: No kinetic data exist for either stage; stage-hold ratios are exposure indices only, not conversion or yield predictors.

Limitation 2: The unprotected liquid-branch reverse path is unresolved across all candidates; no candidate can be executed safely until branch-specific non-return protection is reviewed.

Limitation 3: Pure O_2 is unapproved as a gas source; chemist confirmation, O_2-service review, and materials compatibility assessment are required before execution.

Limitation 4: CV-3301 maximum pressure rating is unconfirmed; cracking pressure (1.0 bar) is not a pressure rating.

Limitation 5: Physics tool (DrFluidics/DrSafety) results are conditional on illustrative unmeasured parameters; no reversal predicted does not establish safe operation.

Limitation 6: Stage 2 inlet/STP apparent residence time (181.8 min) is not a measured liquid exposure; liquid τ in Stage 2 is 1000 min (unconstrained by gas index) and actual photon exposure time is also uncharacterized.

Limitation 7: Stage 2 temperature is not batch-resolved; 25°C assignment is unverified.

Limitation 8: Wavelength mismatch (450/448 nm vs 452 nm target) is unconfirmed as adequate for catalyst excitation.

Limitation 9: FEP coil (Stage 2) max temperature 50°C limits future elevated-temperature Stage 2 exploration.

Limitation 10: Candidates 3, 6, and 11 as-designed gas-first startup simulations did not reach BPR opening pressure; startup operability for those designs is unconfirmed.

Suggested measurement 1: Verify and install branch-specific non-return protection on the liquid line between Stage 1 outlet and the Stage 2 gas-introduction junction; confirm device cracking pressure differential, maximum pressure rating, O_2 compatibility, and low-flow seating before any execution.

Suggested measurement 2: Confirm CV-3301 maximum pressure rating independent of cracking pressure specification; verify O_2 service compatibility.

Suggested measurement 3: Obtain chemist, inventory, and oxygen-service review approval for pure O_2 use before laboratory execution.

Suggested measurement 4: Measure Stage 1 outlet conversion (Giese product) from C1 run to determine whether 100 min liquid τ is sufficient; this single data point determines whether longer-S1 candidates (C3, C6) are needed.

Suggested measurement 5: Measure Stage 2 sulfoxide yield from C1 run; compare to C7 (shorter S2) and C6 (longer S1, same S2) to resolve the stage-limiting factor.

Suggested measurement 6: Record startup pressure traces and startup sequence for the connected two-stage system to validate or falsify the unresolved reverse-path risk and simulation assumptions.

Suggested measurement 7: Confirm gas supply pressure and MFC inlet differential pressure under operating conditions (illustrative model assumed 12 bar supply, 1.0 bar ΔP required).

Suggested measurement 8: Verify Stage 2 operating temperature resolves to 25°C as assigned; the batch protocol does not explicitly state Stage 2 temperature.

Suggested measurement 9: Assess photon penetration adequacy at 450 nm and 448 nm relative to the protocol 452 nm lamp for the iridium photocatalyst absorption band.

Suggested measurement 10: Measure acrylonitrile solubility and stability in EtOH:pH 9 buffer at 0.2 M stock concentration to confirm no precipitation or phase separation at 0.02 mL/min flow.

Source: F5-S1, result.json, scientific_assessment; corresponding prompts and exact response text are retained in llm_calls.jsonl and the numbered response files in the companion archive. The original result and final design data have not been edited.

Figure 5, response Set 2 (F5-S2): selected Candidate 6

Recorded objective: Prioritize high conversion through both stages and a high final yield of sulfoxide 4a. Choose the shortest residence times reasonably justified for this conversion-focused starting design rather than adding a large time excess by default. Keep reactor volumes and flow rates practical for flow screening.

Final-record cross-check: R1: 5 mL, 250.00 min (liquid); R2: 20 mL, 181.82 min (inlet/STP apparent). The model priority was yield_priority. All four specialists reviewed all 12 candidates; the recorded veto list is empty. The selected candidate was preserved by final numerical reconciliation.

Response-format retries: Skeptic, attempt 1. These responses were truncated or invalid JSON and were retried, not treated as extra deliberation rounds. The unsuccessful text remains in the companion archive.

DrChemistry: complete selected-candidate verdict (recorded text; F5-S2, call 1)

Candidate: 6

Recommendation: prefer

Hard violation reported: false

Justification: Stage 1: 5 mL PFA 1 mm / UV-150 450 nm / 250 min (ratio 1.04) — meets batch anchor, supports Giese conversion objective; 1 mm ID consistent with UV-150 photon penetration. Stage 2: 20 mL

FEP / 181.8 min apparent (ratio 0.51), BPR 7 bar. Stronger Stage 1 coverage than Candidates 1,2,4,5,7–11 with better light-path geometry than Candidate 3. Reverse-path and oxygen-service risks unresolved.

Uncertainty 1: Stage 1 250 min under irradiation with acrylonitrile present: unquantified side-reaction or catalyst-fatigue risk.

Uncertainty 2: Stage 2 181.8 min apparent index (51% batch) oxidation adequacy uncharacterized.

Uncertainty 3: Unprotected reverse path on interstage liquid branch unresolved.

Uncertainty 4: CV-3301 pressure rating unverified.

Uncertainty 5: Pure O_2 oxygen-service approval required.

DrKinetics: complete selected-candidate verdict (recorded text; F5-S2, call 2)

Candidate: 6

Recommendation: acceptable

Hard violation reported: false

Justification: Stage 1 uses PFA 5 mL 1 mm ID coil, τ=250 min (ratio 1.04 vs batch). Stage 2 inlet/STP τ=182 min (ratio 0.51). Provides near-batch Stage 1 exposure with standard 1 mm geometry, avoiding large-ID light-penetration concern of Candidate 3. BPR 7 bar. Reverse-path unresolved.

Uncertainty 1: Stage 2 oxidation adequacy at 182 min inlet/STP uncharacterized.

Uncertainty 2: Whether near-batch Stage 1 exposure improves yield over shorter candidates is uncharacterized.

Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Uncertainty 4: CV-3301 pressure rating unconfirmed.

Uncertainty 5: O_2 service and gas-source change unapproved.

DrFluidics: complete selected-candidate verdict (recorded text; F5-S2, call 4)

Candidate: 6

Recommendation: acceptable

Hard violation reported: false

Justification: Stage 1: 5 mL PFA (ID 1 mm), 250 min liquid τ (1.04× batch), 7 bar BPR. Stage 2: 20 mL FEP, 181.8 min inlet/STP τ (0.51× batch). Per tool, as_designed startup window did not reach BPR opening pressure; steady-state not confirmed in that window, though hypothetical_liquid_check steady-state shows no reversal. Unprotected reverse path unresolved. Pure O_2 unapproved. Stage 1 residence time near batch anchor; yield-priority objective better served than candidates with 100 min Stage 1.

Uncertainty 1: Tool BPR_opening_pressure_reached=false in as_designed startup window; steady-state unconfirmed.

Uncertainty 2: Unprotected liquid reverse path st2_mixer→st1_reactor unresolved.

Uncertainty 3: Pure O_2 supply unapproved; O_2-service review required.

Uncertainty 4: Stage 2 inlet/STP τ 0.51× batch; oxidation adequacy uncharacterized.

Uncertainty 5: Photon adequacy and degradation risk at 250 min Stage 1 not established.

DrSafety: complete selected-candidate verdict (recorded text; F5-S2, call 6)

Candidate: 6

Recommendation: acceptable

Hard violation reported: false

Justification: Stage 1: 5 mL PFA ID 1 mm, 250 min liquid hold (ratio 1.04), BPR 7 bar. Stage 2: 20 mL FEP, 181.8 min inlet/STP. Tool as_designed startup did not reach BPR operating pressure in simulated window. Consumes one of two available 5 mL PFA coils (coil_row4). Stage 2 same as C1. Reverse-path and O_2 issues shared.

Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Uncertainty 2: Tool startup simulation did not reach BPR operating pressure; steady-state behaviour unverified.

Uncertainty 3: Pure O_2 gas-source change unapproved; oxygen-service review required.

Uncertainty 4: CV-3301 max pressure rating unconfirmed.

Uncertainty 5: Stage 2 temperature not stage-resolved.

Skeptic: selected-candidate assessment and shared qualifications

Vetoes: []

Assessment / Shared issues / Reverse path: All 12 candidates carry an unresolved unprotected reverse path on the liquid branch (st2_mixer → Stage 1 reactor). No forward-oriented non-return valve is declared on that branch. The CV-3301 check valve sits on the MFC/gas branch only and does not protect the liquid branch. The KHU observation of gas reverse flow at 0.02 mL/min liquid / 0.43 mL/min oxygen confirms the credibility of this risk. Reducing gas flow does not close the unprotected path. This is a persistent engineering review requirement for all candidates, not a hard veto, because it is an unresolved experimental outcome requiring mitigation review rather than a confirmed design-rules violation.

Assessment / Shared issues / Tool status: DrFluidics and DrSafety tool results are conditional on unmeasured parameters (illustrative viscosity 1.0 mPa·s, gas plenum 0.2 mL, MFC ΔP 1.0 bar, pump response 1.0 s). No reversal predicted in the finite scenario window is not proof of safety. Predicted quantity is liquid displacement at the junction, not oxygen reaching Reactor 1. Results for Candidates 3, 6, and 12 (as_designed) did not reach BPR opening pressure in the startup simulation window; their steady-state transient behaviour under operating conditions is not confirmed by these results.

Assessment / Shared issues / Gas source: Pure oxygen (O_2) is chemist-confirmed (Q-GAS-001) but this constitutes a gas-source change from the air specified in the earlier design that was actually used in the KHU experiment. This change is unapproved for oxygen service; chemistry, inventory, and O_2-service compatibility review are required before laboratory use across all candidates.

Assessment / Shared issues / Cv3301: CV-3301 maximum pressure rating is not supplied. Cracking pressure (1.0 bar) is not a pressure rating. Verification required before execution for all candidates.

Assessment / Shared issues / Stage1 conversion: All candidates that use the 2 mL PFA coil at 0.02 mL/min (Candidates 1, 2, 4, 5, 7, 8, 9, 10, 11) deliver a Stage 1 liquid residence time of 100 min (ratio 0.42 vs the 240 min batch anchor). Whether this is adequate for Giese conversion is uncharacterized; kinetics are unknown.

Assessment / Shared issues / Stage2 apparent index: The Stage 2 inlet/STP apparent residence-time index is $V/(Q_{liq} + Q_{gas,STP})$ and is not equivalent to liquid residence time or photon exposure time. Gas uptake, dissolution, slug regime, and actual O_2 delivery to the liquid phase are not modeled in any candidate.

Assessment / Shared issues / Stage2 temperature: Stage 2 batch temperature is not stage-resolved in the protocol. All candidates assign 25 °C to Stage 2; no validated rate correction is available.

Assessment / Shared issues / Wavelength: Target wavelength is 452 nm. Stage 1 uses UV-150 450 nm (2 nm deviation); Stage 2 uses Manual 2 Blue Strip LED at 448 nm (4 nm deviation). No photon-dose adequacy or intensification is established for either source.

Assessment / Shared issues / Acrylonitrile pump: Acrylonitrile is excluded from red pump tubing. All candidates correctly assign Stream A (containing acrylonitrile) to pump_e_series_blue (blue tubing). No violation.

Assessment / Shared issues / Objective policy: Yield-priority policy requires investigation of conversion adequacy at all stage holds. Longer hold times are not inherently better; degradation risks at extended irradiation in the presence of acrylonitrile and Ir photocatalyst are unquantified for any candidate. Neither shorter nor longer holds are proven superior.

Assessment / Candidate specific notes / Candidates 1 2 4: Stage 1 at 100 min (0.42× batch) is the primary yield-priority concern. These candidates provide the shortest Stage 1 exposure in the set and represent legitimate low-end screening brackets, but conversion adequacy is the most uncertain dimension under the yield-priority objective.

Assessment / Candidate specific notes / Candidate 3: Stage 1 ETFE 2.4 mm ID at 250 min meets the batch anchor but introduces uncharacterized light-penetration depth concern in the wider bore. As-designed startup simulation did not reach BPR operating pressure in the simulation window; steady-state behaviour unverified for as-designed configuration.

Assessment / Candidate specific notes / Candidate 6: Stage 1 PFA 5 mL 1 mm ID at 250 min meets the batch anchor with standard bore geometry, avoiding the large-ID photon penetration concern. Preferred over Candidate 3 on geometry grounds under yield priority, subject to unresolved reverse-path and O_2-service review. As-designed startup simulation did not reach BPR operating pressure in simulation window.

Assessment / Candidate specific notes / Candidate 12: Stage 1 PFA 5 mL 0.75 mm ID at 250 min meets the batch anchor with the narrowest bore, potentially maximizing wall-proximity photon delivery. Pressure drop is within rating (0.005 bar calculated). As-designed startup simulation did not reach BPR operating pressure in the simulation window. Plugging risk at 0.75 mm if any precipitate forms is uncharacterized.

Assessment / Candidate specific notes / Candidates 5 7 8 9 10 11: Stage 2 inlet/STP apparent index of 90.9 min (0.25× batch 360 min anchor) is the shortest Stage 2 exposure in the set. Under yield-priority policy this is the highest-risk Stage 2 bracket for oxidation adequacy. These are legitimate short-time screening points but carry the greatest under-oxidation uncertainty.

Assessment / Candidate specific notes / Candidates 8 11: FEP Stage 2 coil (max 50 °C) is within the 25 °C screening temperature but constrains future temperature-elevation experiments. Not a violation at current conditions.

Assessment / Candidate specific notes / Candidate 4: BPR 1.5 bar provides the lowest system back-pressure; startup pressure sequencing is most critical here to prevent gas intrusion during transients. Unresolved reverse path concern is highest-consequence at lowest back-pressure, though no reversal was predicted in the tool's entered-profile steady scenarios.

Assessment / Preferred candidates for yield priority: Candidates 6 and 12 are most aligned with the yield-priority objective: both provide Stage 1 liquid residence time of 250 min (1.04× batch anchor) with standard or narrow 1 mm bore PFA geometry in the UV-150 module, and Stage 2 at 181.8 min inlet/STP index (0.51× batch) in the 20 mL FEP coil at 7 bar. Candidate 6 uses the 1 mm PFA coil (coil_row4); Candidate 12 uses the 0.75 mm PFA coil (coil_row8) with higher potential photon-wall proximity but uncharacterized plugging risk. Candidate 3 is also near-batch in Stage 1 but the 2.4 mm ETFE bore introduces a light-penetration uncertainty that makes Candidates 6 and 12 preferable under yield priority. All three remain subject to the unresolved reverse-path and O_2-service review requirements.

Assessment / No hard violations found: No candidate is vetoed. The unresolved reverse path is a credible engineering risk requiring mitigation before execution but is not a confirmed design-rules hard violation that eliminates any candidate from screening consideration. The gas-source change to pure O_2 requires approval but is chemist-confirmed as the intended reagent. No cross-domain scientific claim conflicts were identified that constitute hard violations.

Requested measurement 1: Branch-specific non-return valve (liquid branch, st2_mixer to Stage 1 outlet): inventory item, cracking pressure compatible with pump operating pressure, O_2 and EtOH/buffer service compatibility, and pressure rating—verified before laboratory execution

Requested measurement 2: CV-3301 maximum pressure rating (distinct from 1.0 bar cracking pressure)—verified before laboratory execution

Requested measurement 3: Pure O_2 gas-source approval: chemistry review, O_2-service compatibility of all wetted materials and tubing, and laboratory safety review before use

Requested measurement 4: Actual gas supply pressure and MFC differential-pressure requirement at operating conditions—measured, not assumed

Requested measurement 5: Startup pressure sequence verification: liquid flow and BPR back-pressure established before gas introduction—procedurally confirmed in laboratory

Requested measurement 6: Stage 2 batch temperature (not stage-resolved in protocol)—confirmed for rate and yield comparison

Requested measurement 7: Stage 1 kinetics under flow conditions: qualitative conversion check at 100 min and 250 min liquid residence times to confirm adequacy for Giese reaction

Requested measurement 8: Stage 2 oxidation kinetics: qualitative conversion check at 90.9 min and 181.8 min inlet/STP apparent index to confirm adequacy for sulfoxide formation

Requested measurement 9: Light penetration adequacy: photon dose delivered by UV-150 450 nm to 2 mL PFA (1 mm ID) at 0.02 mL/min, and by Manual 2 448 nm to Stage 2 coil geometry—not established by equipment listing alone

Requested measurement 10: ETFE 2.4 mm ID coil (Candidates 3, 5, 9): light penetration depth and gas-liquid contacting regime characterization before comparing to 1 mm ID results

Requested measurement 11: PFA 0.75 mm ID coil (Candidate 12): plugging risk assessment with the EtOH/pH 9 buffer/Ir-catalyst solution at 0.02 mL/min

Requested measurement 12: Pressure traces during startup and steady-state operation to validate or replace illustrative tool parameters (liquid compliance, gas plenum volume, MFC response time, pump droop)

Chief: complete recorded selection report

Candidate: 6

Justification: Stage 1: 5 mL PFA 1 mm ID / UV-150 450 nm / 250 min liquid τ (ratio 1.04 vs 240 min batch anchor) — the only 1 mm-bore Stage 1 option that meets the batch anchor, preserving standard photon-penetration geometry. Stage 2: 20 mL FEP / Manual 2 448 nm / 181.8 min inlet/STP apparent index (ratio 0.51 vs 360 min batch), BPR 7 bar, 2.008 equiv O_2 at 0.09 sccm. Under yield-priority policy, Stage 1 conversion adequacy is the primary risk; 250 min at 1 mm bore is the most defensible starting point without inventing kinetics. Stage 2 at 0.51× batch is a sub-batch but reasonable first screening exposure at 7 bar. DrFluidics/DrSafety as-designed startup simulation did not reach BPR opening pressure in the simulation window — this is an unresolved limitation, not a confirmed violation. All shared risks apply (reverse path, O_2 service, CV-3301 rating).

Objective alignment: Yield-priority requires adequate Stage 1 Giese conversion before Stage 2 oxidation can produce sulfoxide 4a. The 1 mm bore PFA coil at 250 min provides the batch-anchored Stage 1 exposure without the light-penetration uncertainty introduced by the 2.4 mm ETFE bore (Candidate 3) or the plugging risk of the 0.75 mm bore (Candidate 12). Stage 2 at 181.8 min inlet/STP is a sub-batch starting bracket; the objective policy does not require matching the full 360 min batch Stage 2 time, only a reasonably justified exposure. No degradation or kinetic superiority is claimed.

Alternative 1 / Candidate: 3

Alternative 1 / Reason not selected: Stage 1 uses ETFE 5 mL 2.4 mm ID at 250 min — same batch-anchor exposure as Candidate 6, but the wider bore introduces uncharacterized light-penetration depth uncertainty in the UV-150 module. Under yield priority, uncharacterized photon delivery is a weaker basis than the established 1 mm geometry. As-designed startup simulation also did not reach BPR operating pressure in the simulation window. No hard violation, but geometry uncertainty tips the trade-off to Candidate 6.

Alternative 2 / Candidate: 12

Alternative 2 / Reason not selected: Stage 1 uses PFA 5 mL 0.75 mm ID at 250 min — meets batch anchor and may improve wall-proximity photon delivery, but the narrower bore carries an uncharacterized plugging risk with the EtOH/buffer/Ir-catalyst solution at 0.02 mL/min. As-designed startup simulation did not reach BPR operating pressure in the simulation window. Stage 2 is identical to Candidate 6. Under yield priority, the plugging risk at 0.75 mm is an additional unresolved experimental uncertainty not present in Candidate 6.

Alternative 3 / Candidate: 1

Alternative 3 / Reason not selected: Stage 1 uses 2 mL PFA at 100 min (ratio 0.42 vs 240 min batch anchor) — a legitimate short-time screening bracket but the lowest Stage 1 exposure. Under yield-priority policy, starting with sub-batch Stage 1 exposure creates the highest risk of insufficient Giese conversion entering Stage 2, undermining the stated objective. Stage 2 is identical to Candidate 6. If Stage 1 kinetics prove fast, Candidate 1 becomes a valid follow-on point.

Answer impact 1 / Question: Q-OBJ-001

Answer impact 1 / Effect: Yield-priority drives selection of Candidate 6 over shorter Stage 1 candidates (1, 2, 4, 5, 7–11). The objective policy does not guarantee higher yield; it selects the most defensible starting exposure for conversion-focused screening.

Answer impact 2 / Question: Q-HYP-001

Answer impact 2 / Effect: The chemist hypothesis that Stage 1 requires sufficient photon delivery and residence time supports choosing 250 min over 100 min for Stage 1. Stage 2 time is treated as a first screening bracket, not an optimized value.

Answer impact 3 / Question: Q-GAS-001

Answer impact 3 / Effect: Pure O_2 confirmed by chemist but unapproved for oxygen service. This requirement applies identically to Candidate 6 and all other candidates; it does not change the relative selection but is a hard prerequisite for execution.

Answer impact 4 / Question: Q-GAS-003

Answer impact 4 / Effect: O_2 introduction at Stage 2 inlet only is satisfied by Candidate 6 as designed. The unprotected liquid reverse path means Stage 1 oxygen-free requirement is not structurally guaranteed and requires engineering mitigation before execution.

Answer impact 5 / Question: Q-HIST-001

Answer impact 5 / Effect: The KHU observation of gas reverse flow at 0.02 mL/min liquid / 0.43 mL/min O_2 confirms the credibility of the unprotected reverse-path risk. Candidate 6 at 0.09 sccm O_2 is a lower gas flow, but reducing gas flow does not close the unprotected liquid-branch path; the risk is not resolved by this change.

Answer impact 6 / Question: Q-PREF-001

Answer impact 6 / Effect: Stream A: substrate 0.1 M, 0.002 mmol/min (1.0 equiv); acrylonitrile 0.2 M, 0.004 mmol/min (2.0 equiv); Ir catalyst 0.0005 M, 0.000001 mmol/min (0.5 mol%); solvent EtOH:pH 9 buffer 5:1 v/v, degassed offline; pump_e_series_blue, 0.02 mL/min. Stream B: O_2 pure, 0.09 sccm inlet/STP (0.012422 mL/min pressure-corrected at 8.01 bar abs), 2.008 equiv inlet/STP; mfc_ffc00. Stage 1: 25 °C, 5 mL PFA 1 mm, 250 min liquid τ. Stage 2: 25 °C, 20 mL FEP 1 mm, 181.8 min inlet/STP apparent index. BPR 7 bar. Unprotected reverse path and O_2-service approval are unresolved prerequisites.

Limitation 1: No kinetic data exist for this Giese/oxidation sequence in flow; 250 min Stage 1 and 181.8 min Stage 2 inlet/STP index are batch-anchored screening starting points, not proven adequate exposures.

Limitation 2: Stage 2 inlet/STP apparent residence-time index is $V/(Q_{liq} + Q_{gas,STP})$ and is not equivalent to liquid residence time or photon exposure time; gas uptake, dissolution, and slug dynamics are not modeled.

Limitation 3: UV-150 450 nm vs. target 452 nm (2 nm deviation) and Manual 2 448 nm (4 nm deviation): photon-dose adequacy is not established for either source.

Limitation 4: Stage 2 batch temperature is not stage-resolved in the protocol; the 25 °C assignment carries no validated rate correction.

Limitation 5: As-designed startup simulation for Candidate 6 did not reach BPR opening pressure in the DrFluidics/DrSafety simulation window; steady-state transient behavior under operating conditions is not confirmed by these tool results.

Limitation 6: Unprotected reverse path (st2_mixer → Stage 1 reactor) is unresolved; no liquid-branch non-return valve is declared. The CV-3301 on the MFC branch does not protect this path.

Limitation 7: Pure O_2 is chemist-confirmed but not approved for oxygen service; chemistry, inventory, and O_2-service compatibility review are required before execution.

Limitation 8: CV-3301 maximum pressure rating is unconfirmed; cracking pressure is not a pressure rating.

Limitation 9: Whether 250 min Stage 1 irradiation with acrylonitrile present leads to side reactions or Ir catalyst fatigue is unquantified.

Limitation 10: Tool results are conditional on unmeasured illustrative parameters; no reversal predicted is not proof of safety.

Suggested measurement 1: Verify and install a forward-oriented, service-verified (O_2 and EtOH/buffer compatible) non-return valve on the liquid branch between Stage 1 reactor outlet and the Stage 2 gas-introduction junction before any execution — this is an unapproved proposal requiring laboratory and safety review.

Suggested measurement 2: Confirm CV-3301 maximum pressure rating (distinct from 1.0 bar cracking pressure) before execution.

Suggested measurement 3: Complete chemistry, inventory, and O_2-service compatibility review for pure O_2 across all wetted materials before execution.

Suggested measurement 4: Measure or verify actual gas supply pressure and MFC differential-pressure requirement at operating conditions to replace illustrative tool assumptions.

Suggested measurement 5: Establish and verify a startup pressure sequence: confirm liquid flow and BPR back-pressure are reached before gas introduction to reduce transient reverse-flow risk.

Suggested measurement 6: Confirm Stage 2 batch temperature from the protocol to allow rate comparison.

Suggested measurement 7: Qualitative Stage 1 conversion check at 100 min and 250 min liquid τ to determine whether 100 min is sufficient (which would make Candidates 1, 2, 4 viable alternatives and reduce reactor volume requirements).

Suggested measurement 8: Qualitative Stage 2 oxidation check at 90.9 min and 181.8 min inlet/STP apparent index to bound the minimum required Stage 2 exposure.

Suggested measurement 9: Measure pressure traces during startup and steady-state to validate or replace illustrative DrFluidics/DrSafety parameters and confirm BPR opening-pressure behavior for Candidate 6 as-designed.

Source: F5-S2, result.json, scientific_assessment; corresponding prompts and exact response text are retained in llm_calls.jsonl and the numbered response files in the companion archive. The original result and final design data have not been edited.

Figure 5, response Set 3 (F5-S3): selected Candidate 1

Recorded objective: Identify a practical two-stage flow design that balances final sulfoxide yield and selectivity with substrate throughput, total processing time, oxygen delivery, and stable operation. Prefer a compact, higher-throughput starting design where chemically justified.

Final-record cross-check: R1: 2 mL, 100.00 min (liquid); R2: 20 mL, 181.82 min (inlet/STP apparent). The model priority was balanced. All four specialists reviewed all 12 candidates; the recorded veto list is empty. The selected candidate was preserved by final numerical reconciliation.

Response-format retries: DrSafety, attempt 1; Skeptic, attempt 1. These responses were truncated or invalid JSON and were retried, not treated as extra deliberation rounds. The unsuccessful text remains in the companion archive.

DrChemistry: complete selected-candidate verdict (recorded text; F5-S3, call 1)

Candidate: 1

Recommendation: prefer

Hard violation reported: false

Justification: Stage 1: 2 mL PFA in UV-150/450 nm, τ_{liq}=100 min (0.42× batch). Stage 2: 20 mL FEP in Manual2/448 nm, τ_{inlet}=181.8 min (0.51× batch), BPR 7 bar. Compact baseline with intermediate stage-time ratios for both steps; interpretable anchor for the screen. Backflow unresolved; oxygen-service review required.

Uncertainty 1: Unprotected reverse path to Stage 1 unresolved; engineering review required before execution.

Uncertainty 2: Stage 2 batch temperature not stage-resolved; 25 °C assumed.

Uncertainty 3: Wavelength offset (450/448 vs 452 nm) photocatalytic efficiency unverified.

Uncertainty 4: Actual liquid exposure and gas-liquid mass transfer in Stage 2 unmeasured.

Uncertainty 5: Offline deoxygenation adequacy under flow unconfirmed.

Uncertainty 6: Pure O_2 oxygen-service compatibility review required.

DrKinetics: complete selected-candidate verdict (recorded text; F5-S3, call 2)

Candidate: 1

Recommendation: acceptable

Hard violation reported: false

Justification: Stage 1: 2 mL PFA / UV-150 450 nm / τ_{liq}=100 min (ratio 0.42 vs 240 min batch). Stage 2: 20 mL FEP / Manual-2 448 nm / τ_{inlet}=181.8 min (ratio 0.51 vs 360 min batch). BPR 7 bar; O_2 0.09 sccm / 2.0078 equiv. Compact baseline with intermediate Stage-1 exposure and moderate Stage-2 apparent index.

No hard violation; unresolved backflow and CV-3301 verification are shared limitations. Suitable as interpretable screening baseline per objective.

Uncertainty 1: Unresolved reverse path: Stage-1 reactor unprotected against upstream O_2 intrusion; engineering review required before execution.

Uncertainty 2: CV-3301 pressure rating and low-flow gas-branch service unconfirmed.

Uncertainty 3: Stage-1 liquid τ 100 min is 0.42× batch; adequacy for Giese step unvalidated.

Uncertainty 4: Stage-2 τ_{inlet} 181.8 min is apparent index only; actual liquid exposure and O_2 uptake unmodeled.

Uncertainty 5: Pure O_2 service review pending for all wetted components.

DrFluidics: complete selected-candidate verdict (recorded text; F5-S3, call 4)

Candidate: 1

Recommendation: prefer

Hard violation reported: false

Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch), UV-150 450 nm, 25°C. Stage 2: 20 mL FEP, 181.8 min inlet/STP τ (0.51× batch), 0.09 sccm O_2 (2.01 equiv), 7 bar BPR. Compact balanced baseline matching objective. Unprotected liquid reverse path and unverified check valve require engineering resolution before execution.

Uncertainty 1: Unprotected reverse path from Stage 2 junction to Stage 1 reactor; no non-return device declared on liquid branch.

Uncertainty 2: CV-3301 cracking pressure, max pressure rating, and service compatibility unverified.

Uncertainty 3: Adequacy of 100 min photon exposure for Giese reaction unestablished; kinetics uncharacterized.

Uncertainty 4: Gas-source change from batch air to pure O_2 requires chemist and safety confirmation.

Uncertainty 5: Stage 2 temperature not stage-resolved in batch protocol; 25°C assumption unvalidated.

Uncertainty 6: Inlet/STP apparent τ is a hydraulic index; actual liquid exposure and O_2 uptake are not modeled.

Uncertainty 7: Offline Ar degassing is the only deoxygenation route; effectiveness in flow not confirmed.

Uncertainty 8: FEP coil max 50°C; no thermal deviation here but no active temperature control confirmed.

DrSafety: complete selected-candidate verdict (recorded text; F5-S3, call 7)

Candidate: 1

Recommendation: acceptable

Hard violation reported: false

Justification: Stage 1 liquid τ=100 min (ratio 0.42 vs batch 240 min); Stage 2 inlet/STP apparent τ=181.8 min (ratio 0.51 vs batch 360 min). BPR 7 bar; pure O_2 0.09 sccm, 2.01 equiv confirmed. Reverse path to Stage 1 unprotected on liquid branch; no liquid-branch check valve declared. Tool predicted no liquid

displacement under entered illustrative profile only—this does not establish safety. CV-3301 pressure rating unverified; pure O_2 service requires chemist and oxygen-service review. Acrylonitrile excluded from red tubing; blue peristaltic confirmed. Stage 2 temperature not stage-resolved in protocol.

Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved; no liquid-branch check valve assigned.

Uncertainty 2: CV-3301 max pressure rating unverified; cracking pressure ≠ pressure rating.

Uncertainty 3: Pure O_2 service compatibility with all wetted materials requires laboratory confirmation.

Uncertainty 4: Tool results conditional on illustrative, uncalibrated assumptions—no measured pressure traces.

Uncertainty 5: Stage 2 batch temperature not stage-resolved; flow temperature deviation consequence unknown.

Uncertainty 6: Stage 1 photon dose adequacy at 450 nm vs 452 nm protocol unconfirmed; light-dose sufficiency unproven.

Uncertainty 7: Actual gas uptake, slug behaviour, and O_2 dissolution not modelled.

Skeptic: selected-candidate assessment and shared qualifications

Vetoes: []

Assessment: No hard violations found across all twelve candidates. All share an identical shared topology finding: the liquid branch from Stage 1 to the Stage 2 mixer junction carries no declared forward-oriented non-return valve, leaving an unprotected reverse path to Stage 1. This is a credible engineering risk requiring resolution before any execution, but it is a shared unresolved uncertainty, not a confirmed hard violation that can be attributed to any individual candidate. The gas branch check valve (CV-3301) requires pressure-rating and O_2-service verification in all candidates. Pure O_2 was confirmed by the chemist (Q-GAS-001) but no gas-source change approval or oxygen-service review has been completed; this must be documented as unapproved pending laboratory review. Tool results (DrFluidics/DrSafety) are conditional on illustrative, uncalibrated assumptions; no reversal predicted in the finite scenario window does not establish safety or close the unprotected reverse path. Candidates 3, 6, and 11 (large-bore ETFE Stage 1 coils assigned to UV-150) have only startup-interval tool results; BPR operating pressure was not reached in those windows, meaning steady-state reverse-path behaviour is uncharacterised by the tool for those three. This is a model limitation, not a confirmed hard failure, and does not constitute a veto. The ETFE 2.4 mm ID coils used in UV-150 have unconfirmed photon penetration and coupling efficiency relative to 1 mm PFA or FEP coils; this is an unresolved experimental uncertainty, not a hard constraint violation. FEP coil max temperature 50 °C is respected at 25 °C for Candidate 8 but leaves no margin for LED heating or exothermic events. Stage 2 temperature is not stage-resolved in the source batch protocol for any candidate; 25 °C is an assumption. Acrylonitrile is correctly excluded from red pump tubing in all candidates. The MFC (mfc_ffc00) is a gas MFC and is not used as a liquid pump. All hold ratios are inlet/STP apparent indices or liquid V/Q; none are equivalent to measured liquid exposure or batch kinetic equivalence. No candidate claims yield superiority; none have been rewritten. All candidates carry SCREEN_OR_EXECUTE_AFTER_REVIEW disposition. Candidate 1 (2 mL PFA / 20 mL FEP, BPR 7 bar) is identified as the preferred interpretable baseline per the objective policy: it provides intermediate stage-time ratios for both steps (0.42× and 0.51× batch), uses 1 mm PFA in Stage 1 for better-defined

photon path, and is the most compact design with moderate O_2 partial pressure. Preference is not a yield claim; it is a screening starting point subject to all shared unresolved requirements.

Requested measurement 1: Confirm liquid-branch non-return protection (e.g. forward-oriented check valve) between Stage 1 outlet and Stage 2 mixer junction before any execution; verify device pressure rating, O_2-service compatibility, and low-flow operability

Requested measurement 2: Verify CV-3301 maximum pressure rating against each candidate BPR setpoint plus coil pressure drop; confirm O_2-service compatibility

Requested measurement 3: Complete chemist and oxygen-service review for pure O_2 use across all wetted materials; document as approved or rejected before laboratory use

Requested measurement 4: Measure actual gas supply pressure and confirm MFC differential pressure requirement under operating conditions

Requested measurement 5: Record pressure traces at Stage 2 inlet, junction, and BPR outlet during startup, steady state, and shutdown to characterise reverse-path risk

Requested measurement 6: Verify UV-150 photon coupling and irradiation uniformity for ETFE 2.4 mm ID coils (Candidates 3, 5, 6, 9, 10, 11) relative to 1 mm PFA reference

Requested measurement 7: Define and document startup sequence (liquid-first pressurisation to BPR setpoint before gas introduction) and validate in laboratory, particularly for Candidates 3, 6, and 11 where tool only completed startup interval

Requested measurement 8: Measure liquid residence time distribution and gas-liquid mass transfer in Stage 2 coil geometries to validate inlet/STP apparent residence time as an exposure index

Requested measurement 9: Confirm offline Ar degassing adequacy and residual O_2 in feed solution under flow conditions

Requested measurement 10: Confirm Stage 2 operating temperature in flow and assess LED heating contribution in both stages

Requested measurement 11: Characterise FEP coil thermal margin under flow conditions with LED irradiation for Candidate 8

Requested measurement 12: Obtain measured yield and conversion data at the selected baseline condition (Candidate 1) before comparing stage-time variants

Chief: complete recorded selection report

Candidate: 1

Justification: Stage 1: 2 mL PFA / UV-150 450 nm / Q_{liq}=0.02 mL/min / τ_{liq}=100 min (0.42× batch 240 min). Stage 2: 20 mL FEP / Manual-2 448 nm / τ_{inlet}=181.8 min inlet/STP (0.51× batch 360 min) / O_2 0.09 sccm at inlet/STP (2.01 equiv) / BPR 7 bar. Compact interpretable baseline with intermediate stage-time ratios for both steps, 1 mm PFA Stage 1 for better-defined photon path, and highest operating pressure among candidates (7 bar), providing greatest O_2 partial-pressure driving force and narrowest in-channel gas volume fraction (0.012 mL/min) relative to other 7-bar candidates with smaller reactors. Feed: substrate 0.1 M / 0.002 mmol/min, acrylonitrile 0.2 M / 0.004 mmol/min, Ir-photocatalyst 0.0005 M / 0.001 mmol/min in EtOH:pH9 buffer 5:1 v/v, degassed offline with Ar. Pure O_2 via mfc_ffc00 at 0.09 sccm

(inlet/STP); chemist confirmed (Q-GAS-001) but oxygen-service and gas-source change require unapproved laboratory review before use. Unprotected reverse path (st2_mixer → st1_reactor) remains unresolved for all candidates; installing a forward-oriented, service-verified non-return valve on the liquid branch between Stage 1 outlet and Stage 2 mixer is an unapproved proposal requiring engineering, chemistry compatibility, inventory, and oxygen-service review. CV-3301 cracking pressure (1.0 bar) is not a pressure rating; maximum pressure rating unconfirmed. Tool results (DrFluidics/DrSafety) predict no liquid displacement under entered illustrative, uncalibrated assumptions within finite scenario windows only; this does not establish safety or close the unprotected reverse path. No yield is claimed.

Objective alignment: Balanced priority requires balancing yield/selectivity potential, substrate throughput, total processing time, oxygen delivery, and stable operation. Candidate 1 provides an interpretable two-point baseline (0.42× Stage 1, 0.51× Stage 2 vs batch anchors) without collapsing either stage to the minimum or extending either beyond the batch anchor, keeping total inlet/STP time at 281.8 min. The 7 bar BPR maximises dissolved-O_2 driving force among candidates with the same reactor geometry and delivers the minimum in-channel gas fraction, which is operationally simpler at 0.02 mL/min liquid flow. The 1 mm PFA coil in Stage 1 provides a more predictable optical path than 2.4 mm ETFE alternatives. Feed throughput is identical across all candidates (0.002 mmol/min substrate, 0.02 mL/min liquid); no candidate has higher measured product throughput. Selection as starting design is chemically justified by optical path confidence and O_2 pressure margin; it is not a yield claim.

Alternative 1 / Candidate: 6

Alternative 1 / Reason not selected: Stage 1: 5 mL PFA / τ_{liq}=250 min (1.04× batch), Stage 2: 20 mL FEP / τ_{inlet}=181.8 min, BPR 7 bar. Extends Stage 1 to near-batch for yield-floor assessment with 1 mm PFA (better optical path than ETFE). Total inlet/STP time 431.8 min is 53% longer than Candidate 1, reducing comparative throughput. Gas-first startup: tool only completed startup interval; BPR not reached, so steady-state reverse-path behaviour is uncharacterised by the tool for this candidate. Over-irradiation degradation at 250 min is an untested hypothesis. Useful as a near-batch Stage 1 companion after Candidate 1 baseline is established, not as the starting design under the balanced objective.

Alternative 2 / Candidate: 7

Alternative 2 / Reason not selected: Stage 1: 2 mL PFA / τ_{liq}=100 min, Stage 2: 10 mL PFA / τ_{inlet}=90.9 min (0.25× batch 360 min), BPR 7 bar. Total inlet/STP time 190.9 min. Compact design with 1 mm PFA in both stages. Stage 2 apparent index is the shortest at 7 bar; whether 90.9 min provides adequate oxidation exposure is experimentally unresolved and represents a greater yield risk than the more moderate 181.8 min of Candidate 1. Stage 2 gas-liquid uptake and actual liquid contact at this short index are unmodelled. Useful as a short-Stage-2 data point after baseline but not preferred as the starting design when oxidation adequacy is unknown.

Alternative 3 / Candidate: 2

Alternative 3 / Reason not selected: Identical reactor geometry to Candidate 1 (2 mL PFA / 20 mL FEP) but BPR 2.5 bar. In-channel gas volume increases to 0.028 mL/min vs 0.012 mL/min in Candidate 1; lower O_2 partial pressure reduces dissolved-O_2 driving force. Inlet/STP τ identical (181.8 min) because it is STP-based, not pressure-corrected. Effect of lower pressure on oxidation selectivity is uncharacterised. Unprotected reverse path identical topology; lower BPR provides a narrower pressure margin at startup.

Useful as a BPR-contrast companion to Candidate 1 but not preferred as the starting design when higher O_2 partial pressure is operationally simpler and more informative.

Answer impact 1 / Question: Q-GAS-001

Answer impact 1 / Effect: Pure O_2 confirmed by chemist; mfc_ffc00 delivers 0.09 sccm (inlet/STP) = 2.01 equiv. Gas-source change from batch air to pure O_2 is not approved; oxygen-service compatibility review for all wetted materials, inventory items, and CV-3301 pressure rating required before laboratory execution. This requirement applies equally to all twelve candidates.

Answer impact 2 / Question: Q-GAS-002

Answer impact 2 / Effect: Minimum 2.0 equiv O_2 met at 0.09 sccm (2.01 equiv at inlet/STP); grid rounding accounts for 0.01 excess. Applies to all candidates identically; equivalents are inlet/STP delivered molar ratios, not confirmed uptake.

Answer impact 3 / Question: Q-GAS-003

Answer impact 3 / Effect: O_2 introduced only at Stage 2 inlet in all candidates; Stage 1 is gas-free. Unprotected reverse path on liquid branch from Stage 2 junction to Stage 1 reactor remains unresolved and requires engineering review before execution.

Answer impact 4 / Question: Q-HYP-001

Answer impact 4 / Effect: Chemist hypothesis that Stage 1 may maintain sufficient sulfide formation at higher throughput and that batch time is a reference not a fixed target is respected: 100 min Stage 1 (0.42× batch) is an exploration of shorter exposure, not a guaranteed equivalent. Yield and kinetics uncharacterised; validation required experimentally.

Answer impact 5 / Question: Q-OBJ-001

Answer impact 5 / Effect: Balanced objective implemented: Candidate 1 selected as compact, higher-throughput starting design with intermediate stage-time ratios, 1 mm PFA optical path, and 7 bar BPR for O_2 delivery. Preference is a screening starting point, not a yield or selectivity claim.

Answer impact 6 / Question: Q-HIST-001

Answer impact 6 / Effect: Qualitative observation of gas reverse flow in previous experiment is acknowledged; the unprotected liquid-branch reverse path is flagged as unresolved for all candidates. Reducing gas flow alone (0.09 sccm vs previously reported 0.43 mL/min) does not close the unprotected reverse path. No measured pressure traces or yield from the historical experiment inform candidate selection.

Answer impact 7 / Question: Q-PREF-001

Answer impact 7 / Effect: Reported: substrate 0.1 M / 0.002 mmol/min (1.0 equiv), acrylonitrile 0.2 M / 0.004 mmol/min (2.0 equiv), Ir-catalyst 0.0005 M / 0.001 mmol/min (0.5 mol%), O_2 0.09 sccm at inlet/STP (2.01 equiv, 0.012 mL/min pressure-corrected at 8.01 bar absolute). Stage 1: 25°C, 2 mL PFA, τ_{liq}=100 min. Stage 2: 25°C (assumed, not stage-resolved), 20 mL FEP, τ_{inlet}=181.8 min inlet/STP. Conditional backflow: unprotected liquid reverse path unresolved; CV-3301 pressure rating unconfirmed. Missing: measured pressure traces, startup sequence, gas supply pressure, yield data.

Shared assessment / Unprotected reverse path: All twelve candidates share an identical topological finding: no forward-oriented non-return valve is declared on the liquid branch between Stage 1 reactor outlet and the Stage 2 mixer junction. This is a credible engineering risk under gas-introduction, startup, pump-interruption, and shutdown scenarios. The reverse path to Stage 1 is structurally open regardless of BPR setpoint, gas flow rate, or reactor geometry. Installing a confirmed, service-verified, O_2-compatible non-return valve on the liquid branch is an unapproved proposal requiring engineering design review, chemistry compatibility assessment, inventory verification, and oxygen-service confirmation before any laboratory use. A valve on the MFC gas branch (CV-3301) does not protect the liquid branch.

Shared assessment / Gas check valve: CV-3301 cracking pressure is 1.0 bar; this concerns the pressure difference across the valve, not BPR setpoint minus cracking pressure. Maximum pressure rating is unconfirmed; verification required before execution.

Shared assessment / Pure o2 approval: Pure O_2 use confirmed by chemist (Q-GAS-001) but oxygen-service compatibility with all wetted materials, tubing, fittings, and safety accessories is not approved. This requires laboratory review and documented confirmation before any execution. The gas-source change from batch air to pure O_2 is unapproved.

Shared assessment / Physics tool limitations: DrFluidics and DrSafety results are conditional on illustrative, uncalibrated parameters (viscosity 1.0 mPa·s, compliance 0.01 mL/bar, gas plenum 0.2 mL, etc.; no measured KHU values). No reversal predicted in finite scenario windows does not establish safety, close the unprotected reverse path, or constitute proof of safe operation. Predicted liquid displacement means displacement at the junction model node, not proof that O_2 reaches Reactor 1.

Shared assessment / Startup interval only: Candidates 3, 6, and 11 (ETFE 2.4 mm ID Stage 1 coils): the tool only completed the startup interval for the as_designed profile; BPR opening pressure was not reached. Steady-state reverse-path behaviour is uncharacterised by the tool for these three candidates. This is a model limitation, not a confirmed hard failure or veto.

Shared assessment / Etfe wide bore photon path: Candidates using ETFE 2.4 mm ID coils (3, 5, 9, 10, 11) have unconfirmed photon penetration and irradiation uniformity in UV-150 relative to 1 mm PFA/FEP coils. This is an unresolved experimental uncertainty.

Shared assessment / Gas stage residence time: τ_{inlet}/STP is V_R / (Q_{liq} + $Q_{gas,STP}$) and is an apparent hydraulic index only. It is not equivalent to liquid residence time or to batch hold time. Actual liquid exposure, gas-liquid mass transfer, O_2 uptake, and slug-flow behaviour are not modelled.

Shared assessment / Temperature: Stage 2 batch temperature is not stage-resolved in the source protocol. 25°C is an assumed screening value for all candidates. No temperature compensation, rate correction, or yield consequence is validated.

Shared assessment / Wavelength: UV-150 / 450 nm and Manual-2 / 448 nm differ from protocol 452 nm. Photocatalytic efficiency at these offsets is unconfirmed; no optical intensification or light-dose adequacy is established.

Shared assessment / Throughput: All candidates have identical feed throughput: 0.02 mL/min liquid, 0.002 mmol/min substrate. Product throughput is unmeasured and cannot be compared.

Shared assessment / Yield: No yield, conversion, or kinetic data exist for any candidate. No kinetic ranking, yield claim, or conversion floor is made or implied.

Limitation 1: Unprotected reverse path from Stage 2 junction to Stage 1 reactor is unresolved for all candidates; no execution before engineering review.

Limitation 2: CV-3301 maximum pressure rating unconfirmed; cracking pressure is not a pressure rating.

Limitation 3: Pure O_2 oxygen-service and gas-source change approval pending; all candidates require laboratory review.

Limitation 4: Tool results are conditional on illustrative uncalibrated assumptions; no reversal in finite windows does not establish safety.

Limitation 5: τ_{inlet}/STP is a hydraulic index; actual liquid exposure, O_2 uptake, and gas-liquid mass transfer are unmodelled.

Limitation 6: Stage 2 temperature not stage-resolved; 25°C is an assumption with unknown yield consequence.

Limitation 7: Wavelength offsets (450/448 nm vs 452 nm) photocatalytic efficiency unconfirmed.

Limitation 8: Offline Ar degassing adequacy under flow conditions unconfirmed.

Limitation 9: Feed throughput is identical across all candidates; product throughput is unmeasured.

Limitation 10: No yield, conversion, or kinetic data support selection; preference is for screening starting point only.

Limitation 11: ETFE 2.4 mm ID coil photon penetration in UV-150 uncharacterised.

Limitation 12: Batch hold times are screening anchors, not guaranteed flow equivalents.

Suggested measurement 1: Confirm and install a forward-oriented, service-verified, O_2-compatible non-return valve on the liquid branch between Stage 1 outlet and Stage 2 mixer junction; verify cracking pressure, pressure rating, and low-flow operability before execution.

Suggested measurement 2: Verify CV-3301 maximum pressure rating against candidate BPR setpoint plus coil pressure drop; confirm O_2-service and chemical compatibility.

Suggested measurement 3: Complete chemist and oxygen-service review for pure O_2 across all wetted materials; document as approved before laboratory use.

Suggested measurement 4: Measure and record supply pressure at MFC inlet and confirm MFC differential pressure requirement under operating conditions.

Suggested measurement 5: Record pressure traces at Stage 2 inlet, mixer junction, and BPR outlet during startup, steady-state, and shutdown to characterise reverse-path behaviour.

Suggested measurement 6: Define and validate a liquid-first startup sequence (liquid flow established and system pressure confirmed at BPR setpoint before gas introduction); validate in laboratory particularly for Candidates 3, 6, and 11.

Suggested measurement 7: Measure actual product yield and Stage 1 intermediate conversion at Candidate 1 baseline conditions before screening stage-time variants.

Suggested measurement 8: Measure liquid residence time distribution and gas-liquid mass transfer coefficient in Stage 2 (20 mL FEP, 1 mm ID, 0.02 mL/min + 0.09 sccm O_2, 7 bar) to validate τ_{inlet}/STP as an exposure index.

Suggested measurement 9: Verify UV-150 irradiance and photon penetration depth for ETFE 2.4 mm ID coils relative to 1 mm PFA reference coil.

Suggested measurement 10: Confirm offline Ar degassing residual O_2 concentration in feed solution under flow-startup conditions.

Suggested measurement 11: Measure Stage 2 temperature under LED irradiation and flow to confirm 25°C assumption and assess LED heating contribution.

Suggested measurement 12: Characterise FEP coil temperature under LED irradiation and exothermic conditions for Candidate 8 to assess 50°C margin.

Source: F5-S3, result.json, scientific_assessment; corresponding prompts and exact response text are retained in llm_calls.jsonl and the numbered response files in the companion archive. The original result and final design data have not been edited.

Figure 6: thioester formation and amidation

Chemistry and stage coupling. The reports treated thioester formation and subsequent benzylamine aminolysis as one telescoped process. Stage 1 determines the amount and composition of intermediate entering Stage 2. Benzylamine was kept out of the activation feed and introduced between reactors. The council compared shorter, near-batch and extended holds against the 30 min + 30 min batch procedure. It discussed incomplete activation, incomplete aminolysis and possible prolonged-heating side reactions as uncertainties, not established rate laws or measured degradation.

Feed and equipment review. The fixed activation feed contained acid (0.5 M), DPDTC (0.525 M) and DMAP (0.05 M); the separate benzylamine feed was 2.1 M, all in 2-MeTHF. The volumetric feed ratio $Q_B/Q_A = 0.25$ gives 1.05 equivalents of benzylamine. DrFluidics noted that Stage 2 time must use $Q_A + Q_B$, not Q_A alone. The reviewers addressed BLUE pump-tubing compatibility, independent pump channels, a real mixing junction and thermal/pressure limits. Final allocation identifies Pump A and Pump B as separate operations and assigns the P-713 T-mixer. Earlier narrative claims that the mixer was unassigned are not descriptions of that final topology; physical availability and assembly still require confirmation.

Kinetic and operating uncertainty. The proposals retained 95 deg C and a 7 bar(g) BPR. Their pressure rationale was maintaining liquid-phase operation, not proof of stability for an uncharacterized mixture. Omission of the batch cooling step was treated as a direct-transfer hypothesis to evaluate. Solubility of the concentrated feeds, precipitation on mixing, pump accuracy and intermediate/product stability were listed as follow-up checks. No gas transient-tool call is present in these Figure 6 records; that later feature must not be retroactively attributed to them.

Outcome across response sets. Set 1 selected Candidate 4 as a near-batch screening reference: two 5 mL coils, $Q_A = 0.16$ and $Q_B = 0.04$ mL/min, with 31.25 and 25.00 min stage times. Sets 2 and 3 selected Candidate 3: 10 mL then 5 mL, $Q_A = 0.28$ and $Q_B = 0.07$ mL/min, with 35.71 and 14.29 min stage times. The latter raises acid feed throughput from 0.08 to 0.14 mmol/min but leaves shorter aminolysis exposure as an explicit experimental question. It does not demonstrate higher product throughput or establish a yield-maximizing design.

Interpretation of the decision. In Set 1, the Chief deferred Candidate 3 because its shorter second stage increased uncertainty relative to the near-batch reference. In Set 2, the same asymmetric allocation was preferred as an informative test of a longer activation stage combined with shorter aminolysis. In Set 3, substrate feed throughput was emphasized. This is an inspectable difference in decision priorities, not proof that one rationale is chemically optimal. A final-yield measurement alone cannot identify the rate-limiting stage; stage-resolved measurements would be needed. Later experimental results were not present as yield evidence in these council requests and are not retrospectively inserted into their reasoning.

Interpretation notes for the unedited reports

Numerical transcription: in the Set 1 Chief answer_impacts entry for Q-PREF-001, benzylamine molar flow is written as 0.02 mmol/min. The final feed and independent arithmetic give 2.1 mol/L x 0.04 mL/min = 0.084 mmol/min, which is 1.05 times the acid feed of 0.08 mmol/min. For Sets 2/3, the corresponding amine rate is 0.147 mmol/min and acid rate is 0.14 mmol/min. The copied council text remains unedited; these corrected values govern the reporting.

Inference limits: wording such as adequate exposure, confirmed photon geometry, or a single run distinguishing the rate-limiting step is not accepted as an experimentally demonstrated conclusion. A nominal hold near the batch duration does not establish equal conversion in flow. In particular, the Set 2 claim that incomplete Stage 1 conversion must have another cause because its hold exceeds batch is not justified. Changes in mixing, temperature distribution, feed state and residence-time distribution remain possible. The selected designs are therefore described as screening hypotheses.

Inventory reconciliation: references to a missing mixer or one pump label belong to the council's view of the candidate record. The final allocation explicitly assigns the interstage P-713 mixer and separate stream-pump operations. This resolves the reported assignment at the software-record level, not by claiming that the council physically installed equipment or verified the laboratory configuration.

Figure 6, response Set 1 (F6-S1): selected Candidate 4

Recorded objective: Develop a safe, reproducible, and inventory-compatible two-stage telescoped flow process that maximizes the final amide yield while avoiding unnecessarily long residence times. Design the two stages as one connected process, because Stage 1 determines the composition entering Stage 2.

Final-record cross-check: R1: 5 mL, 31.25 min (liquid); R2: 5 mL, 25.00 min (liquid). The model priority was balanced. All four specialists reviewed all 12 candidates; the recorded veto list is empty. The selected candidate was preserved by final numerical reconciliation.

All six council report calls completed without a recorded response-format retry.

DrChemistry: complete selected-candidate verdict (recorded text; F6-S1, call 1)

Candidate: 4

Recommendation: prefer

Hard violation reported: false

Justification: Both stages near batch anchor (τ_1=31.3 min ratio 1.04, τ_2=25.0 min ratio 0.83); closest symmetric match to batch 30/30 min protocol. Both ETFE 5 mL coils. Natural baseline reference candidate. Throughput 0.08 mmol/min.

Uncertainty 1: Stage 2 ratio 0.83 is below batch; aminolysis completeness at 25 min unknown.

Uncertainty 2: Equivalence between batch stirred and flow plug-flow exposure not established.

Uncertainty 3: DPDTC solubility unconfirmed.

DrKinetics: complete selected-candidate verdict (recorded text; F6-S1, call 2)

Candidate: 4

Recommendation: prefer

Hard violation reported: false

Justification: Stage 1: 31.3 min (1.04× batch), Stage 2: 25.0 min (0.83× batch). Closest to the batch protocol in both stages; serves as the primary flow baseline. Both ETFE 5 mL coils, uniform geometry. Essential reference point for the screen.

Uncertainty 1: Stage 2 at 25 min is 0.83× batch; adequacy for full aminolysis uncharacterized.

Uncertainty 2: Two independent BLUE pump channels required.

DrFluidics: complete selected-candidate verdict (recorded text; F6-S1, call 3)

Candidate: 4

Recommendation: prefer

Hard violation reported: false

Justification: Stage 1 τ31.3 min (1.04×), Stage 2 τ25.0 min (0.83×); both stages near batch anchor. Single reactor type (ETFE 5 mL × 2) simplifies setup. Serves as the primary near-batch reference condition for a balanced screen.

Uncertainty 1: Stage 2 at 25 min is slightly below batch anchor; amide yield adequacy uncharacterized.

Uncertainty 2: Mixer connector selection not explicitly assigned.

Uncertainty 3: Two independent pump channels required under single pump_e_series_blue ID.

DrSafety: complete selected-candidate verdict (recorded text; F6-S1, call 4)

Candidate: 4

Recommendation: prefer

Hard violation reported: false

Justification: Near-batch holds both stages (31 min S1, 25 min S2) at 95 °C. Single reactor type (coil_row11 ×2). Serves as primary batch-reference screen. Stream B 0.04 mL/min within range. No hard violations.

Uncertainty 1: S2 25 min below 30 min batch — amidation completeness uncertain.

Uncertainty 2: Peristaltic accuracy at 0.04 mL/min requires verification.

Skeptic: selected-candidate assessment and shared qualifications

Vetoes: []

Shared assessment / Common issues 1: All candidates assign both Stream A and Stream B to 'pump_e_series_blue' as a single equipment_id, but two independent peristaltic channels are physically required. Laboratory must confirm two separate BLUE pump tubing channels are allocated before any candidate is executed.

Shared assessment / Common issues 2: No candidate explicitly assigns a T-mixer or Y-mixer for Stream B introduction at Stage 2. A union fitting is not a three-port mixer. Mixer selection must be confirmed from available inventory (P-632, P-713, P-513) before execution.

Shared assessment / Common issues 3: 2-MeTHF is listed in excluded_chemicals for all RED pump tubing. All candidates correctly route 2-MeTHF through BLUE pump tubing; this is consistent and not a violation.

Shared assessment / Common issues 4: PFA coils (coil_row3, coil_row4) have max_temperature_C of 100 °C; operating at 95 °C leaves only 5 °C margin. Thermostat stability at this margin requires laboratory verification for candidates 2, 5, 8, 9, 10, 11.

Shared assessment / Common issues 5: Benzylamine Stream B concentration of 2.1 M in 2-MeTHF is not confirmed as a soluble, stable, homogeneous solution; laboratory must verify before use.

Shared assessment / Common issues 6: DPDTC solubility at 0.525 M in 2-MeTHF at the operating temperature is unconfirmed across all candidates.

Shared assessment / Common issues 7: All achieved stage hold ratios are V/Q liquid indices, not proven measures of chemical exposure equivalence to batch; kinetics are entirely uncharacterized.

Shared assessment / Common issues 8: No BPR_available list items are present in the hard constraints; BPR assignments (7 bar) use bpr_row6 (fixed, 1 unit available). Laboratory must confirm this unit is unallocated.

Shared assessment / Common issues 9: Candidates 11 and 12 carry label mismatches: target stage times of 120/30 and 120/120 min respectively are not achieved; actual holds are 62.5/20 and 62.5/50 min. These are not hard violations but must be corrected in experimental records to avoid confusion.

Shared assessment / Common issues 10: Extended Stage 1 and/or Stage 2 holds at 95 °C introduce a plausible but entirely untested degradation risk for DPDTC, the thioester intermediate, and the amide product. This is a hypothesis, not an established constraint, and does not veto any candidate.

Shared assessment / Common issues 11: No gas stages, photochemistry, or analogy kinetic data are applicable to this chemistry. No yield, conversion, or residence-time superiority can be claimed for any candidate.

Candidate reviews 1 / Candidate: 4

Candidate reviews 1 / Recommendation: prefer

Candidate reviews 1 / Hard violation reported: false

Candidate reviews 1 / Justification: τ_1=31.3 min (1.04×), τ_2=25.0 min (0.83×). Nearest match to batch protocol; natural baseline. Uniform reactor geometry (ETFE 5 mL ×2). No hard violations.

Candidate reviews 1 / Uncertainty 1: Stage 2 at 0.83× batch anchor; amidation adequacy uncharacterized.

Candidate reviews 1 / Uncertainty 2: Stage 2 mixer not assigned.

Requested measurement 1: Confirm two independent BLUE peristaltic pump channels are available and allocated before any candidate is executed.

Requested measurement 2: Confirm BPR 7 bar (bpr_row6, quantity 1) is unallocated for the planned screen runs.

Requested measurement 3: Assign and install an explicit T-mixer or Y-mixer from inventory (P-632, P-713, or P-513) for Stream B introduction at the Stage 1/Stage 2 junction in every candidate.

Requested measurement 4: Verify thermostat stability at 95 °C for PFA coils (coil_row3, coil_row4) operating within 5 °C of their 100 °C maximum limit (affects candidates 2, 5, 8, 9, 10, 11).

Requested measurement 5: Confirm DPDTC solubility and homogeneity at 0.525 M in 2-MeTHF at ambient and 95 °C.

Requested measurement 6: Confirm benzylamine at 2.1 M in 2-MeTHF forms a stable, homogeneous solution at operating conditions.

Requested measurement 7: Verify E-series BLUE pump accuracy and pulsation behaviour at 0.02–0.03 mL/min (Stream B in candidates 5, 9, 10, 11, 12).

Requested measurement 8: Confirm simultaneous availability of both coil_row12 units for candidate 7, and both coil_row11 units for candidates 1 and 12.

Requested measurement 9: Correct label mismatches for candidates 11 (target 120/30 min, achieved 62.5/20 min) and 12 (target 120/120 min, achieved 62.5/50 min) in experimental records before execution.

Requested measurement 10: Assess thioester intermediate and amide product stability at 95 °C across the range of Stage 1 hold times (16.7–62.5 min) and Stage 2 hold times (14.3–100 min) through analytical sampling, as no stability data are currently available.

Chief: complete recorded selection report

Candidate: 4

Justification: Stage 1 τ=31.3 min (1.04× batch) and Stage 2 τ=25.0 min (0.83× batch) are the nearest achievable match to the batch 30/30 min protocol using a single uniform reactor type (ETFE 5 mL × 2, coil_row11). This provides the most interpretable baseline against which all other candidates are compared. Both holds are adequate relative to the batch anchor without unnecessarily extending exposure. No hard violations; no label mismatch; Stream B at 0.04 mL/min is above the peristaltic minimum.

Objective alignment: The objective requires maximising amide yield while avoiding unnecessarily long residence times. Candidate 4 anchors the screen at the batch reference condition without extending either stage beyond what the batch protocol demonstrated, satisfying both constraints simultaneously. It is the natural starting point for a balanced screen: shorter candidates (C1, C2, C3) test whether the batch times are unnecessarily long; longer candidates (C6, C7) test whether they are insufficient. Without kinetic data, C4 is the most defensible single selection for a first screen run because any yield gap relative to the batch result directly quantifies the flow penalty or equivalence, rather than conflating a time variable with a geometric variable.

Alternative 1 / Candidate: 3

Alternative 1 / Reason not selected: Stage 1 τ=35.7 min (1.19×) with Stage 2 τ=14.3 min (0.48×). Stage 2 is substantially below the batch anchor; with no kinetic data, incomplete amidation at 14.3 min is a plausible unresolved risk that makes this a higher-uncertainty candidate than C4 as the primary run. It is a scientifically valuable asymmetric contrast run after C4 results are in hand.

Alternative 2 / Candidate: 2

Alternative 2 / Reason not selected: Stage 1 τ=16.7 min (0.56×) and Stage 2 τ=33.3 min (1.11×). Stage 1 is well below the batch anchor; incomplete thioester formation in Stage 1 propagates directly into Stage 2 and cannot be remedied by extended aminolysis time. PFA coil operates within 5 °C of its temperature limit, adding a thermostat-stability requirement. C2 is informative as an asymmetric contrast candidate once C4 establishes the baseline.

Alternative 3 / Candidate: 6

Alternative 3 / Reason not selected: Stage 1 τ=62.5 min (2.08×) is more than double the batch anchor with an untested degradation risk for DPDTC and the thioester intermediate at 95 °C. Stage 2 at 25 min is also below batch. Total time 87.5 min conflicts with the objective to avoid unnecessarily long residence times unless shorter candidates show incomplete Stage 1 conversion.

Answer impact 1 / Question: Q-OBJ-001

Answer impact 1 / Effect: C4 directly serves the dual objective: near-batch holds avoid unnecessary extension, and the uniform geometry simplifies interpretation of any yield measurement against the batch result.

Answer impact 2 / Question: Q-HYP-001

Answer impact 2 / Effect: The chemist hypothesis that the cooling step is a handling artefact is addressed by C4 running both stages at 95 °C without interstage cooling; whether this changes yield relative to the batch cooling step cannot be inferred and requires measurement.

Answer impact 3 / Question: Q-HIST-001

Answer impact 3 / Effect: No historical yield data are available; C4 will generate the first flow yield reference point. This does not change the selection rationale but means no yield prediction is possible before execution.

Answer impact 4 / Question: Q-PREF-001

Answer impact 4 / Effect: Reporting requirements are fully met: Stream A at 0.5 M / 0.08 mmol·min^{-1} (1.0 equiv), Stream B at 2.1 M / 0.02 mmol·min^{-1} (1.05 equiv) in 2-MeTHF via BLUE pump; no gas stages. No yield is predicted.

Shared assessment / Common issues all candidates 1: Both Stream A and Stream B are assigned to 'pump_e_series_blue' as a single equipment_id; two independent BLUE peristaltic channels must be physically confirmed and allocated before any candidate is executed.

Shared assessment / Common issues all candidates 2: No candidate explicitly assigns a T- or Y-mixer for Stream B introduction at the Stage 1/Stage 2 junction. A union fitting is not a three-port mixer. An inventory mixer (P-632, P-713, or P-513) must be selected and installed.

Shared assessment / Common issues all candidates 3: BPR 7 bar (bpr_row6, fixed, quantity 1) must be confirmed as unallocated for the planned screen runs.

Shared assessment / Common issues all candidates 4: DPDTC solubility at 0.525 M in 2-MeTHF at ambient and 95 °C is unconfirmed across all candidates.

Shared assessment / Common issues all candidates 5: Benzylamine at 2.1 M in 2-MeTHF: homogeneity and stability at operating conditions unconfirmed.

Shared assessment / Common issues all candidates 6: PFA coils (coil_row3, coil_row4) operating at 95 °C are within 5 °C of their 100 °C maximum; thermostat stability must be verified for candidates 2, 5, 8, 9, 10, 11.

Shared assessment / Common issues all candidates 7: All stage hold times are V/Q liquid indices, not proven measures of chemical exposure equivalence to the batch; kinetics are entirely uncharacterized.

Shared assessment / Common issues all candidates 8: Candidates 11 and 12 carry label mismatches (targets 120/30 and 120/120 min; achieved 62.5/20 and 62.5/50 min); these must be corrected in experimental records before execution.

Shared assessment / Common issues all candidates 9: Extended holds at 95 °C (Stage 1 >35 min or Stage 2 >33 min) carry an untested degradation hypothesis for DPDTC, the thioester intermediate, and the amide product. This is a hypothesis, not a hard constraint, and does not veto any candidate.

Shared assessment / Common issues all candidates 10: No yield, conversion, or residence-time superiority can be claimed for any candidate; no analogy kinetic data are applicable.

Limitation 1: Stage 2 τ=25 min is 0.83× the batch anchor; whether aminolysis is complete at this hold is uncharacterized and cannot be inferred from available data.

Limitation 2: Equivalence between batch stirred-vessel exposure and flow plug-flow exposure at the same nominal time is not established.

Limitation 3: Throughput (0.08 mmol·min^{-1} limiting feed) is a feed throughput; product throughput depends on uncharacterized conversion and yield.

Limitation 4: Elimination of the interstage cooling step relative to the batch procedure is an untested change; its effect on Stage 2 yield is unknown.

Limitation 5: DPDTC and benzylamine solution stability and solubility at operating conditions are unconfirmed.

Suggested measurement 1: Confirm two independent E-series BLUE peristaltic pump channels are available and allocated.

Suggested measurement 2: Select and install a T- or Y-mixer from inventory for Stream B introduction at the Stage 2 inlet.

Suggested measurement 3: Confirm BPR 7 bar (bpr_row6) is unallocated.

Suggested measurement 4: Verify DPDTC solubility at 0.525 M in 2-MeTHF at ambient temperature and at 95 °C.

Suggested measurement 5: Verify benzylamine at 2.1 M in 2-MeTHF is homogeneous and stable at operating conditions.

Suggested measurement 6: After C4 execution, collect inline or offline analytical data (HPLC or UV) on the Stage 1 effluent to quantify thioester formation before committing to the asymmetric candidates (C2, C3) or extended candidates (C6, C7).

Suggested measurement 7: Assess thioester intermediate and amide product stability at 95 °C by sampling at Stage 1 exit and Stage 2 exit across the range of tested hold times.

Suggested measurement 8: Compare C4 amide yield against the batch result to quantify the flow/batch equivalence and determine whether the cooling step omission has any detectable effect.

Source: F6-S1, result.json, scientific_assessment; corresponding prompts and exact response text are retained in llm_calls.jsonl and the numbered response files in the companion archive. The original result and final design data have not been edited.

Figure 6, response Set 2 (F6-S2): selected Candidate 3

Recorded objective: Prioritize high conversion through both stages and a high final amide yield. Choose the shortest residence times reasonably justified for this conversion-focused starting design rather than adding a large time excess by default. Keep reactor volumes and flow rates practical for flow screening.

Final-record cross-check: R1: 10 mL, 35.71 min (liquid); R2: 5 mL, 14.29 min (liquid). The model priority was yield_priority. All four specialists reviewed all 12 candidates; the recorded veto list is empty. The selected candidate was preserved by final numerical reconciliation.

All six council report calls completed without a recorded response-format retry.

DrChemistry: complete selected-candidate verdict (recorded text; F6-S2, call 1)

Candidate: 3

Recommendation: prefer

Hard violation reported: false

Justification: S1 hold 35.7 min (ratio 1.19) exceeds batch anchor — supports high thioester conversion before amine addition. S2 hold 14.3 min (ratio 0.48) is sub-batch but aminolysis with benzylamine may be fast; this asymmetry is a reasonable exploratory hypothesis under yield priority. Higher throughput than C2.

Uncertainty 1: S2 aminolysis completeness at 14.3 min unverified — could limit final amide yield.

Uncertainty 2: Extended S1 thioester stability at 95 °C uncharacterized.

DrKinetics: complete selected-candidate verdict (recorded text; F6-S2, call 2)

Candidate: 3

Recommendation: prefer

Hard violation reported: false

Justification: Stage 1 τ=35.7 min (ratio 1.19) exceeds batch anchor, supporting thioester conversion. Stage 2 τ=14.3 min (ratio 0.48) is sub-batch; risk of incomplete aminolysis. Complementary to candidate 2; together these bracket stage-time sensitivity. Higher throughput (0.14 mmol/min) than candidate 2.

Uncertainty 1: Stage 2 amide conversion at 0.48× batch time uncharacterized.

Uncertainty 2: Thioester stability over 35.7 min Stage 1 hold uncharacterized.

DrFluidics: complete selected-candidate verdict (recorded text; F6-S2, call 3)

Candidate: 3

Recommendation: prefer

Hard violation reported: false

Justification: Stage 1: 35.7 min (ratio 1.19×) in 10 mL ETFE; Stage 2: 14.3 min (ratio 0.48×). Inverted allocation: extended Stage 1 supports thioester formation but Stage 2 is well sub-batch. Higher throughput (0.14 mmol/min). Useful for mapping Stage 1 sensitivity. Stage 2 conversion adequacy is the key experimental question.

Uncertainty 1: Stage 2 at 0.48× batch may be insufficient for full amide conversion.

Uncertainty 2: Extended Stage 1 at 95°C raises unquantified thioester degradation risk.

DrSafety: complete selected-candidate verdict (recorded text; F6-S2, call 4)

Candidate: 3

Recommendation: prefer

Hard violation reported: false

Justification: Stage 1: 35.7 min (1.19×) exceeds batch reference, supporting thioester conversion. Stage 2: 14.3 min (0.48×) is sub-batch but amide formation from a pre-formed thioester may be faster. Practical flow rates; single-platform E-series.

Uncertainty 1: Stage 2 sub-batch; amide conversion adequacy uncharacterized.

Uncertainty 2: Whether thioester is stable during 35.7 min hold is uncharacterized.

Skeptic: selected-candidate assessment and shared qualifications

Vetoes: []

Shared assessment / Chemistry: Two-stage DPDTC-mediated amide formation; kinetics uncharacterized; no measured flow yield data. Batch anchor: 30 min each stage at 95 °C. Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M, DPDTC 0.525 M, DMAP 0.05 M in 2-MeTHF. Stream B: benzylamine 2.1 M in 2-MeTHF, introduced at Stage 2 T-mixer. All candidates use E-series BLUE pump (2-MeTHF compatible); RED pump correctly excluded. BPR 7 bar selected; all PFA and ETFE coils rated ≥ 95 °C and ≥ 34 bar — no temperature or pressure hard violations detected across any candidate. No gas stage, no light source, no inline degasser required.

Shared assessment / Stream b minimum flow flag: Candidates 5, 8, 9, 10, 11, 12 set Stream B at 0.02 mL/min, which equals the E-series peristaltic minimum (0.02 mL/min). Stoichiometric delivery accuracy

at the absolute minimum is unconfirmed and should be flagged for laboratory verification before executing these candidates.

Shared assessment / Benzylamine solubility: Stream B benzylamine at 2.1 M in 2-MeTHF is unconfirmed; precipitation risk is uncharacterized for all candidates equally.

Shared assessment / Thioester stability: Prolonged Stage 1 holds at 95 °C (candidates 6–12) carry uncharacterized thioester intermediate degradation risk — this is a hypothesis, not an established fact.

Shared assessment / Product degradation: Extended Stage 2 holds at 95 °C (candidates 5, 7, 8, 9, 10, 11, 12) carry uncharacterized amide product degradation risk — hypothesis only.

Shared assessment / Coil row12 dual use: Candidate 7 uses two coil_row12 units simultaneously; inventory quantity is 2 — at capacity, confirmed permissible but requires laboratory confirmation of simultaneous availability.

Shared assessment / Hold ratio note: All hold ratios are V/Q liquid indices, not measured conversion equivalents. Physical exposure equivalence to batch is not established for any candidate.

Shared assessment / No hard violations found: No candidate breaches a hard constraint (temperature rating, pressure rating, pump chemical exclusion, inventory quantity, or minimum flow). No veto is warranted.

Candidate assessments 1 / Candidate: 3

Candidate assessments 1 / Cross domain verdict: no_additional_veto

Candidate assessments 1 / Justification: S1 τ=35.7 min (1.19×, 10 mL ETFE coil_row12), S2 τ=14.3 min (0.48×, 5 mL ETFE coil_row11). Inverted asymmetry; Chemistry/Kinetics prefer, Safety prefer. Higher throughput (0.14 mmol/min). S2 sub-batch amide conversion is the key experimental unknown. No hard violations.

Candidate assessments 1 / Uncertainty 1: S2 amide conversion at 0.48× batch uncharacterized.

Candidate assessments 1 / Uncertainty 2: Thioester stability over 35.7 min at 95 °C uncharacterized.

Requested measurement 1: Benzylamine solubility at 2.1 M in 2-MeTHF at 95 °C (all candidates)

Requested measurement 2: E-series BLUE peristaltic pump delivery accuracy at 0.02 mL/min (candidates 5, 8, 9, 10, 11, 12)

Requested measurement 3: E-series BLUE peristaltic pump delivery accuracy at 0.03 mL/min (candidate 2)

Requested measurement 4: DPDTC thioester intermediate stability in 2-MeTHF at 95 °C as a function of hold time (relevant to all candidates; critical for candidates 6, 7, 8, 9, 10, 12)

Requested measurement 5: Amide product stability in 2-MeTHF at 95 °C over Stage 2 hold duration (candidates 5, 7, 8, 9, 10, 11, 12)

Requested measurement 6: Stage 1 thioester conversion as a function of residence time at 95 °C to anchor kinetic rationale for stage-time selection

Requested measurement 7: Stage 2 amide conversion as a function of residence time at 95 °C to confirm aminolysis is complete within sub-batch holds

Requested measurement 8: Simultaneous availability of two coil_row11 or two coil_row12 units for candidates 4, 7, 9

Requested measurement 9: Heating module compatibility confirmation for PFA coil_row3 at 95 °C (candidates 2, 5, 11)

Chief: complete recorded selection report

Candidate: 3

Justification: Stage 1 τ=35.7 min (ratio 1.19×) exceeds the batch anchor, supporting adequate thioester formation before amine addition. Stage 2 τ=14.3 min (ratio 0.48×) is sub-batch but serves the objective's 'shortest reasonably justified' instruction by testing whether aminolysis is fast relative to the batch hold. This asymmetry is the most informative first-run design under yield-priority: it protects the upstream intermediate-forming step while probing whether Stage 2 requires the full 30 min. Highest limiting-feed throughput among the preferred candidates (0.14 mmol/min). Equipment assignment is clean: coil_row12 (ETFE 10 mL, 95°C rated) for Stage 1, coil_row11 (ETFE 5 mL, 95°C rated) for Stage 2, single E-series BLUE platform, BPR 7 bar.

Objective alignment: Yield-priority requires adequate Stage 1 conversion to generate thioester for Stage 2. Candidate 3 is the only preferred candidate that exceeds the batch anchor in Stage 1 (1.19×) while applying the shortest-justified-time instruction to Stage 2 (0.48×). If Stage 2 result is low, extending Stage 2 is the obvious next step (Candidate 4 or 2). If Stage 1 result shows incomplete thioester, the Stage 1 hold is already above batch and the problem lies elsewhere. This asymmetry distinguishes the rate-limiting step in a single run.

Alternative 1 / Candidate: 4

Alternative 1 / Reason not selected: S1 31.25 min (1.04×) and S2 25 min (0.83×) both near-batch. Valuable as reference but does not test whether either stage can be shortened, and S2 is still sub-batch without a mechanistic reason. Both ETFE coil_row11 units consumed simultaneously (at inventory capacity). Less informative than Candidate 3 as a first run because it does not probe Stage 2 speed at all.

Alternative 2 / Candidate: 2

Alternative 2 / Reason not selected: S1 16.7 min (0.56×) and S2 33.3 min (1.11×). Complementary asymmetry to Candidate 3 but S1 is sub-batch, risking insufficient thioester yield that would confound interpretation of any S2 result. Better as a second run after Candidate 3 confirms S2 kinetics. Lower throughput (0.06 mmol/min) than Candidate 3.

Answer impact 1 / Question: Q-OBJ-001

Answer impact 1 / Effect: Candidate 3 directly operationalizes the 'shortest reasonably justified' instruction for Stage 2 while protecting Stage 1 conversion above the batch anchor. If Stage 2 at 14.3 min is sufficient, the objective is met efficiently; if not, Stage 2 time is the actionable variable.

Answer impact 2 / Question: Q-HYP-001

Answer impact 2 / Effect: Result will test the chemist hypothesis that Stage 1 conversion governs amide yield: if Stage 1 τ > batch and final amide yield is low, Stage 2 kinetics are implicated. If yield is high, Stage 2 can be shortened further in subsequent runs.

Answer impact 3 / Question: Q-HIST-001

Answer impact 3 / Effect: No measured data available; outcome of Candidate 3 will constitute the first quantitative anchor for both stage-time sensitivity and flow yield.

Shared assessment / All candidates: No hard constraint violations found across any candidate. All coils are 95°C and ≥34 bar rated. E-series BLUE pump is 2-MeTHF compatible; RED pump correctly excluded. BPR 7 bar confirmed available. Stream B benzylamine at 2.1 M in 2-MeTHF is unconfirmed for solubility at 95°C across all candidates. All hold ratios are V/Q liquid indices; physical exposure equivalence to batch is not established. No gas stage, no light source, no degasser required.

Shared assessment / Minimum flow flag: Candidates 5, 8, 9, 10, 11, 12: Stream B at 0.02 mL/min equals the declared E-series minimum. Stoichiometric delivery accuracy at this limit is unconfirmed and requires laboratory verification before executing these candidates.

Shared assessment / Degradation hypotheses: Extended Stage 1 holds >30 min at 95°C (Candidates 6, 7, 8, 9, 10, 12) and extended Stage 2 holds >30 min at 95°C (Candidates 5, 7, 8, 9, 10, 11, 12) carry uncharacterized intermediate and product degradation risk. These are untested hypotheses, not established facts.

Shared assessment / Coil inventory at capacity: Candidates 4, 9 use two coil_row11 simultaneously (qty 2, permissible). Candidate 7 uses two coil_row12 simultaneously (qty 2, permissible). Laboratory confirmation of simultaneous availability required.

Candidate justifications / 1: S1 17.9 min (0.60×), S2 14.3 min (0.48×). Both sub-batch; conversion adequacy in both stages uncharacterized. Useful extreme short-time anchor only.

Candidate justifications / 2: S1 16.7 min (0.56×) sub-batch; S2 33.3 min (1.11×) exceeds anchor. Complementary to C3; lower throughput and S1 risk.

Candidate justifications / 3: SELECTED. S1 35.7 min (1.19×) protects thioester step; S2 14.3 min (0.48×) tests shortest-justified amide hold. Highest preferred-candidate throughput.

Candidate justifications / 4: S1 31.3 min (1.04×), S2 25 min (0.83×). Near-batch both stages; strong reference but does not probe Stage 2 minimum time. Two coil_row11 at inventory limit.

Candidate justifications / 5: S1 25 min (0.83×), S2 50 min (1.67×). Extended S2 degrades risk (hypothesis); Stream B 0.02 mL/min at E-series minimum.

Candidate justifications / 6: S1 62.5 min (2.08×), S2 25 min (0.83×). S1 far exceeds batch without kinetic justification; useful boundary only.

Candidate justifications / 7: S1 62.5 min, S2 50 min; both well above batch. Two coil_row12 at inventory limit. Boundary exploration only.

Candidate justifications / 8: S2 100 min (3.33×) extreme overkill; contradicts 'shortest justified' objective. Stream B at E-series minimum.

Candidate justifications / 9: S1 62.5 min, S2 50 min; identical ratios to C7, smaller coils. Stream B at E-series minimum. Redundant with C7.

Candidate justifications / 10: S1 62.5 min ETFE, S2 50 min PFA 1 mm. Same ratios as C9; RTD difference in narrow bore uncharacterized.

Candidate justifications / 11: S1 25 min (0.83×), S2 50 min (1.67×). Same ratios as C5; all-PFA narrow bore. Stream B at E-series minimum.

Candidate justifications / 12: S1 125 min (4.17×) far exceeds batch; directly contradicts objective. Lowest throughput. Extreme boundary only.

Limitation 1: No measured flow kinetics or yield data exist; hold ratios are V/Q indices, not conversion equivalents.

Limitation 2: Thioester stability at 35.7 min/95°C in Stage 1 is uncharacterized; extended hold may or may not be beneficial.

Limitation 3: Stage 2 aminolysis rate in flow at 95°C is unknown; 14.3 min may be insufficient.

Limitation 4: Stream B benzylamine at 2.1 M in 2-MeTHF at 95°C: solubility unconfirmed.

Limitation 5: Peristaltic pump (E-series) delivery accuracy at 0.07 mL/min (Stream B, Candidate 3) is unconfirmed but above minimum.

Limitation 6: No measured product degradation data for either stage; degradation risk at extended holds is hypothesis only.

Suggested measurement 1: Stage 2 amide conversion as a function of residence time at 95°C (10–30 min range) to determine whether 14.3 min Stage 2 is sufficient — this is the primary decision gate for Candidate 3.

Suggested measurement 2: Stage 1 thioester conversion at 35.7 min/95°C to confirm Stage 1 is not rate-limiting at this hold.

Suggested measurement 3: Benzylamine solubility at 2.1 M in 2-MeTHF at 95°C before any run.

Suggested measurement 4: DPDTC thioester intermediate stability at 95°C as a function of hold time (10–120 min) to bound degradation risk for longer-Stage-1 candidates.

Suggested measurement 5: E-series BLUE pump delivery accuracy at 0.02–0.03 mL/min before executing Candidates 2, 5, 8, 9, 10, 11, 12.

Suggested measurement 6: Heating module compatibility confirmation for PFA coil_row3 at 95°C before executing Candidates 2, 5, 11.

Source: F6-S2, result.json, scientific_assessment; corresponding prompts and exact response text are retained in llm_calls.jsonl and the numbered response files in the companion archive. The original result and final design data have not been edited.

Figure 6, response Set 3 (F6-S3): selected Candidate 3

Recorded objective: Identify a practical two-stage flow design that balances final amide yield with substrate throughput, total processing time, and stable operation. Prefer a compact, higher-throughput starting design where chemically justified.

Final-record cross-check: R1: 10 mL, 35.71 min (liquid); R2: 5 mL, 14.29 min (liquid). The model priority was balanced. All four specialists reviewed all 12 candidates; the recorded veto list is empty. The selected candidate was preserved by final numerical reconciliation.

All six council report calls completed without a recorded response-format retry.

DrChemistry: complete selected-candidate verdict (recorded text; F6-S3, call 1)

Candidate: 3

Recommendation: acceptable

Hard violation reported: false

Justification: Stage 1 τ=35.7 min (ratio 1.19) exceeds batch anchor; Stage 2 τ=14.3 min (ratio 0.48) sub-batch. Throughput matches C1 at 0.14 mmol/min. Uses 10+5 mL ETFE. Tests whether extended thioester activation improves amide stage performance, with compact Stage 2.

Uncertainty 1: Whether prolonged Stage 1 at 95 °C degrades thioester intermediate is uncharacterized.

Uncertainty 2: Sub-batch Stage 2 aminolysis adequacy is uncharacterized.

DrKinetics: complete selected-candidate verdict (recorded text; F6-S3, call 2)

Candidate: 3

Recommendation: prefer

Hard violation reported: false

Justification: Highest feed throughput (0.14 mmol/min) with Stage 1 above batch (35.7 min, 1.19×), giving thioester formation headroom. Stage 2 shortened (14.3 min, 0.48×). Practical starting point balancing throughput and Stage 1 yield adequacy; Stage 2 risk is the key experimental question.

Uncertainty 1: Stage 2 aminolysis at 14.3 min (0.48× batch) may be incomplete; requires experimental confirmation.

Uncertainty 2: Thioester stability during extended Stage 1 hold is uncharacterized.

DrFluidics: complete selected-candidate verdict (recorded text; F6-S3, call 3)

Candidate: 3

Recommendation: prefer

Hard violation reported: false

Justification: Highest substrate feed throughput (0.14 mmol/min), equal to C1. Stage 1 at 35.7 min (1.19× batch) exceeds batch anchor — more conservative thioester formation. Stage 2 at 14.3 min (0.48× batch) is shorter. Balanced design favoring Stage 1 completion. Requires two different ETFE coils (10+5 mL).

Uncertainty 1: Stage 2 sub-batch exposure for amide coupling; aminolysis completion at 14.3 min unconfirmed.

Uncertainty 2: coil_row12 (10 mL ETFE, qty 2) and coil_row11 (5 mL ETFE, qty 2) inventory must be confirmed as simultaneously available.

DrSafety: complete selected-candidate verdict (recorded text; F6-S3, call 4)

Candidate: 3

Recommendation: prefer

Hard violation reported: false

Justification: Highest substrate throughput (0.14 mmol/min), equal to Candidate 1. Stage 1 τ=35.7 min (1.19×batch) in 10 mL ETFE coil—exceeds batch anchor, supporting thioester completeness. Stage 2 τ=14.3 min (0.48×batch)—sub-batch for aminolysis. Good screening complement to Candidate 1.

Uncertainty 1: Stage 2 sub-batch aminolysis: amide yield at 14.3 min unvalidated.

Uncertainty 2: Stream B at 2.1 M benzylamine in 2-MeTHF: solubility unconfirmed.

Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Uncertainty 4: Uses two ETFE coils of different volumes; both within inventory.

Skeptic: selected-candidate assessment and shared qualifications

Vetoes: []

Shared assessment / Common issues 1 / Issue: Direct hot interstage transfer at 95°C without batch cooling step

Shared assessment / Common issues 1 / Scope: all candidates

Shared assessment / Common issues 1 / Status: unresolved chemist hypothesis — not a hard constraint; requires chemist confirmation before execution

Shared assessment / Common issues 2 / Issue: Stream B benzylamine concentration 2.1 M in 2-MeTHF

Shared assessment / Common issues 2 / Scope: all candidates

Shared assessment / Common issues 2 / Status: solubility unconfirmed; no measured evidence; requires laboratory verification before use

Shared assessment / Common issues 3 / Issue: 2-MeTHF excluded from E-series RED pump tubing; all candidates correctly assign Stream A and B to BLUE pump tubing only

Shared assessment / Common issues 3 / Scope: all candidates

Shared assessment / Common issues 3 / Status: no violation detected

Shared assessment / Common issues 4 / Issue: Interstage T-mixer for Stream B injection not explicitly enumerated in design realizations

Shared assessment / Common issues 4 / Scope: all candidates

Shared assessment / Common issues 4 / Status: PEEK or ETFE T-mixers are available in inventory; assignment must be confirmed at setup; not a hard violation

Shared assessment / Common issues 5 / Issue: Kinetics entirely uncharacterized; no measured flow yield, conversion, or residence-time superiority data exist

Shared assessment / Common issues 5 / Scope: all candidates

Shared assessment / Common issues 5 / Status: all stage hold ratios are screening anchors only; no yield prediction or ranking is supportable

Shared assessment / Common issues 6 / Issue: Feed throughput (limiting reagent mmol/min) is computed from stream A flow × concentration; product throughput is unmeasured and cannot be compared

Shared assessment / Common issues 6 / Scope: all candidates

Shared assessment / Common issues 6 / Status: throughput comparisons are feed-side only

Shared assessment / Common issues 7 / Issue: BPR 7 bar selected across all candidates; 2-MeTHF bp ~80°C at 1 atm, operation at 95°C requires sufficient back-pressure to suppress boiling; 7 bar BPR provides adequate headroom

Shared assessment / Common issues 7 / Scope: all candidates

Shared assessment / Common issues 7 / Status: no hard violation; confirm BPR setting before operation

Shared assessment / Common issues 8 / Issue: Analogy records retrieved are from different reaction classes and are not usable as kinetic evidence for this DPDTC-mediated amide formation

Shared assessment / Common issues 8 / Scope: all candidates

Shared assessment / Common issues 8 / Status: confirmed non-applicable; no kinetic inference drawn

Candidate reviews 1 / Candidate: 3

Candidate reviews 1 / Council recommendation: prefer

Candidate reviews 1 / Hard violation reported: false

Candidate reviews 1 / Justification: Highest feed throughput (0.14 mmol/min), equal to C1. S1 τ=35.7 min (1.19×) exceeds batch anchor, providing thioester formation headroom. S2 τ=14.3 min (0.48×) is sub-batch. Requires 10 mL ETFE (coil_row12) + 5 mL ETFE (coil_row11); both available. Key experimental question is S2 aminolysis adequacy.

Candidate reviews 1 / Uncertainty 1: S2 aminolysis at 14.3 min (0.48× batch) may be incomplete; uncharacterized.

Candidate reviews 1 / Uncertainty 2: Thioester stability during extended S1 at 95°C is an untested hypothesis.

Priority recommendations / First screen 1: 3

Priority recommendations / First screen 2: 1

Priority recommendations / Rationale: C3 and C1 share the highest feed throughput (0.14 mmol/min) and compact ETFE geometry. C3 provides conservative S1 exposure (1.19× batch) to support thioester formation while probing compressed S2; C1 is the aggressive short-S1/short-S2 complement. Together they bracket the high-throughput region most consistent with the balanced objective.

Priority recommendations / Baseline reference: 4

Priority recommendations / Baseline rationale: C4 provides the most interpretable near-batch baseline (S1 1.04×, S2 0.83×) at 0.08 mmol/min if either C1 or C3 shows yield deficiency attributable to sub-batch exposure.

Requested measurement 1: Benzylamine solubility in 2-MeTHF at 2.1 M and at 95°C before pressurized operation

Requested measurement 2: E-series peristaltic pump delivery accuracy and pulsation characterization at 0.02–0.07 mL/min for 2-MeTHF streams

Requested measurement 3: Chemist confirmation that direct hot interstage transfer at 95°C without intermediate cooling is chemically and operationally acceptable

Requested measurement 4: T-mixer assignment for Stream B injection at Stage 2 inlet confirmed from inventory before setup

Requested measurement 5: Thioester intermediate stability at 95°C as a function of hold time (to assess C6, C7, C9, C11 extended S1 risk)

Requested measurement 6: Amide product stability at 95°C as a function of hold time (to assess C5, C7, C8, C10, C12 extended S2 risk)

Requested measurement 7: Confirmation that coil_row11 (qty=2) and coil_row12 (qty=2) are simultaneously available for candidates requiring two coils of the same type

Requested measurement 8: BPR back-pressure setting verified sufficient to suppress 2-MeTHF boiling at 95°C before operation

Chief: complete recorded selection report

Candidate: 3

Justification: C3 delivers the highest feed throughput (0.14 mmol/min), equal to C1, while providing Stage 1 τ=35.7 min (1.19× batch anchor) that offers conservative thioester formation headroom before the amide-coupling stage. Stage 2 τ=14.3 min (0.48× batch) is the key experimental question. Compact 10+5 mL ETFE geometry; both reactors within inventory. No hard violations. Stream B at 0.07 mL/min is above E-series minimum, reducing pulsation risk versus lower-throughput candidates.

Objective alignment: Balanced priority requires maximum feed throughput, stable operation, and adequate stage exposure. C3 meets throughput parity with C1 while reducing the Stage 1 thioester-completion risk that C1 accepts by holding both stages sub-batch. The trade-off is accepting a sub-batch Stage 2 (0.48×) in exchange for S1 adequacy at maximum throughput; this is chemically justified as a starting screen because aminolysis of activated thioesters is generally faster than thioester formation, but that hypothesis is uncharacterized and requires experimental confirmation. Total time 50 min versus C1's 32 min reflects the S1 extension, not an inherent optimum.

Alternative 1 / Candidate: 1

Alternative 1 / Reason not selected: C1 shares the highest feed throughput (0.14 mmol/min) and is the more aggressive compact screen (S1 τ=17.9 min, 0.60×; S2 τ=14.3 min, 0.48×). Both stages are sub-batch, leaving thioester formation adequacy uncharacterized at S1. C3 is preferred as the first screen because it retains throughput parity while reducing S1 risk. C1 is the recommended co-first screen if C3 shows adequate S2 yield, to test whether S1 can be further compressed.

Alternative 2 / Candidate: 4

Alternative 2 / Reason not selected: C4 provides the most interpretable near-batch baseline (S1 τ=31.3 min, 1.04×; S2 τ=25 min, 0.83×) at 0.08 mmol/min, which is 43% lower feed throughput than C3. The balanced objective prefers the higher-throughput C3 as a starting screen; C4 is designated the baseline reference if C3 or C1 show yield deficiency attributable to sub-batch S2 exposure. Near-batch times are not inherently optimal in flow; C4 does not supersede C3 on yield grounds absent measured evidence.

Answer impact 1 / Question: Q-OBJ-001

Answer impact 1 / Effect: C3's maximum-throughput, conservative-S1 design directly addresses the balance of yield adequacy and throughput; sub-batch S2 is the unresolved element that experimental data will resolve.

Answer impact 2 / Question: Q-HYP-001

Answer impact 2 / Effect: C3 operationalizes the chemist hypothesis that direct hot interstage transfer is acceptable and that S1 completeness is the governing constraint; both remain unvalidated. S1 extended beyond batch anchor partially hedges the thioester-formation risk.

Answer impact 3 / Question: Q-HIST-001

Answer impact 3 / Effect: No historical yield data are available; candidate ranking is based solely on throughput, stage exposure ratios, and equipment feasibility. Measured yield from C3 would become the primary evidence for subsequent stage-time decisions.

Answer impact 4 / Question: Q-PREF-001

Answer impact 4 / Effect: Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M / DPDTC 1.05 equiv / DMAP 0.1 equiv in 2-MeTHF; molar flow 0.14 mmol/min substrate; Q=0.28 mL/min, pump_e_series_blue. Stream B: benzylamine 2.1 M in 2-MeTHF; 1.05 equiv relative to substrate; molar flow 0.147 mmol/min; Q=0.07 mL/min, pump_e_series_blue. No gas stages; Q_gas_sccm=0 for all stages. BPR 7 bar.

Shared assessment / Common uncertainties 1: Direct hot interstage transfer at 95°C without batch cooling: chemist confirmation required before execution for all candidates.

Shared assessment / Common uncertainties 2: Benzylamine 2.1 M in 2-MeTHF solubility unconfirmed; laboratory verification required before pressurized operation.

Shared assessment / Common uncertainties 3: 2-MeTHF excluded from E-series RED pump tubing; all candidates correctly use BLUE pump tubing only — no violation.

Shared assessment / Common uncertainties 4: Interstage T-mixer for Stream B injection not explicitly assigned in design realizations; PEEK/ETFE T-mixers are available in inventory and must be confirmed at setup.

Shared assessment / Common uncertainties 5: Kinetics entirely uncharacterized; no yield, conversion, or residence-time superiority can be inferred from stage hold ratios.

Shared assessment / Common uncertainties 6: Feed throughput (mmol/min) is computed from Stream A flow × concentration; product throughput is unmeasured.

Shared assessment / Common uncertainties 7: BPR 7 bar selected for all candidates; 2-MeTHF bp ~80°C at 1 atm — 7 bar provides adequate headroom to suppress boiling at 95°C; confirm before operation.

Shared assessment / Common uncertainties 8: Analogy records retrieved are from different reaction classes and provide no kinetic evidence for this DPDTC-mediated amide formation.

Limitation 1: No measured flow yield, conversion rate, or kinetic data exist; C3 preference is based on throughput parity and stage-exposure reasoning, not proven superiority.

Limitation 2: Stage 2 τ=14.3 min (0.48× batch) for aminolysis is sub-batch; whether this is sufficient cannot be inferred without measurement.

Limitation 3: Thioester intermediate stability at 95°C for 35.7 min is an untested hypothesis; degradation risk cannot be excluded.

Limitation 4: Direct hot interstage transfer replacing batch cooling is a chemist hypothesis, not an established fact.

Limitation 5: Benzylamine 2.1 M concentration in 2-MeTHF is unconfirmed; solubility failure would require stream reformulation across all candidates.

Limitation 6: Feed throughput is not product throughput; measured yield from C3 may differ from batch yield at any stage-time combination.

Suggested measurement 1: Confirm benzylamine solubility in 2-MeTHF at 2.1 M and at 95°C before pressurized operation.

Suggested measurement 2: Chemist must confirm that direct hot interstage transfer at 95°C without intermediate cooling is chemically and operationally acceptable before any candidate is executed.

Suggested measurement 3: Verify E-series peristaltic pump delivery accuracy and pulsation at 0.07 mL/min (Stream B, C3) and at 0.02–0.03 mL/min (lower-throughput candidates) in 2-MeTHF.

Suggested measurement 4: Assign and confirm T-mixer from inventory for Stream B injection at Stage 2 inlet before setup.

Suggested measurement 5: Measure amide yield from C3 at Stage 2 τ=14.3 min; if sub-batch S2 yield is deficient, execute C4 as the near-batch baseline reference before lengthening S2.

Suggested measurement 6: If C3 yield is adequate, execute C1 to determine whether S1 can be further compressed to τ=17.9 min at maximum throughput.

Suggested measurement 7: Characterize thioester intermediate stability at 95°C as a function of hold time before executing C6, C7, C9, or C11 with extended S1 >35 min.

Suggested measurement 8: Confirm simultaneous availability of coil_row12 (10 mL ETFE, qty=2) and coil_row11 (5 mL ETFE, qty=2) for candidates requiring both coils of the same type.

Source: F6-S3, result.json, scientific_assessment; corresponding prompts and exact response text are retained in llm_calls.jsonl and the numbered response files in the companion archive. The original result and final design data have not been edited.

What these records demonstrate

The reports show how chemical interpretation, stage exposure, equipment constraints and operating uncertainties were communicated to a final selector. They also expose limits: repeated warnings were not equivalent to installed mitigations; some narrative numbers were wrong; and similar briefs could produce the same selection. The records support an auditable design-assistance workflow, not a claim that discussion guarantees the highest yield or eliminates the need for experimental and engineering review. The six runs contain four distinct selected physical configurations: two for each chemistry.

8 Laboratory implementation, experimental procedures, and analytical data

This section documents the KHU implementation of the photoredox Giese addition/oxidation and DPDTC-mediated amidation cases. Tables S15–S16 report the measured outcomes, and Tables S17–S18 connect the implemented equipment and nominal feeds to the design proposals. The photographs, spectra, experimental procedures, and grouped response-set entries are retained from the laboratory record. Grouped sets are not treated as independent measurements.

8.1 General materials, instrumentation, and analytical methods

All reagents were purchased from commercial suppliers (Acros Organics, Alfa Aesar, Tokyo Chemical Industry, and Sigma-Aldrich) and used without further purification unless otherwise noted. Selected compounds were purified by flash column chromatography on silica gel 60 (0.040–0.063 mm) prior to use. Analytical thin-layer chromatography (TLC) was performed on silica gel 60 F254 glass plates (Merck/Supelco, 20 × 20 cm). Visualization was achieved by UV irradiation (254 or 365 nm) or staining (anisaldehyde, ninhydrin, or $KMnO_4$) followed by heating when appropriate.

Flow reactions were performed using a Vapourtec E-series system (part no. 50-1319) equipped with a Blue V3 pump tube (part no. 50-1301). For the photochemical reaction (Figure 5), the system was fitted with an organometallic reagent kit (part no. 50-1311). Reactor 1 was a UV–150 photochemical reactor equipped with an LED power supply (part no. 50-1453) and irradiated with a 450 nm LED (part no. 50-1448). The effluent then passed through a second coil reactor irradiated with a blue LED strip. Reactor 1 was maintained at 25 °C using an E-Series cooling module (part no. 50-1314), while reactor 2 was maintained at 25 °C using a water bath and fan cooling. O_2 gas was delivered via a Bronkhorst FLEXI-FLOW compact mass flow controller (FF-C00). For the amide coupling reaction (Figure 6), both reactor coils were held at 95 °C using a water bath. Both setups were fitted with a 7 bar cartridge back-pressure regulator.

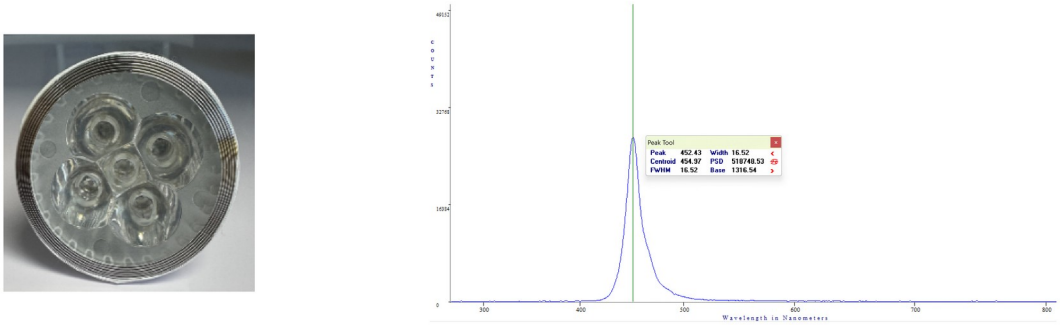
All isolated products were characterized by ^{1}H and ^{13}C NMR spectroscopy and LRMS. Known products were identified by comparison with published NMR data. Copies of all ^{1}H, ^{13}C NMR spectra are provided in the Supporting Information. NMR spectra were recorded at ambient temperature on a JEOL JNM-ECZ500R FT-NMR spectrometer (^{1}H, 500 MHz; $^{13}C\{^{1}H\}$, 126 MHz). Chemical shifts are reported relative to residual solvent peaks ($CDCl_3$ δ = 7.26 ppm for ^{1}H, 77.16 ppm for $^{13}C\{^{1}H\}$; DMSO-d_6, δ = 2.50 ppm for ^{1}H, 39.52 ppm for $^{13}C\{^{1}H\}$). All ^{13}C NMR spectra were broadband ^{1}H-decoupled. Coupling constants (J) are given in Hertz (Hz). Signal multiplicities: br (broad), s (singlet), d (doublet), t (triplet), q (quartet), p (quintet), m (multiplet). Low-resolution mass spectra (LRMS) were recorded on an Advion expressionS CMS spectrometer using electrospray ionization (ESI) in positive-ion mode.

The 7 bar values identify the installed BPR cartridge, not the measured system pressure. Tables S15–S16 report the laboratory pressure ranges separately. The pressure reference (gauge or absolute), sensor position, and calibration were not specified. The reported 95 °C water-bath arrangement is retained pending laboratory confirmation.

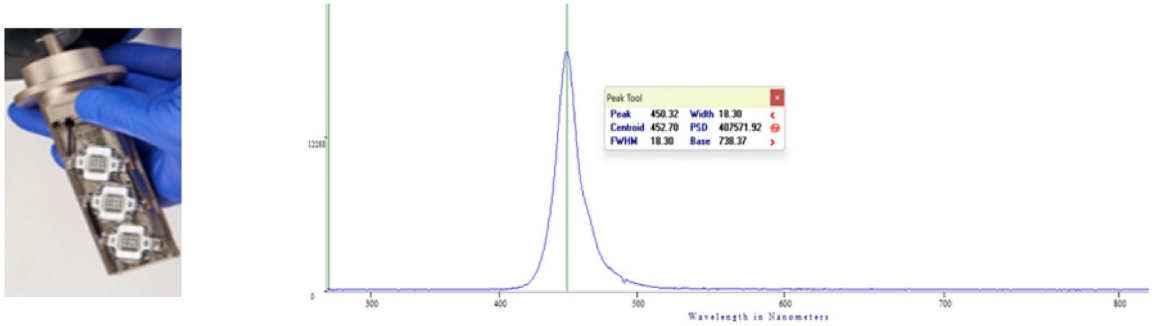
8.2 Irradiation sources and recorded specifications

The batch source comprised two MR16 blue LEDs; flow R1 used the UV-150 450 nm LED and R2 a separate blue strip. Figure S18 combines the supplied photographs and emission spectra. The caption records electrical-input and radiant-power specifications separately; neither is a measured photon flux at the fluid.

a **Batch: two MR16 5 W lamps**



b **R1: Vapourtec 450 nm LED**



c **R2: blue strip LED**

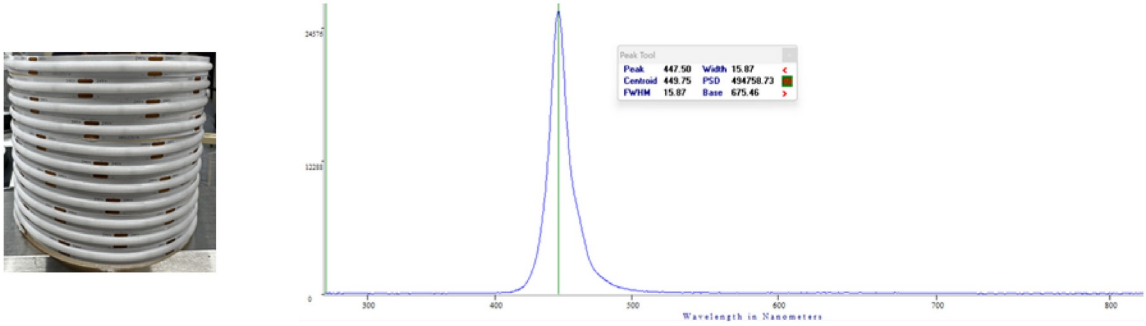


Figure S18. Irradiation sources supplied by KHU: (a) two MR16 5 W batch lamps; (b) Vapourtec 450 nm LED for R1; (c) blue strip LED for R2. The original photographs and emission traces are retained. Reported peak wavelengths are 452.4, 450.32 and 447.50 nm, respectively. Electrical inputs are 5 W per MR16 lamp, 60 W for the Vapourtec source, and 10 W for the strip. The Vapourtec radiant-power specification is 24 W; none of these values establishes irradiance at the reacting fluid.

8.3 Photoredox Giese addition and oxidation: preparation and operation

The sequence converts (((4-methoxyphenyl)thio)methyl)trimethylsilane (1a) and acrylonitrile (2a) through sulfide 3a to sulfoxide 4a, following Park et al. [11]. Main-text Figure 5 shows the connected configuration; Figure S19 shows KHU's apparatus.

A Vapourtec E-Series flow system (part no. 50-1319) equipped with a Blue V3 pump tube (part no. 50-1301) was used. Reactor 1 was a UV–150 photochemical reactor fitted with a 2 or 5 mL coil of PFA tubing (0.04 in. i.d.) and irradiated with a 450 nm LED (part no. 50-1448; rated electrical input, 60 W). The reactor temperature was maintained at 25 ± 1 °C. Reactor 2 comprised a 20 mL coil of FEP tubing (0.04 in. i.d.) maintained at 25 ± 1 °C using a water bath and fan cooling and irradiated with a blue LED strip.

A solution of 1a (0.2 mmol, 1.0 equiv), 2a (0.4 mmol, 2.0 equiv), and $Ir(dF(CF_3)ppy)_2(dtbpy)PF_6$ (0.001 mmol, 0.5 mol%) in EtOH/pH 9.00 buffer (bicarbonate-based buffer, 5:1, v/v; 1a, 0.10 M) was loaded into a 2.0 mL sample loop and injected into a carrier stream of the same solvent mixture. The liquid flow rate was 0.020 mL min–1. Oxygen was introduced through a T-mixer between the two reactors using a mass-flow controller set to 0.090 mL min–1. A 7 bar cartridge back-pressure regulator was installed downstream of reactor 2, and the recorded system pressure was 5.1–5.6 bar.

The Stage 1 liquid feed therefore contains nominal concentrations of 0.100 M 1a, 0.200 M acrylonitrile, and 0.000500 M iridium photocatalyst. The 2.0 mL sample loop contains 0.200 mmol 1a and corresponds to a nominal injection duration of 100 min at 0.020 mL min^{-1}. This is a finite sample-loop experiment in a continuously flowing carrier, not evidence of indefinitely sustained reagent-fed operation. Exact leading/trailing collection windows and the degassing/argon-handling procedure used during the implemented runs were not specified in the new methods. The frozen design input retains an oxygen-free, argon-prepared first stage; oxygen is introduced only at the interstage T-mixer.

At the stated settings, nominal Stage 1 times are 2/0.020 = 100 min and 5/0.020 = 250 min. The supplied artwork labels Stage 2 as "181 min"; recalculation from the stated settings gives the nominal inlet-reference index 20/(0.020 + 0.090) = 181.82 min. On the proposal's STP convention (273.15 K, 1 atm), 0.090 mL min^{-1} of pure O_2 corresponds to 0.004015 mmol min^{-1}, or 2.008 equivalents relative to 0.00200 mmol min^{-1} of 1a. These are calculated proposal-basis values; the actual MFC reference temperature/pressure must be confirmed. Gas compression, dissolution, consumption, and phase holdup prevent this index from being identified with physical gas or liquid residence time. No tracer or residence-time-distribution measurement was supplied.

An aliquot of the collected reactor effluent was analyzed by ^1H NMR spectroscopy using 1,3-benzodioxole as an internal standard to determine the NMR yield of 4a. The remaining effluent was concentrated under reduced pressure and purified by flash column chromatography on silica gel. KHU reports 97% NMR yield for the grouped Sets 1/3 entry (2 mL R1), and 98% NMR yield for Set 2 (5 mL R1). The recorded system pressures are 5.1–5.4 and 5.2–5.6 bar, respectively. Replicate counts and quantitative-NMR integration/calibration records were not supplied.

Table S15. Giese addition/oxidation operating points and yield. Yield is determined by ^{1}H NMR. Sets 1/3 share one supplied entry; no independent replicate count is inferred. Stage 2 is a nominal inlet/STP time index.

Parameter	KHU sets 1 / 3	KHU set 2
Yield (%)	97	98
R1 / R2 volume (mL)	2 / 20	5 / 20
Liquid flow (mL/min)	0.020	0.020
O_2 controller setting (mL/min)	0.090	0.090
R1 time / R2 index (min)	100 / 181.82	250 / 181.82
Temperature (both stages)	25 ± 1 °C	25 ± 1 °C
Recorded pressure (bar)	5.1–5.4	5.2–5.6
BPR cartridge	7 bar	7 bar

The literature flow reference in main-text Figure 5 is 90% NMR yield (Park et al. [11], Table 4 entry 5). Literature and KHU configurations differ in gas delivery, irradiation and other operating conditions; the reference provides context rather than a matched experimental control.

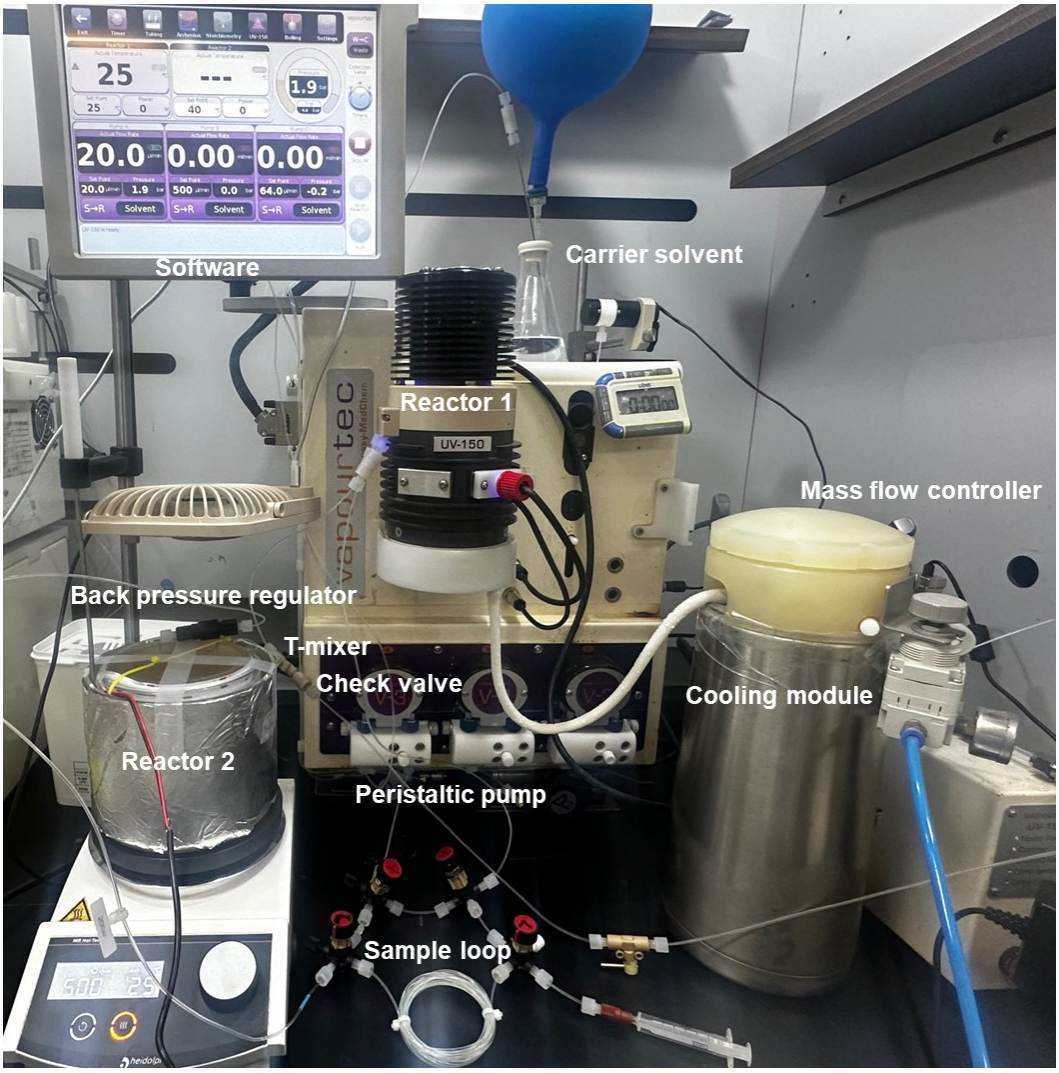


Figure S19. Actual KHU photochemical flow apparatus, as supplied. The labeled photograph identifies the E-series system, UV-150 reactor, downstream coil, MFC, T-mixer, check valve, BPR, cooling module, carrier solvent, and sample loop. The image is not a pressure calibration or a record of startup dynamics; exact check-valve location/orientation and a pressure trace remain to be confirmed.

8.4 DPDTC-mediated amidation: preparation and operation

The sequence converts 3-methyl-4-nitrobenzoic acid (1b) through a 2-pyridyl thioester to N-benzyl-3-methyl-4-nitrobenzamide (3b), following Saunders et al. [12]. The intermediate is not isolated. Main-text Figure 6 shows the connected configuration; Figure S20 shows KHU's apparatus.

A Vapourtec E-Series flow system (part no. 50-1319) equipped with a Blue V3 pump tube (part no. 50-1301) was used with two pump channels. Reactor 1 comprised a 5 or 10 mL coil of ETFE tubing (0.093 in. i.d.), and reactor 2 comprised a 5 mL coil of ETFE tubing (0.093 in. i.d.). Both reactors were maintained at 95 °C in a water bath.

A solution of 1b (1.0 mmol, 1.0 equiv), DPDTC (1.05 mmol, 1.05 equiv), and DMAP (0.1 mmol, 0.10 equiv) in 2-MeTHF (1b, 0.50 M) was loaded into a 2.0 mL sample loop and injected into a carrier stream of 2-MeTHF. The liquid flow rate through reactor 1 was 0.160 or 0.280 mL min–1. At a T-mixer downstream of reactor 1, this stream was combined with a solution of 2b (1.05 mmol, 1.05 equiv) in 2-MeTHF (2b, 2.10 M), delivered through the second pump channel at 0.040 or 0.070 mL min–1, respectively. The combined stream then passed through reactor 2. A 7 bar cartridge back-pressure regulator was installed downstream of reactor 2, and the recorded system pressure was 5.2–5.8 bar.

Nominal stock concentrations in Feed A are 0.500 M acid, 0.525 M DPDTC, and 0.0500 M DMAP. A 2.0 mL loop contains 1.00 mmol acid and is displaced in 12.50 or 7.14 min at the two stated flow rates. The benzylamine stock has a nominal concentration of 2.10 M; the specified 1.05 mmol quantity corresponds to 0.500 mL of that stock. The required total feed volume for priming, overlap with the dispersed sample plug, and collection cannot be reconstructed from this nominal reagent amount alone. The new methods do not specify the second-feed start/stop timing or collection window, which are important for interpreting finite-loop stoichiometry.

At Set 1 flows, the acid and benzylamine nominal molar feeds are 0.0800 and 0.0840 mmol min^{-1}. At Sets 2/3 flows, they are 0.140 and 0.147 mmol min^{-1}. Both give 1.05 equivalents of benzylamine. The combined stream has acid-equivalent concentration 0.400 M and amine-feed-equivalent concentration 0.420 M before allowing for reaction; these are mixing calculations, not measured thioester or product concentrations. R1 times are 31.25 and 35.7143 min; R2 times are 25.00 and 14.2857 min. Actual intermediate conversion was not measured in the supplied data.

An aliquot of the collected effluent was analyzed by ^{1}H NMR spectroscopy using 1,3-benzodioxole as an internal standard. The remaining effluent was concentrated by rotary evaporation and purified by flash column chromatography on silica gel. Set 1 gave 86% NMR yield; Sets 2/3 share a reported NMR yield of 68%. The recorded system pressures were 5.2–5.3 and 5.5–5.8 bar, respectively. Both coils were described as operating at 95 °C without an intervening cooling operation. Individual-run identifiers and repeat statistics were not supplied.

Table S16. DPDTC-mediated amidation operating points and yield. Yield is determined by ^{1}H NMR. Sets 2/3 share one supplied entry. Stage times are nominal liquid-volume/flow calculations.

Parameter	KHU set 1	KHU sets 2 / 3
Yield (%)	86	68
R1 / R2 volume (mL)	5 / 5	10 / 5
Feed A / B (mL/min)	0.160 / 0.040	0.280 / 0.070
R1 / R2 time (min)	31.25 / 25.00	35.71 / 14.29
Temperature (both stages)	95 °C	95 °C
Recorded pressure (bar)	5.2–5.3	5.5–5.8
BPR cartridge	7 bar	7 bar

The literature reference in main-text Figure 6 is 98% NMR yield in batch (Saunders et al. [12], Table 1 entry 5). This measurement basis matches the KHU yield determinations, but the operating mode and conditions differ; the comparison does not isolate the effect of batch-to-flow translation.

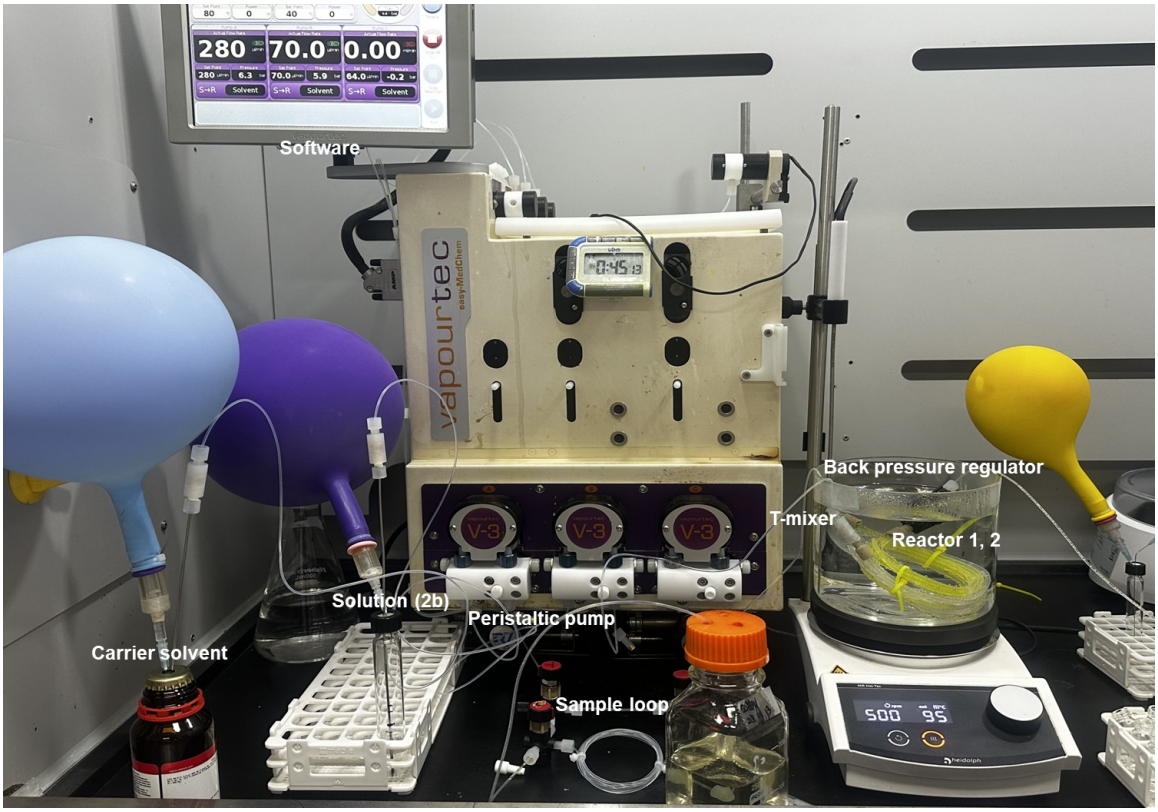
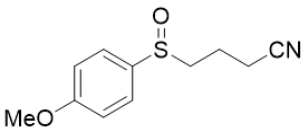


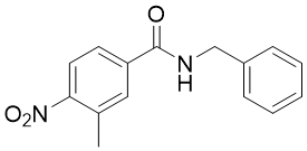
Figure S20. Actual KHU two-stage amidation apparatus, as supplied. Two E-series pump channels deliver the sample-loop/carrier stream and the benzylamine stream. The photograph identifies the sample loop, T-mixer, two reactor coils, BPR, and solution containers. It documents the assembled apparatus, not a complete time-resolved collection or calibration record.

8.5 Product characterization and supplied spectra

The characterization below is transcribed from KHU's supplied records with punctuation and section-reference corrections only. Reported masses, yields, shifts, couplings, and observed LRMS values are retained. Figures S21–S24 are product-characterization spectra; they are not presented as the missing raw quantitative-NMR yield calculations or as independent repeat measurements.



4-((4-methoxyphenyl)sulfinyl)butanenitrile (4a) was synthesized according to the standard flow procedure for Figure 5 (Section 8.3) and purified by flash column chromatography on silica gel (n-hexane:EtOAc = 1:4) to afford 4a as a colourless oil (42.0 mg, 94% isolated yield). Rf = 0.20 (n-hexane:EtOAc = 1:4). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.56–7.53 (m, 2H), 7.05–7.02 (m, 2H), 3.85 (s, 3H), 2.95–2.90 (m, 2H), 2.58–2.46 (m, 2H), 2.19–1.94 (m, 2H); $^{13}C\{^{1}H\}$ NMR (126 MHz, $CDCl_3$) δ 162.32, 133.66, 125.93, 118.60, 115.11, 55.70, 54.90, 18.67, 16.66; LRMS (ESI) m/z: calcd for $C_{11}H_{14}NO_2S$ $[M+H]^+$: 224.1; found: 224.4; calcd for $C_{11}H_{13}NO_2SK$ $[M+K]^+$: 262.0, found: 261.8. [11]



N-Benzyl-3-methyl-4-nitrobenzamide (3b) was synthesized according to the standard flow procedure for Figure 6 (Section 8.4) and purified by flash column chromatography on silica gel (n-hexane:EtOAc = 1:3) to afford 3b as a pale-yellow solid (227.0 mg, 84% isolated yield). Rf = 0.36 (n-hexane:EtOAc = 1:3). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.95 (d, J = 8.4 Hz, 1H), 7.77 (s, 1H), 7.68 (dd, J = 8.4, 2.0 Hz, 1H), 7.38–7.29 (m, 5H), 6.67–6.60 (br s, 1H), 4.62 (d, J = 5.4 Hz, 2H), 2.60 (br s, 3H); $^{13}C\{^{1}H\}$ NMR (126 MHz, $CDCl_3$) δ 165.64, 150.89, 138.33, 137.66, 134.16, 131.85, 129.02, 128.08, 128.01, 125.46, 125.03, 44.49, 20.44; LRMS (ESI): calcd for $C_{15}H_{15}N_2O_3$ $[M+H]^+$: 271.1, found: 271.4; calcd for $C_{15}H_{14}N_2O_3Na$ $[M+Na]^+$: 293.1, found: 293.1. [12-13]

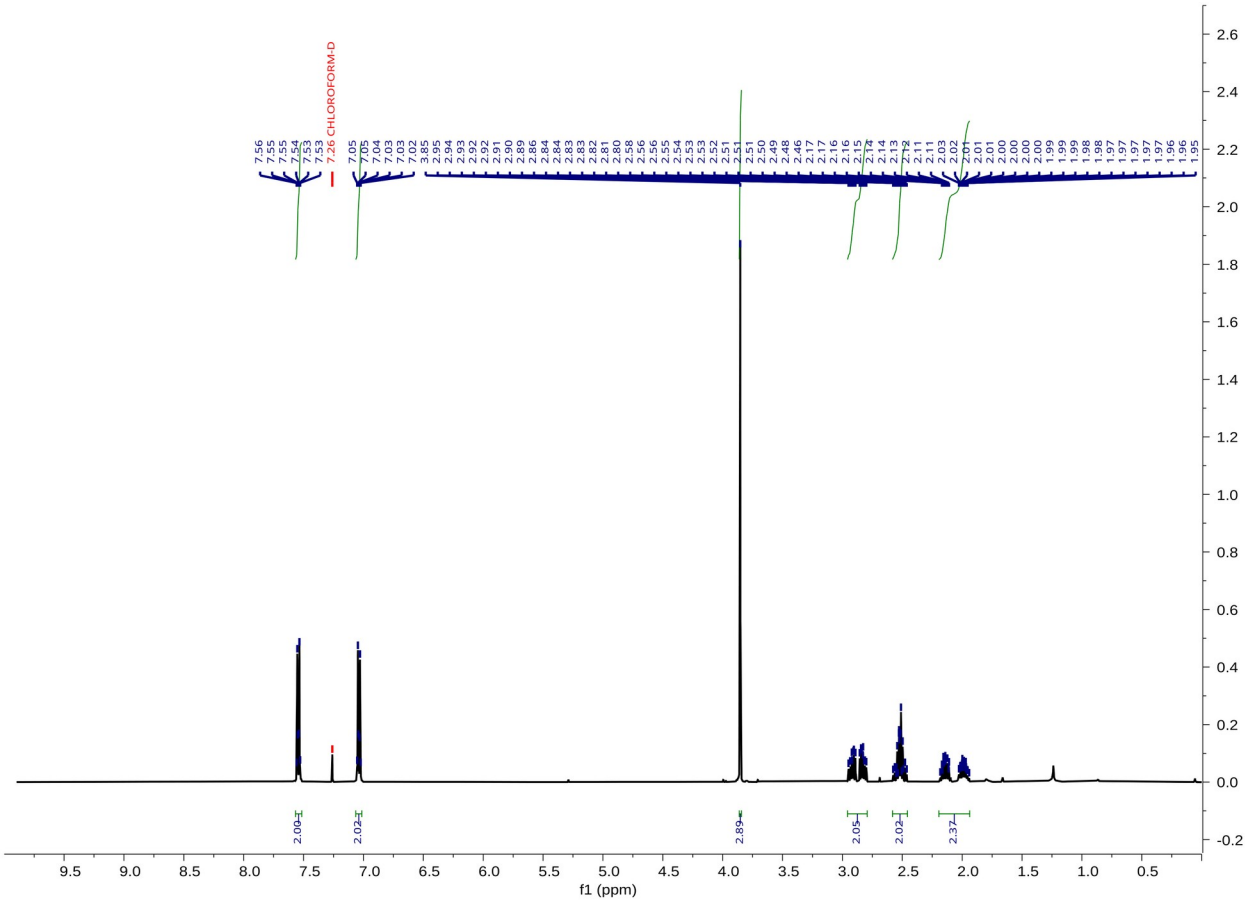


Figure S21. 4a: ^{1}H NMR (500 MHz, $CDCl_3$). Supplied KHU product-characterization spectrum, reproduced without peak reassignment, baseline correction, or deletion of signals.

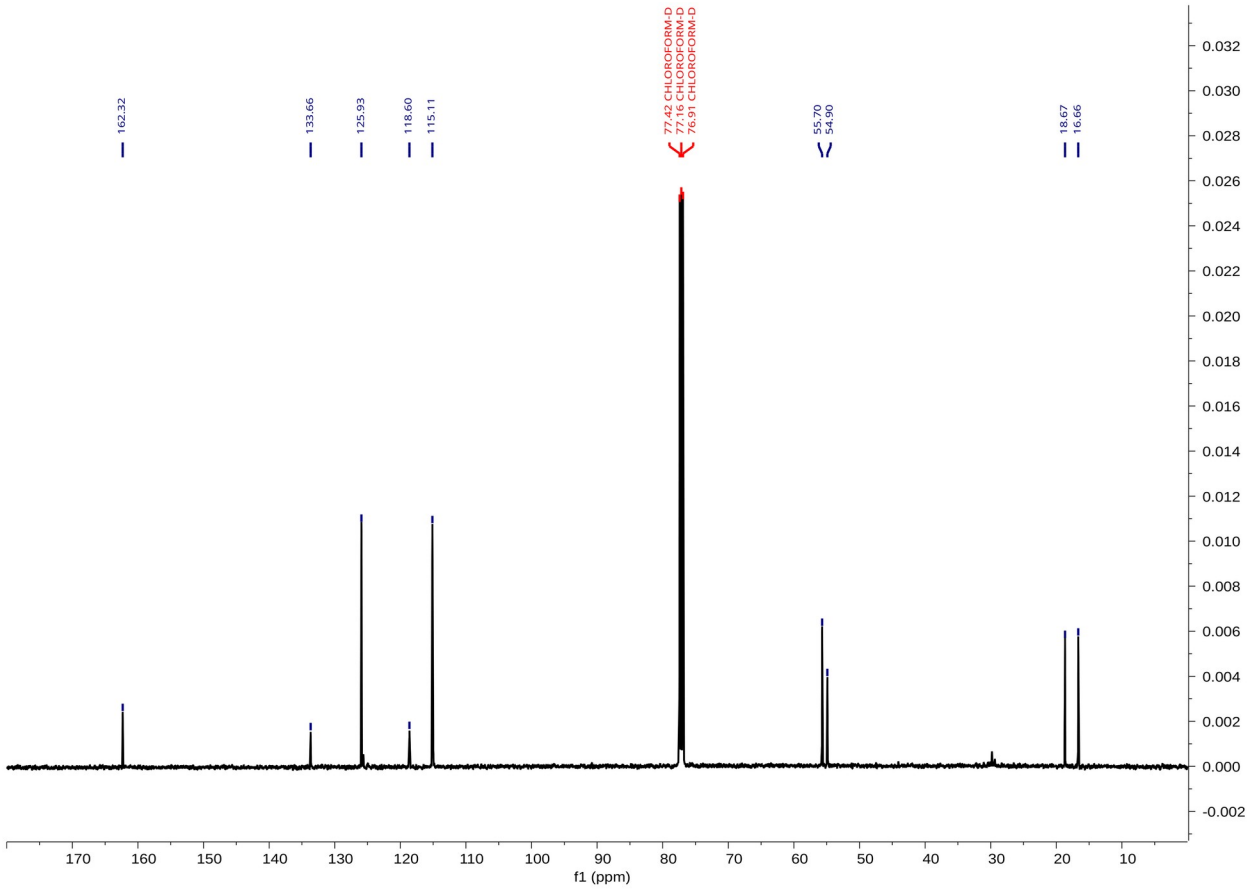


Figure S22. 4a: $^{13}C\{^{1}H\}$ NMR (126 MHz, $CDCl_3$). Supplied KHU product-characterization spectrum, reproduced without peak reassignment, baseline correction, or deletion of signals.

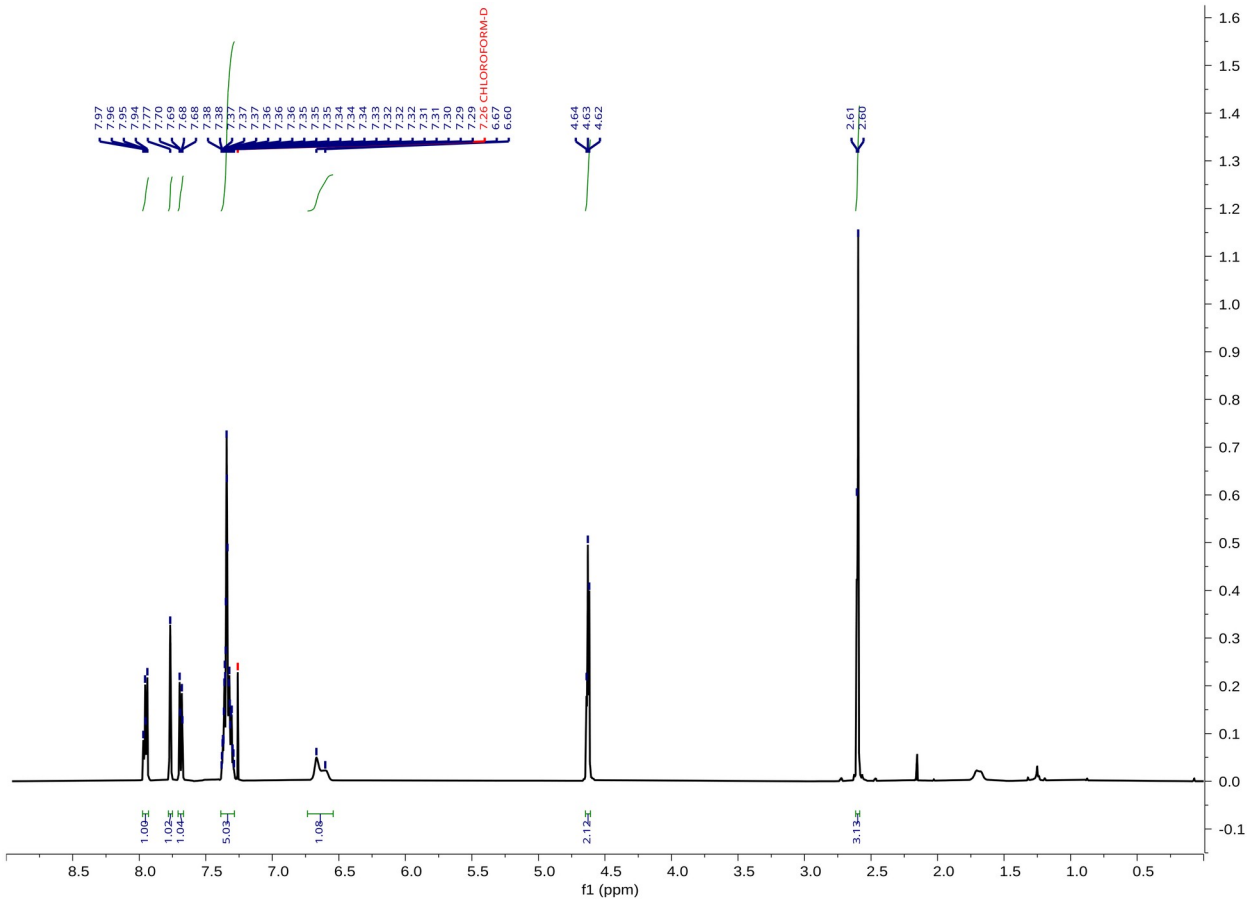


Figure S23. 3b: ^{1}H NMR (500 MHz, $CDCl_3$). Supplied KHU product-characterization spectrum, reproduced without peak reassignment, baseline correction, or deletion of signals.

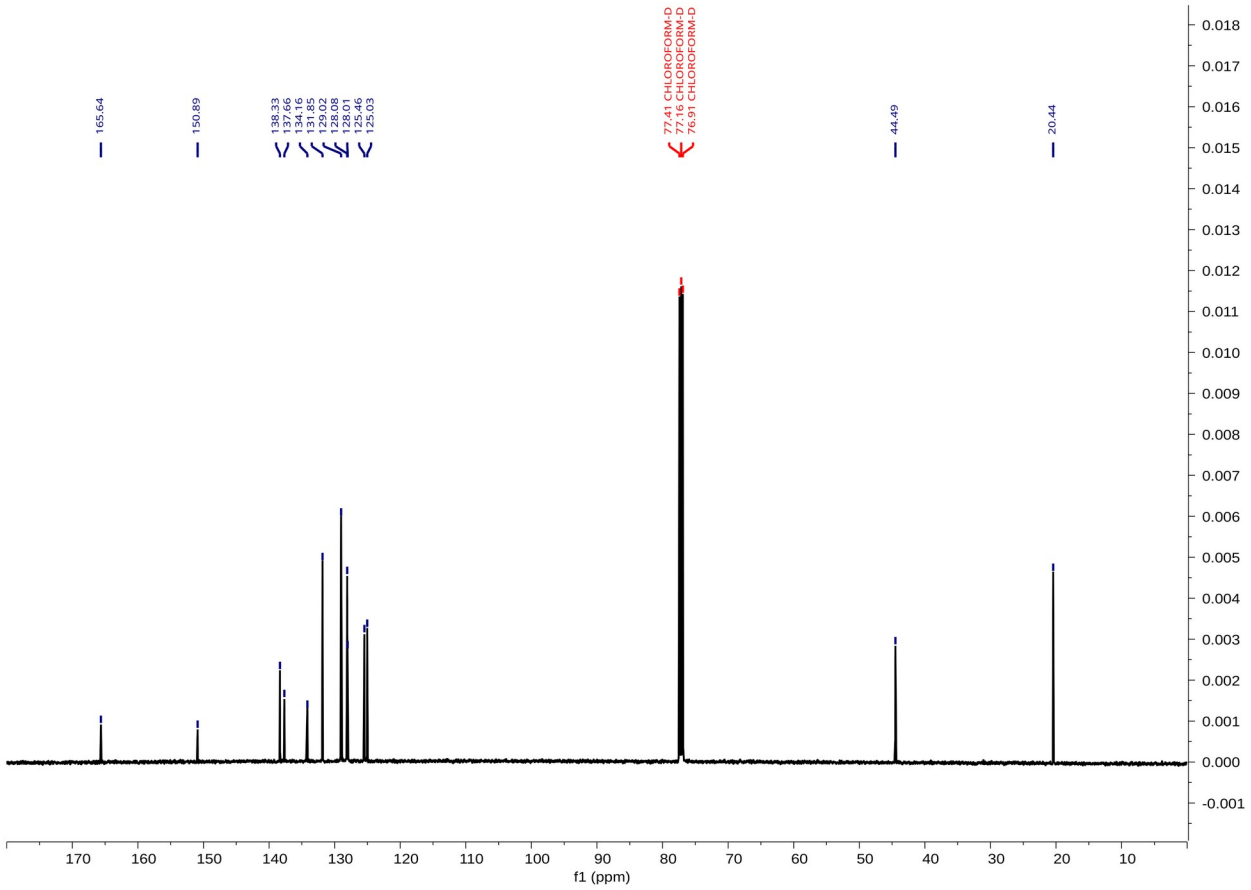


Figure S24. 3b: $^{13}C\{^{1}H\}$ NMR (126 MHz, $CDCl_3$). Supplied KHU product-characterization spectrum, reproduced without peak reassignment, baseline correction, or deletion of signals.

8.6 Implementation reconciliation and calculation basis

Table S17. Proposal-to-implementation reconciliation. Agreement of selected settings does not imply that unreported operational details were verified.

Feature	FlowPilot proposal / input	KHU implementation / remaining distinction
Pump configuration	Use compatible channels, avoid unnecessary mixing of automated systems.	Both cases used E-series pumps; **Figure 6** used two channels of the same system.
Photochemical reactor/light pairing	UV-150-compatible PFA first coil; separately illuminated FEP second coil.	PFA 2/5 mL with Vapourtec 450 nm LED; FEP 20 mL with blue strip.
Oxygen placement and quantity	Pure O_2 only after Stage 1; 0.090 mL/min at proposal STP convention.	O_2 introduced at the interstage T-mixer; same controller setpoint. MFC reference conditions/calibration not supplied.
Backflow protection	Gas-branch check valve included in revised proposal.	Photo labels a check valve; exact branch, orientation, and startup pressure trace need confirmation. No empirical failure probability is claimed.
Pressure	7 bar(g) design/BPR target.	7 bar cartridge installed; reported pressure 5.1–5.6 bar (**Figure 5**), 5.2–5.8 bar (**Figure 6**). Sensor basis/location unreported.
Feed introduction	Nominal component concentrations and pump settings.	Finite 2.0 mL sample-loop injection. Pump rates do not establish a constant concentration throughout the dispersed plug.
Amidation interstage operation	Direct amine addition, no intermediate isolation.	Two heated coils and interstage T-mixer, both at 95 °C. Water bath reported; no cooling operation described.
Tubing dimensions	Nominal inventory IDs: 1.00 mm and 2.40 mm.	Specified 0.04 in = 1.016 mm and 0.093 in = 2.3622 mm. Use actual tubing dimensions for future hydraulic calculations.
Performance assessment	Three response sets per chemistry, some identical conditions.	Two distinct implemented settings per chemistry. Shared-set rows are not documented independent repeats.

Table S18. Nominal component feeds from reported stock concentrations and pump settings. Molar feed is $C \times Q$ (M × mL/min = mmol/min). Values describe a stock-filled portion of the sample plug; no constant concentration or intermediate conversion is assumed outside that interval.

Case / set	Feed component	C (M)	Q (mL/min)	Molar feed (mmol/min)
Figure 5, all	Substrate 1a	0.100	0.020	0.00200
Figure 5, all	Acrylonitrile 2a	0.200	0.020	0.00400
Figure 5, all	Ir photocatalyst	0.000500	0.020	0.0000100
Figure 5, all	O_2 (STP calculation)	Gas	0.090	0.004015
Figure 6, 1	Acid 1b	0.500	0.160	0.0800
Figure 6, 1	DPDTC	0.525	0.160	0.0840
Figure 6, 1	DMAP	0.0500	0.160	0.00800
Figure 6, 1	Benzylamine 2b	2.10	0.040	0.0840
Figure 6, 2/3	Acid 1b	0.500	0.280	0.140
Figure 6, 2/3	DPDTC	0.525	0.280	0.147
Figure 6, 2/3	DMAP	0.0500	0.280	0.0140
Figure 6, 2/3	Benzylamine 2b	2.10	0.070	0.147

The literature benchmarks in main-text Figures 5 and 6 use verified NMR yields. They provide context rather than controlled comparisons: feed concentrations, irradiation, reactor size, reagent addition and gas delivery differ. No causal yield advantage or intensification factor is inferred. For the gas–liquid stage, the inlet-reference index is not a measurement of operating-pressure residence time or phase holdup.

The earlier KHU backflow report concerned pure oxygen at 0.43 mL min^{-1} and liquid at 0.020 mL min^{-1}, despite the original proposal's air-based feed calculation. The revised point retained pure oxygen at a reduced controller setting of 0.090 mL min^{-1}. A gas-branch check valve resists reverse liquid entry toward the MFC; it does not necessarily prevent gas entering the upstream liquid branch. The assembled photograph identifies a check valve, but the exact branch, orientation, and startup-pressure history remain unconfirmed. No backflow probability or verified hydraulic protection is inferred from the reduced gas setting.

8.7 Outstanding experimental metadata

The available record does not include raw quantitative-NMR files and yield-calculation worksheets (including internal-standard amount and purity), independent repeat identifiers, exact collection windows, or second-feed timing. Gas-controller reference conditions and calibration, pressure-sensor reference and location, the implemented oxygen-free feed preparation, and check-valve branch and orientation remain to be confirmed. Original NMR and LRMS instrument exports, the specified tubing dimensions, and the reported 95 °C bath arrangement also require final laboratory confirmation. These gaps are retained explicitly; no measurement or uncertainty has been inferred to fill them.

9 References

1. Pan, Y.; Xiao, Q.; Zhao, F.; Li, Z.; Liu, J.; Ullah, S.; Lim, K. H.; Huang, T.; Yu, Z.; Li, C.; Zhang, D.; Xue, Q.; Chen, Q.; Kawi, S.; Wang, Y.; Luo, G. Chat-microreactor: A large-language-model-based assistant for designing continuous flow systems. *Chemical Engineering Science* 2025, *311*, 121567. https://doi.org/10.1016/j.ces.2025.121567.

2. Ruan, Y.; Lu, C.; Xu, N.; He, Y.; Chen, Y.; Zhang, J.; Xuan, J.; Pan, J.; Fang, Q.; Gao, H.; Shen, X.; Ye, N.; Zhang, Q.; Mo, Y. An automatic end-to-end chemical synthesis development platform powered by large language models. *Nature Communications* 2024, *15*(1), 10160. https://doi.org/10.1038/s41467-024-54457-x.

3. Wang, Q.; Wang, Y.; Gong, X. An LLM-Based Intelligent Agent and Its Application in Making the Lanolin Saponification Process Greener. *Pharmaceuticals* 2026, *19* (2), 264. https://doi.org/10.3390/ph19020264.

4. Tian, X.; Du, W.; Yang, S.; Hu, H.; Xin, H.; Qu, S.; Ye, K. From Text to Simulation: A Multi-Agent LLM Workflow for Automated Chemical Process Design. *Proceedings of the AAAI Conference on Artificial Intelligence* 2026, *40* (35), 29705-29713. https://doi.org/10.1609/aaai.v40i35.40215.

5. Yang, Y.; Li, R.; Ma, J.; Zhang, K.; Liu, Q.; Han, J.; Bu, Y.; Zhou, J.; Lian, D.; Li, X.; Chen, E. CeProAgents: A Hierarchical Agents System for Automated Chemical Process Development. *arXiv* 2026, 2603.01654v1. Preprint. https://arxiv.org/abs/2603.01654v1.

6. Zhang, T. Harness as an Asset: Enforcing Determinism via the Convergent AI Agent Framework (CAAF). *arXiv* 2026, 2604.17025v3. Preprint. https://arxiv.org/abs/2604.17025v3.

7. Tu, J.; Sang, L.; Cheng, H.; Ai, N.; Zhang, J. Continuous Hydrogenolysis of N-Diphenylmethyl Groups in a Micropacked-Bed Reactor. Organic Process Research & Development 2020, 24, 59-66. https://doi.org/10.1021/acs.oprd.9b00416

8. Thomson, C. G.; Banks, C.; Allen, M.; Barker, G.; Coxon, C. R.; Lee, A.-L.; Vilela, F. Expanding the Tool Kit of Automated Flow Synthesis: Development of In-line Flash Chromatography Purification. Journal of Organic Chemistry 2021, 86, 14079-14094. https://doi.org/10.1021/acs.joc.1c01151

9. Fuchs, M.; Goessler, W.; Pilger, C.; Kappe, C. O. Mechanistic Insights into Copper(I)-Catalyzed Azide-Alkyne Cycloadditions using Continuous Flow Conditions. Advanced Synthesis & Catalysis 2010, 352, 323-328. https://doi.org/10.1002/adsc.200900726

10. Lu, Y.; Yan, Z.; Jiang, Y.; He, H.; Zhou, J.; Gao, S.; Xing, D. α-Bromination of Aldehydes by Continuous Flow Chemistry and Its Application to the Flow Synthesis of 2-Aminothiazoles. Organic Process Research & Development 2025, 29, 2047–2055. https://doi.org/10.1021/acs.oprd.5c00076.

11. Park, J.; Kim, S. H.; Cho, J.-Y.; Atriardi, S. R.; Kim, J.-Y.; Mardhiyah, H.; Park, B. Y.; Woo, S. K. Batch and Flow Synthesis of Sulfides and Sulfoxides Using Green Solvents and Oxidant through Visible-Light Photocatalysis. Green Chemistry 2025, 27, 3284-3292. https://doi.org/10.1039/D4GC05769D.

12. Saunders, J. M.; Oceguera Nava, E.; Li, J.; Wong, M. J.; Freiberg, K. M.; Lipshutz, B. H. Flow-to-Flow Technology: Amide Formation in the Absence of Traditional Coupling Reagents Using DPDTC. ACS Sustainable Chemistry & Engineering 2025, 13, 6646-6655. https://doi.org/10.1021/acssuschemeng.5c00914.

13. Wang, A.; Xie, Y.; Wang, J.; Shi, D.; Yu, H. Atom-economic amide synthesis by using an iron-substituted polyoxometalate catalyst. Chemical Communications 2022, 58, 1127–1130. https://doi.org/10.1039/D1CC05417A.